

**K. E. T's V. G. Vaze College of Arts, Science and Commerce
(Autonomous), Mulund (E)**

Roll No.: - A046

Name: - Manasvi Mahesh Taral

PRACTICAL NO: - 01

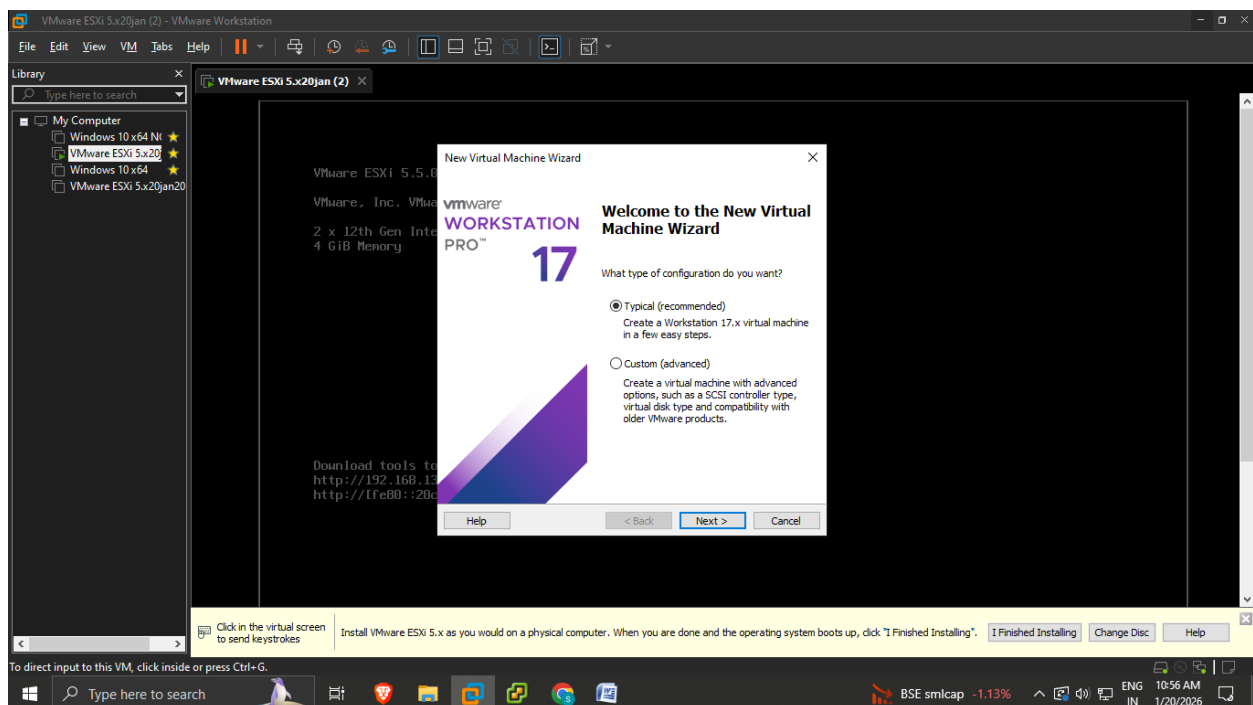
Date: - 20/01/2026

1. Implement Windows/ Linux Cluster.

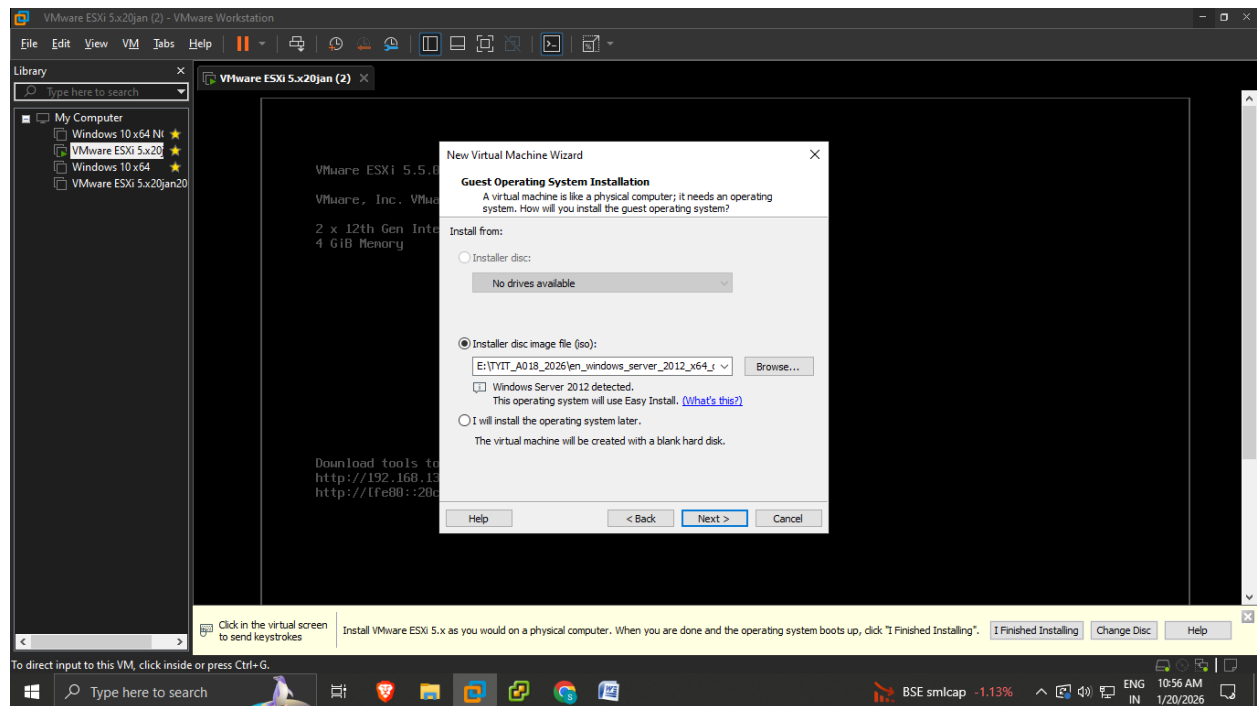
Windows Server Failover Clustering (WSFC) is a high-availability feature that groups independent servers (nodes) together to ensure that critical applications and services remain available during hardware or software failures. This technology is core Components of Modern data centers. Windows Server Failover Clustering uses “Heartbeat” signals for continuous health checks and when a node fails, these signals trigger automatic failover, moving applications/services to a healthy node for high availability and minimal downtime.

STEPS

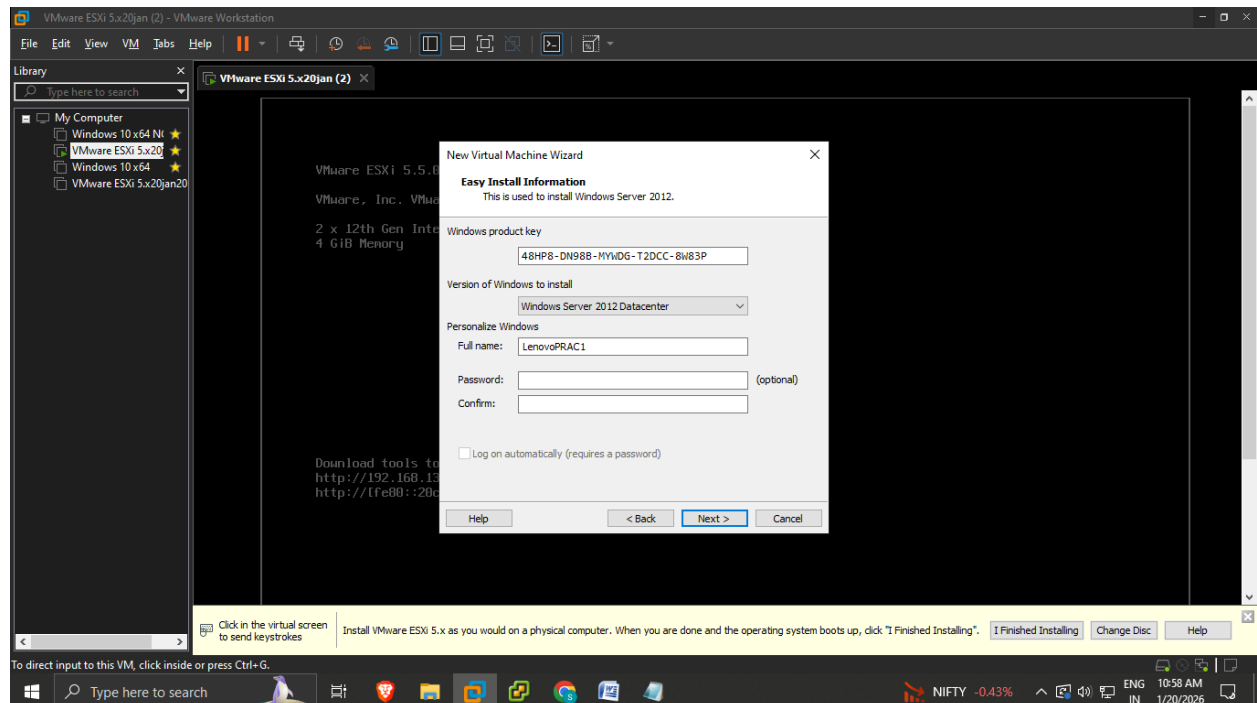
Create New Virtual Machine.

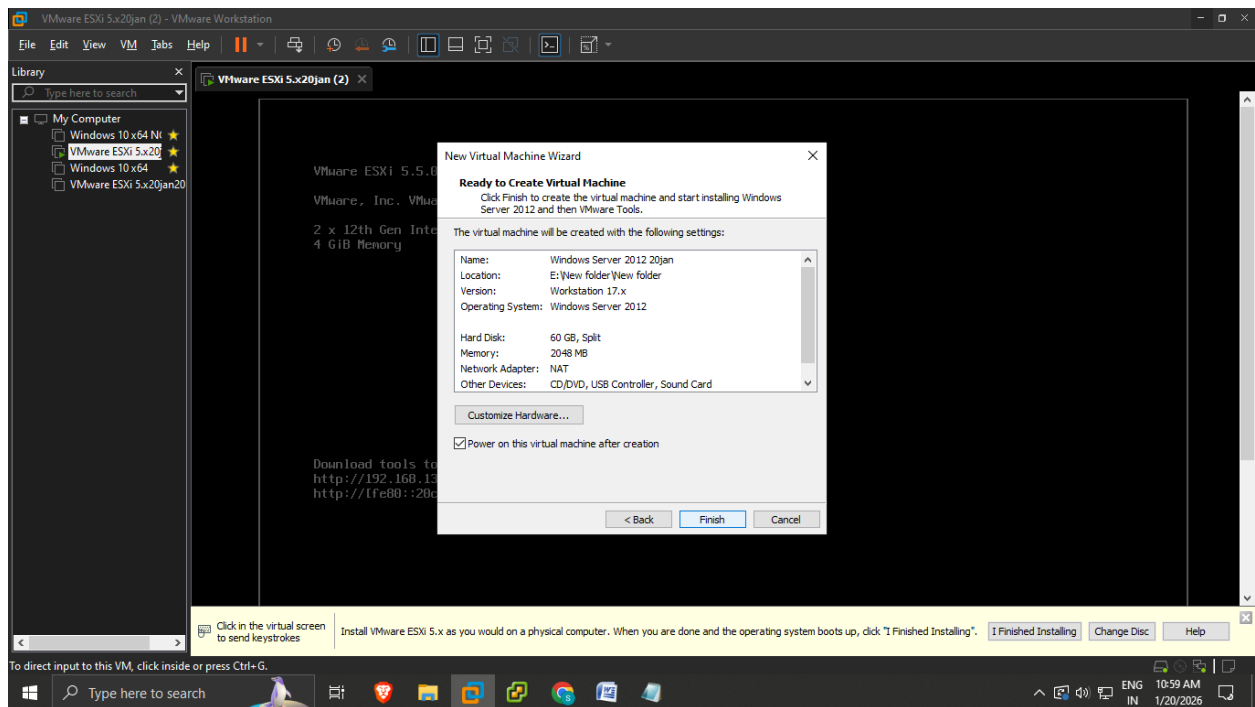


Browse windows server 2012 ISO File.



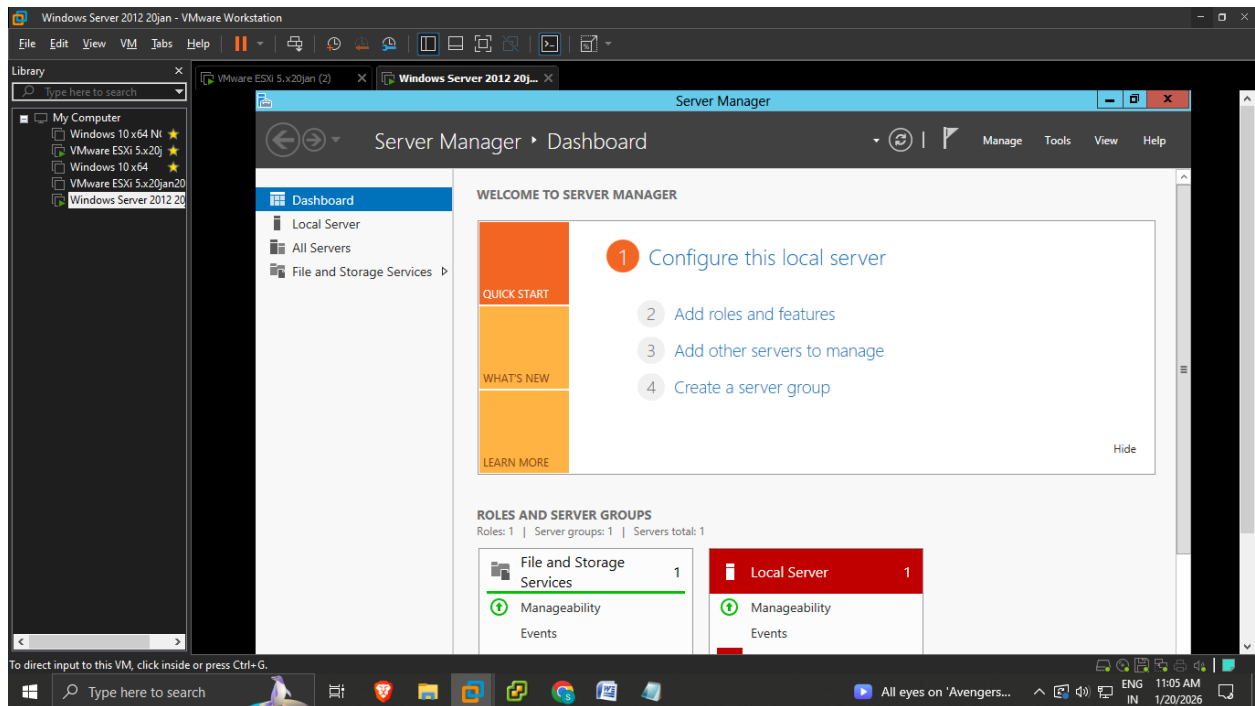
Write Key.

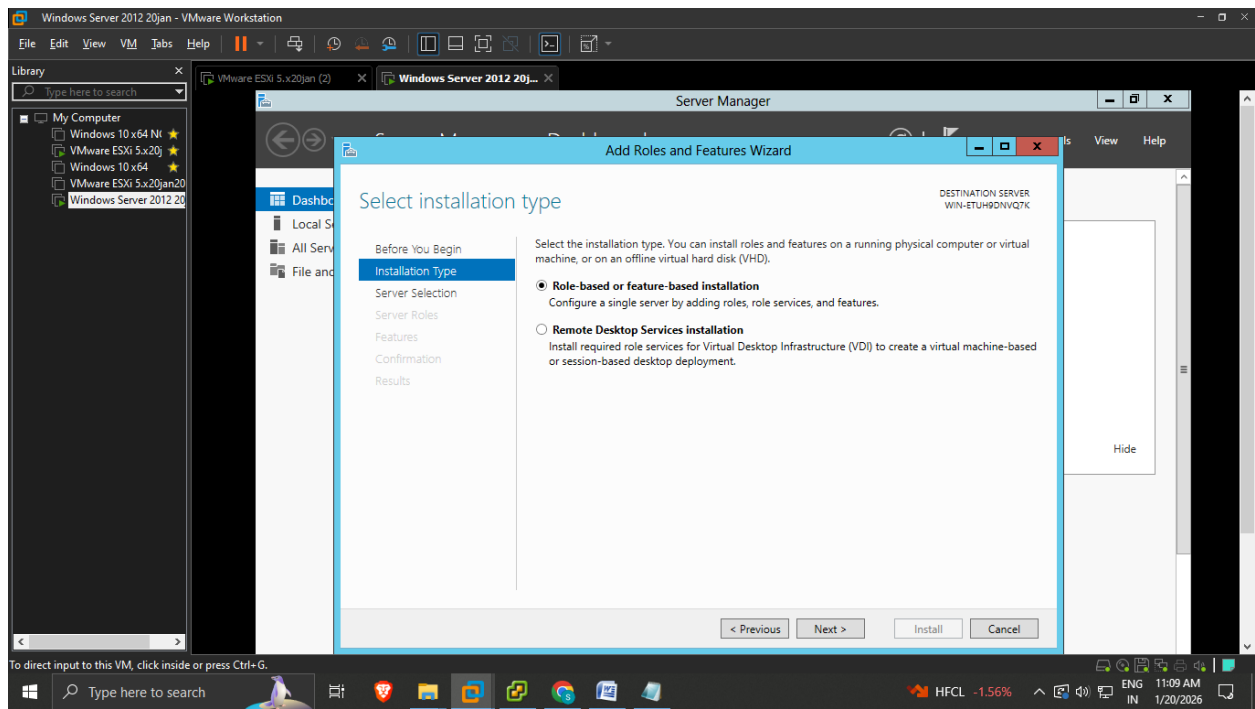
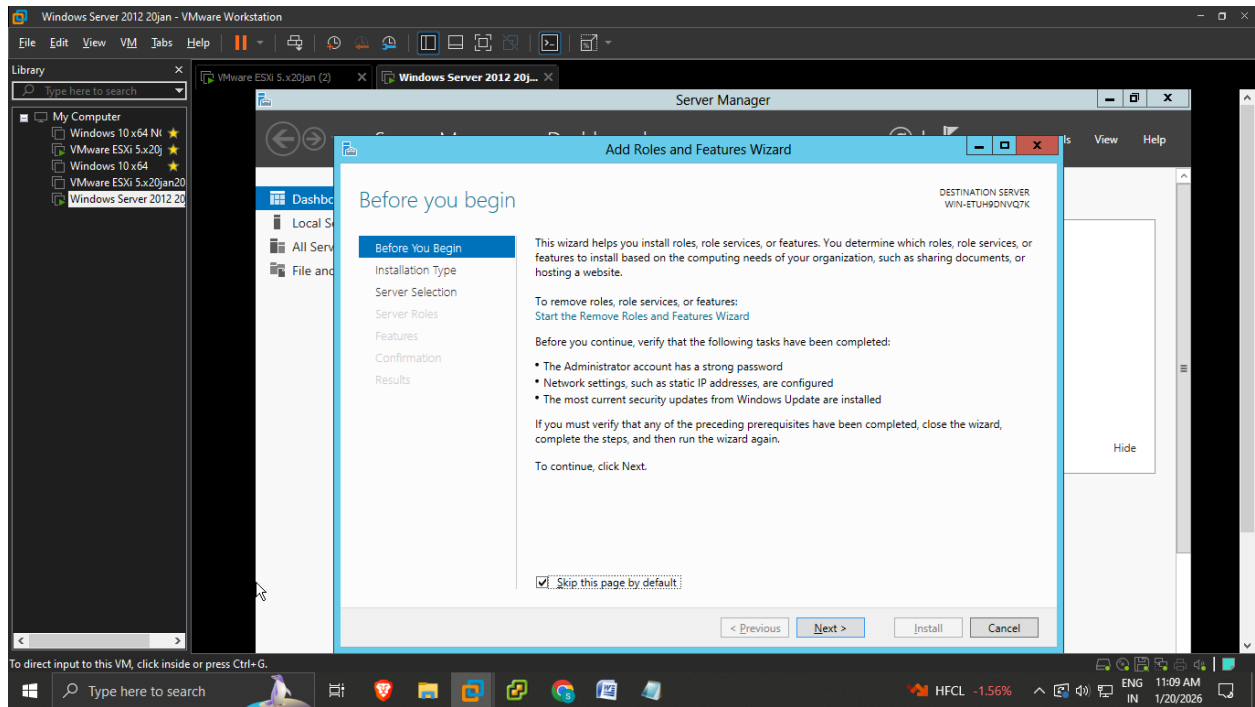


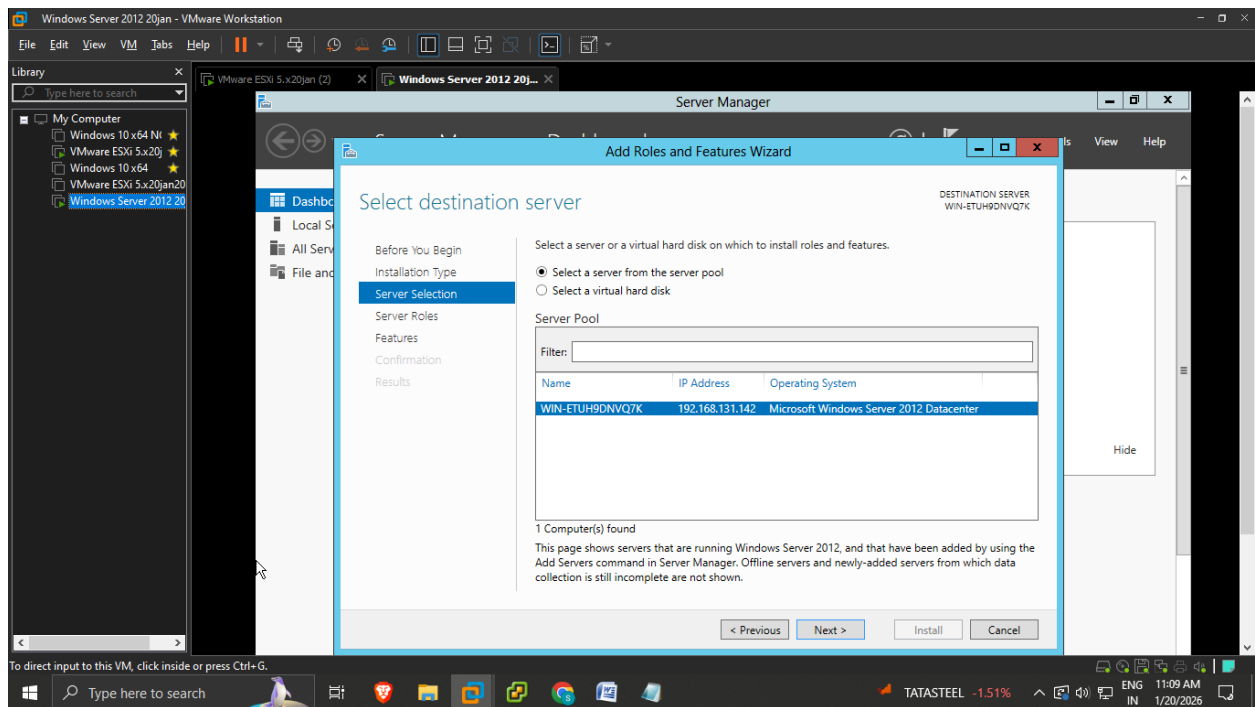


Installation Complete

Click on Add roles and Features.



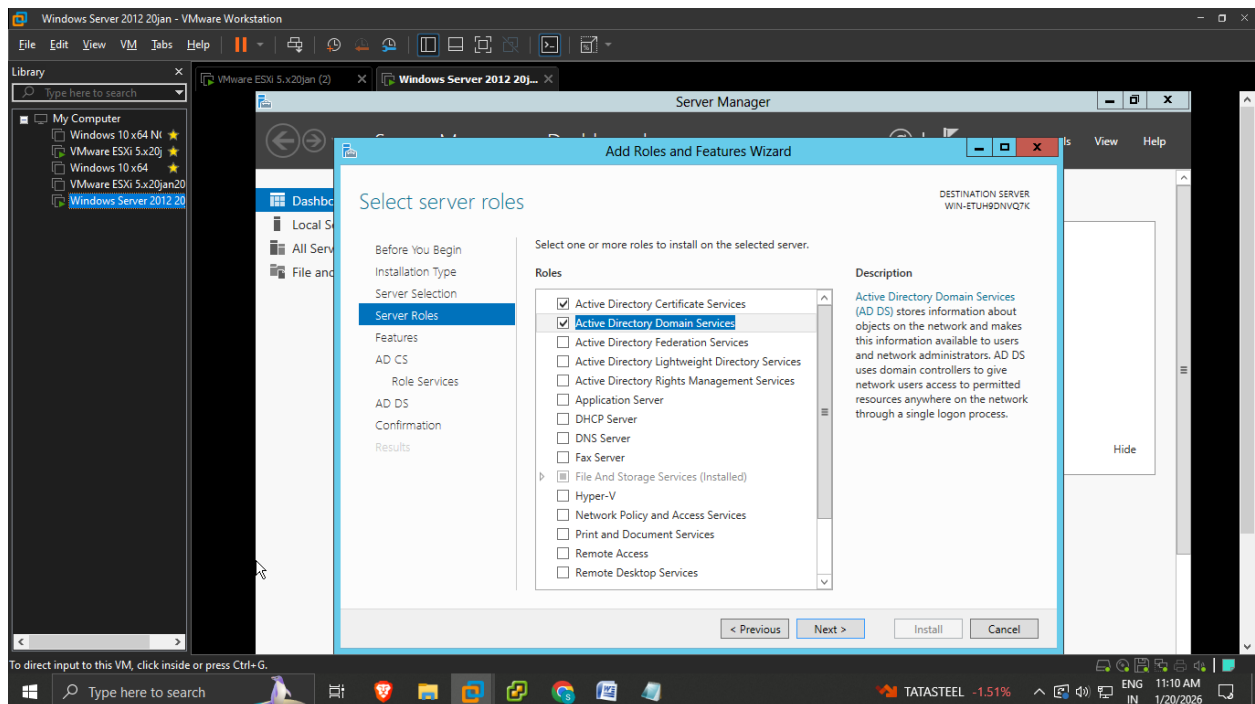


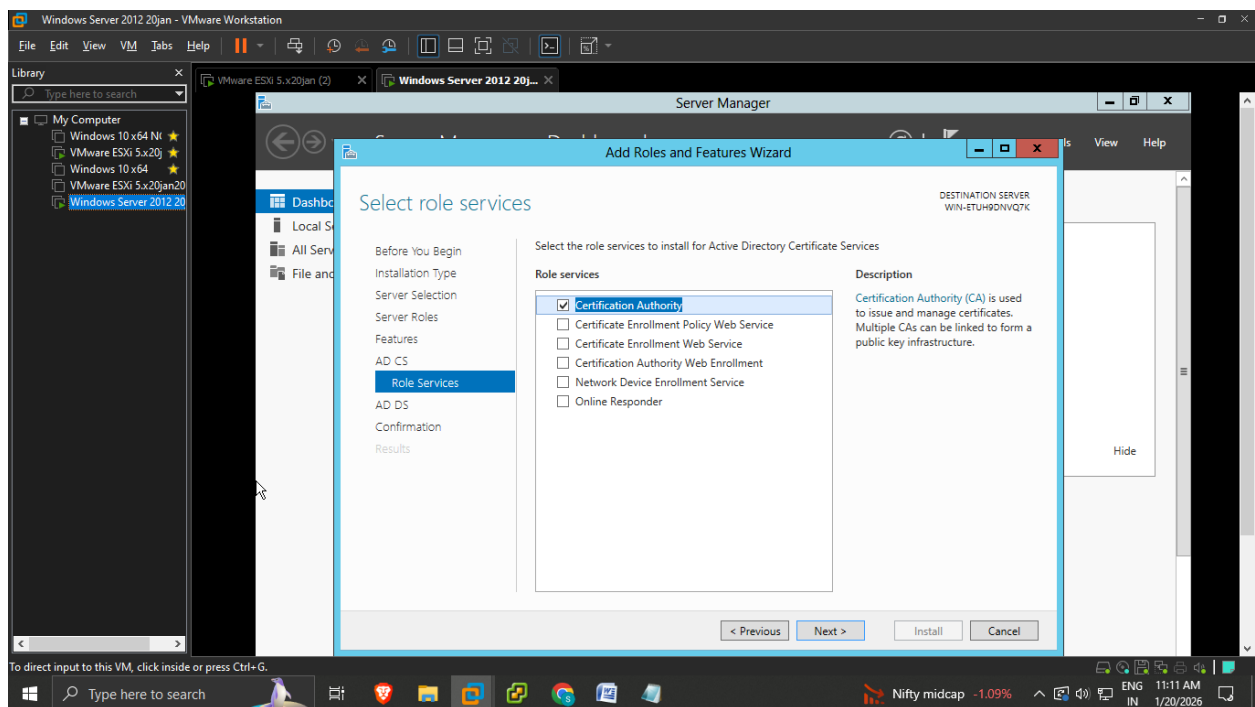
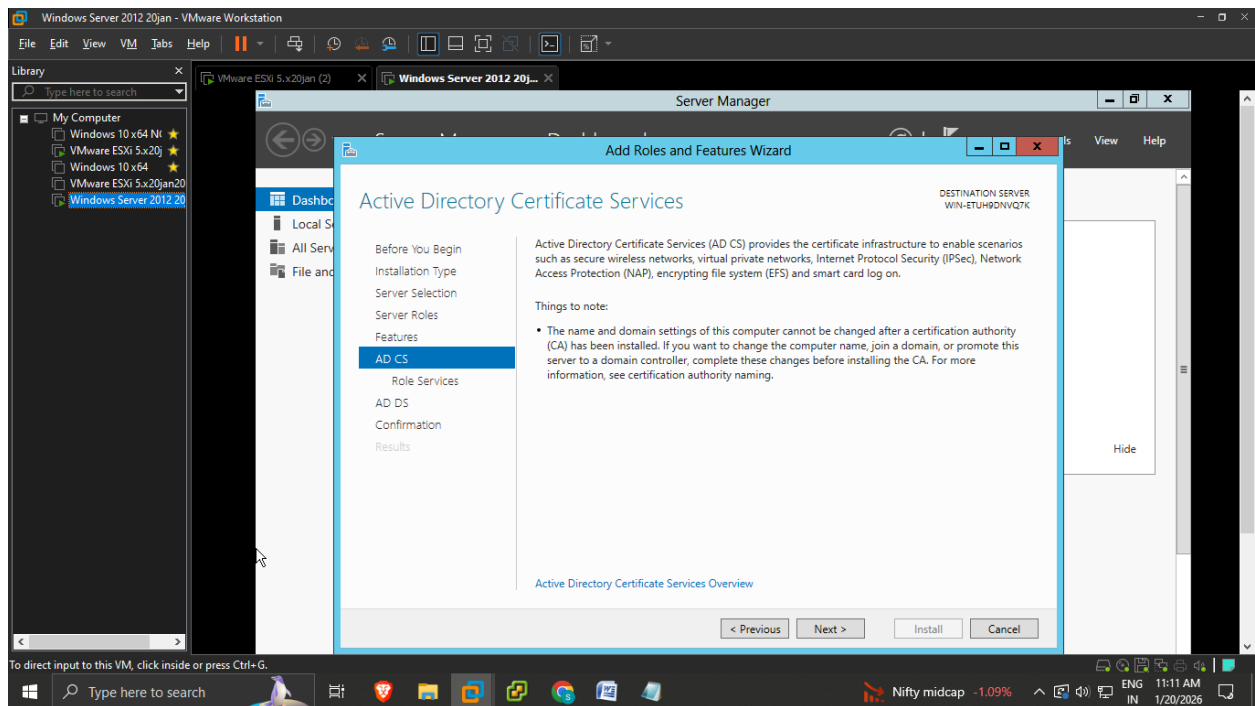


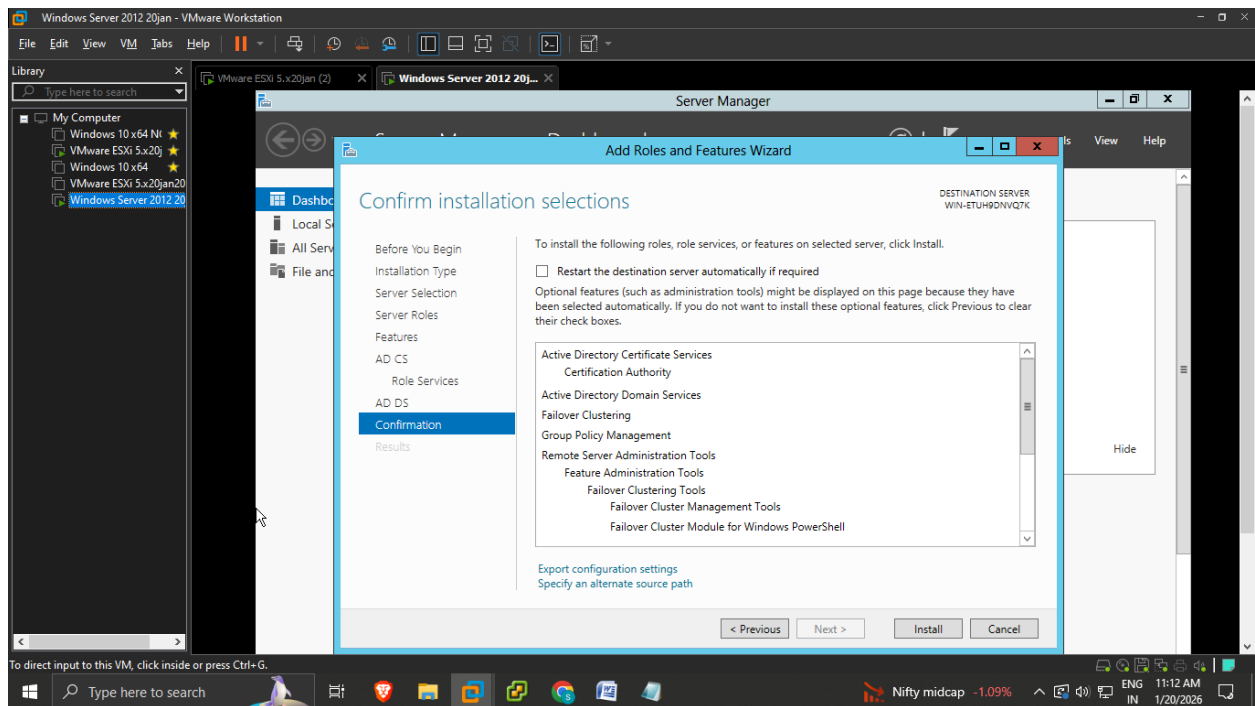
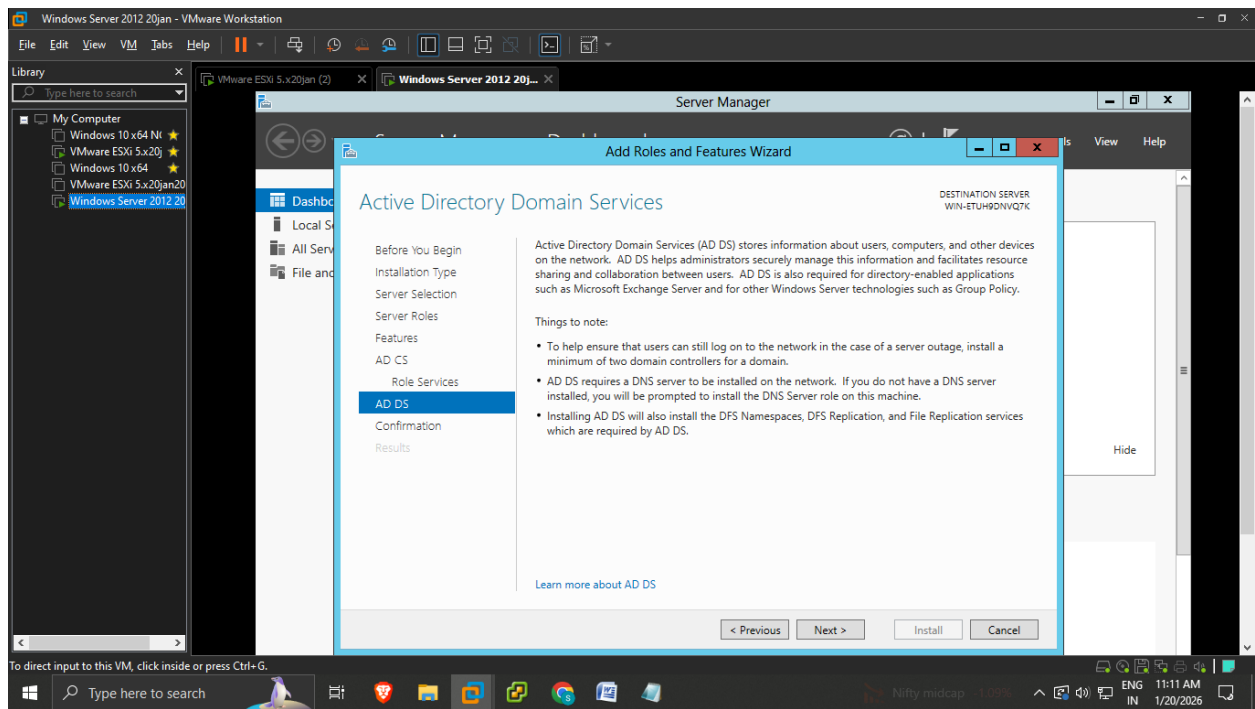
Adding Roles:

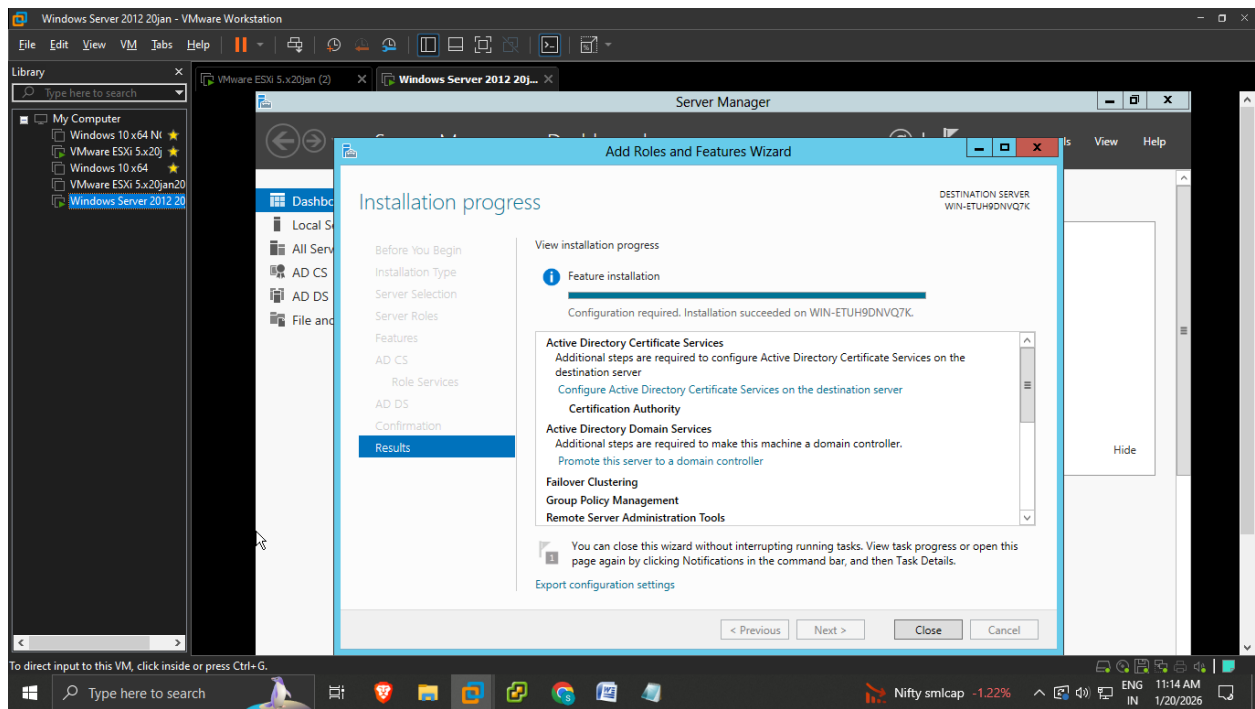
- 1) Active Directory Certificate Services
- 2) Active Directory Domain Services

Adding Features: Failover Cluster



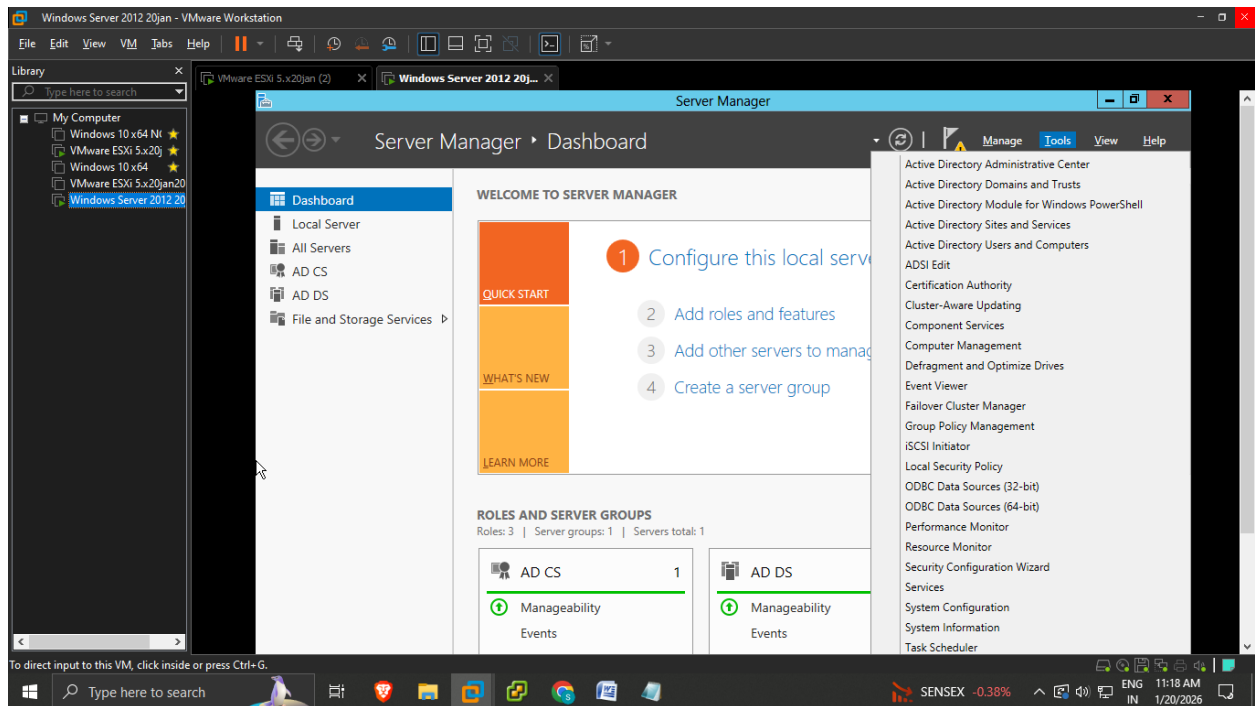


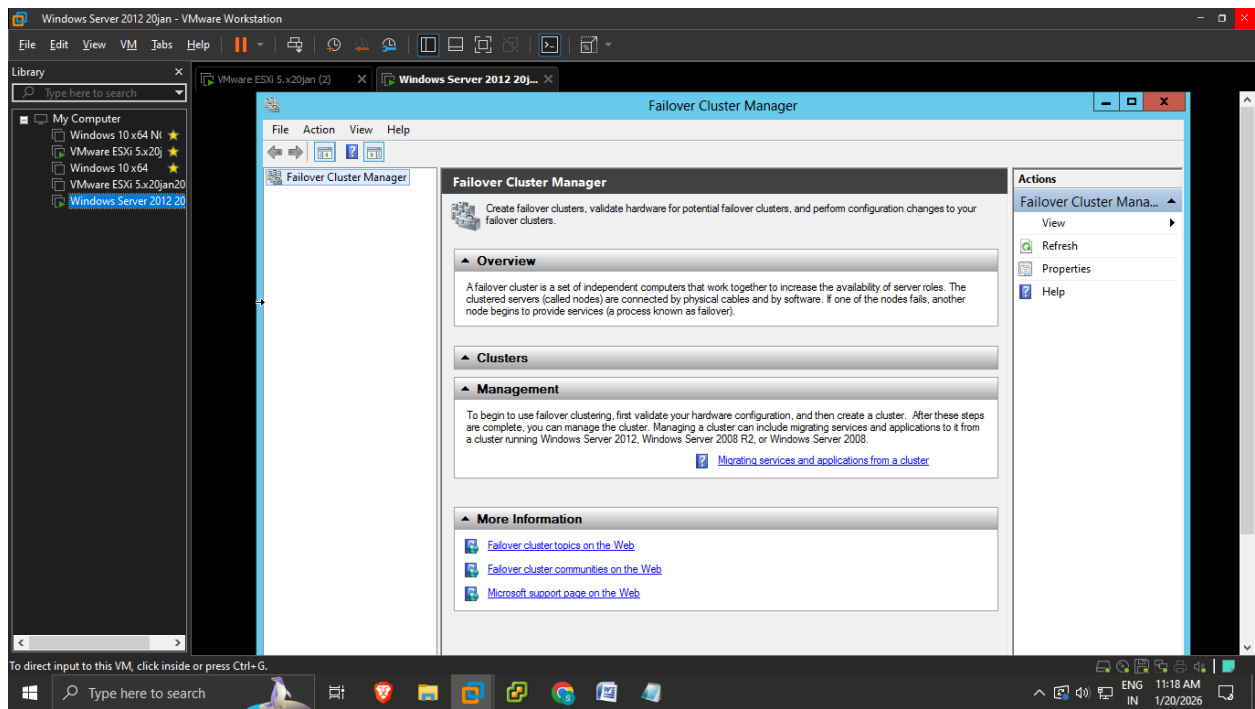




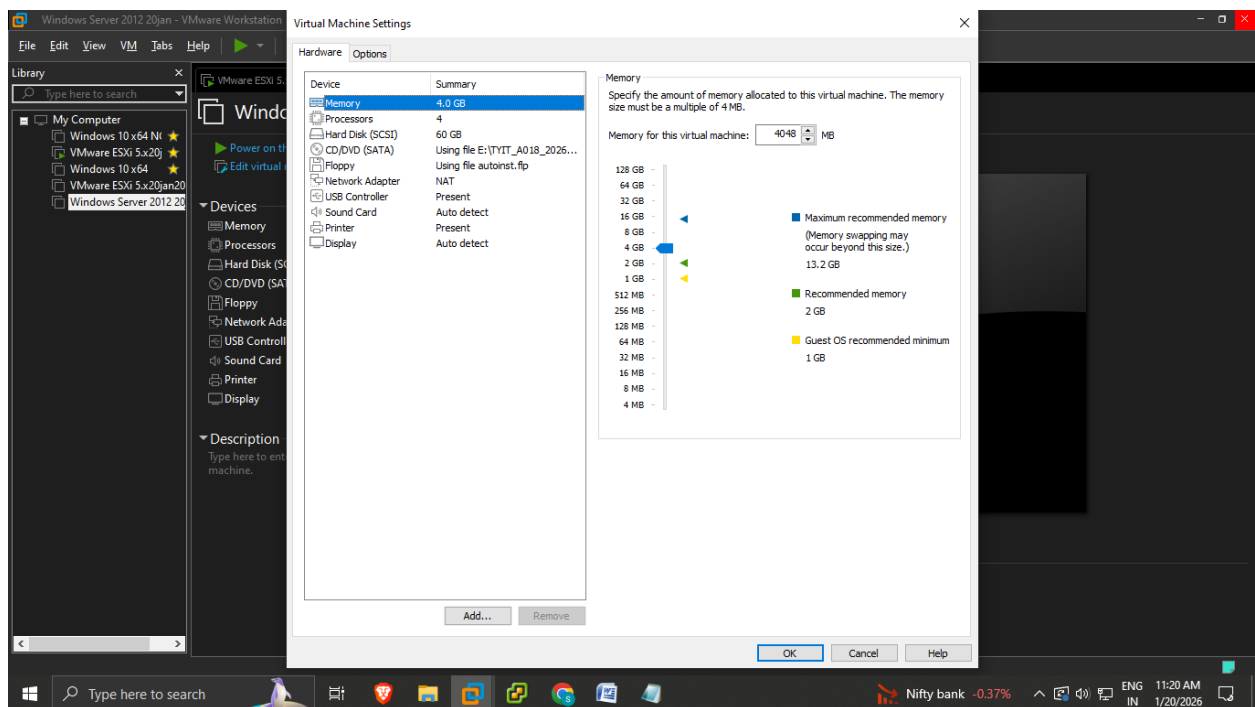
Installation Done

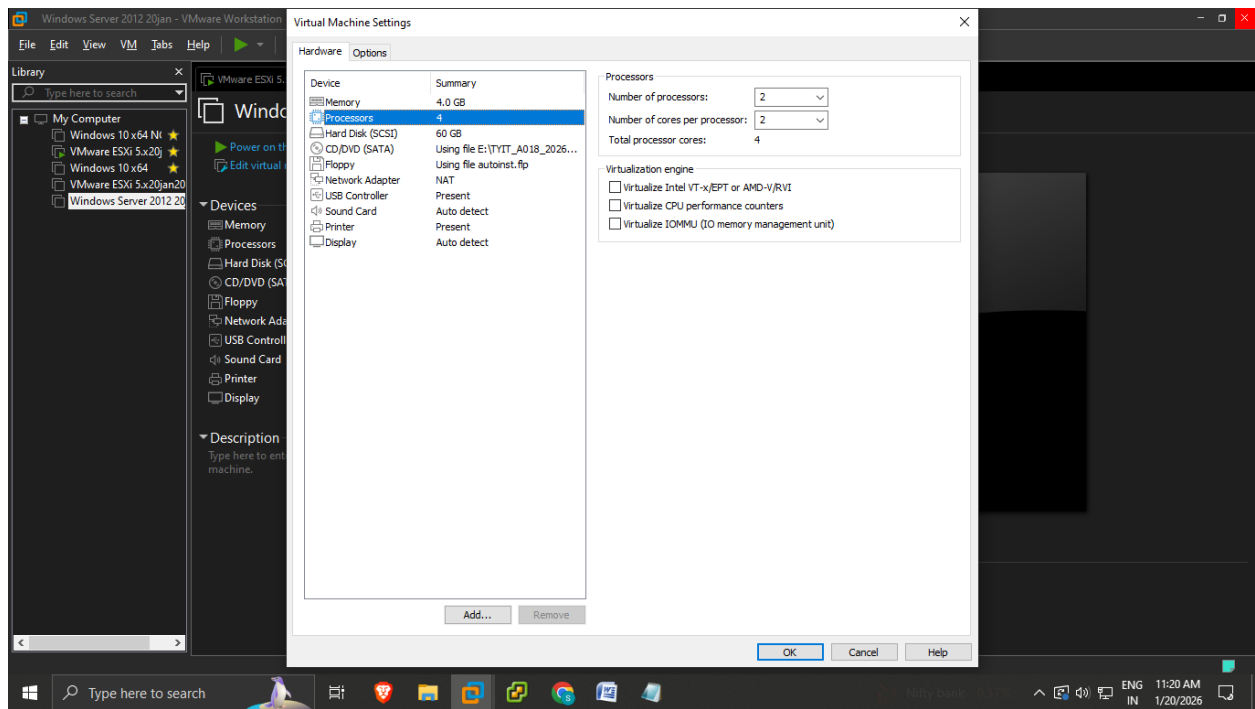
Click on Tools tab and click on failover cluster manager.





Power off and go in the settings and change memory and processors.





Power on machine.

PRACTICAL NO: - 03

Date: - 07/01/2026

3. Implementing private cloud with Xen Server.

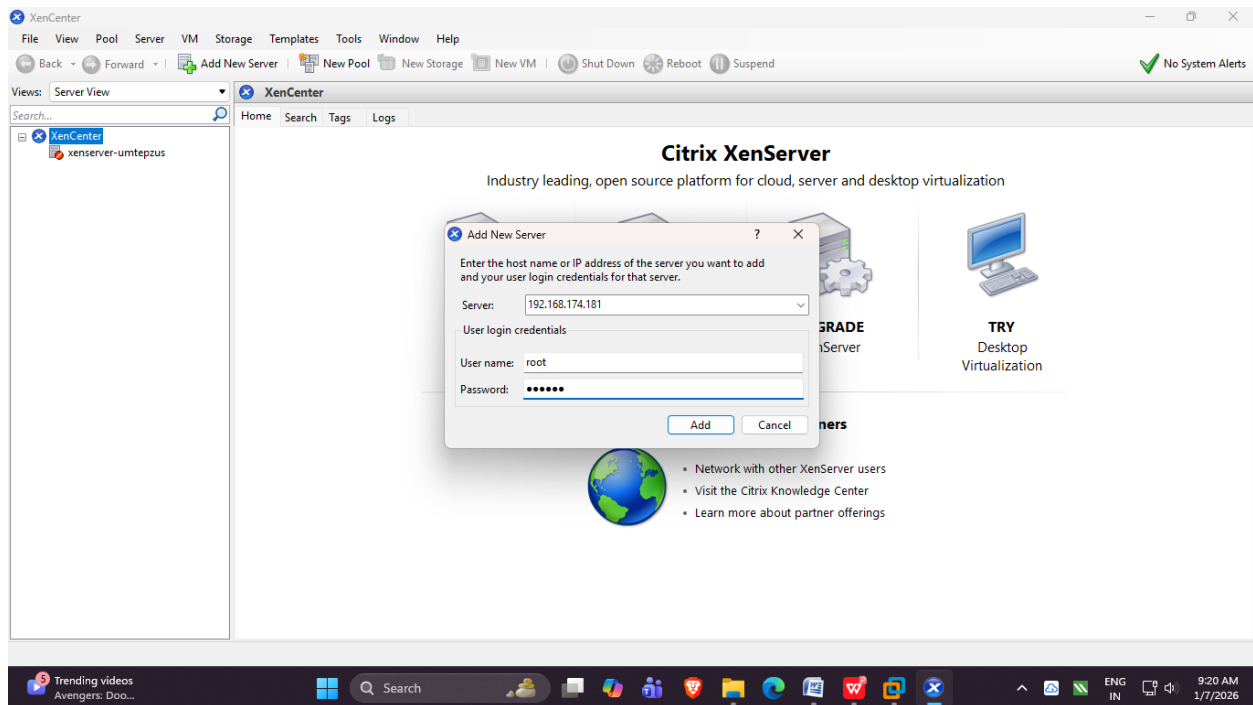
XenServer is a virtualization software that lets one physical server run many virtual machines at the same time. It uses the Xen hypervisor and runs directly on the hardware, which makes it fast and efficient. It was developed by Citrix and is used in data centers to save resources and manage servers easily. XenServer supports features like moving virtual machines without stopping them, backups, and high availability. It is now called Citrix Hypervisor.

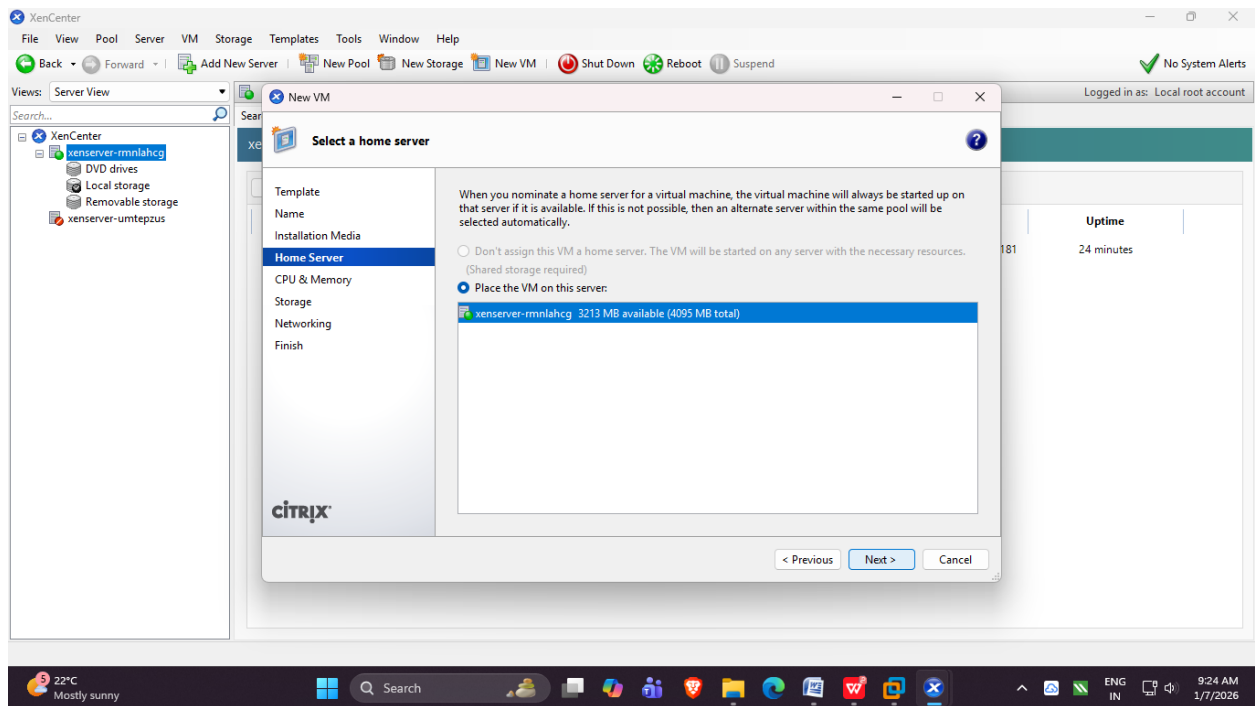
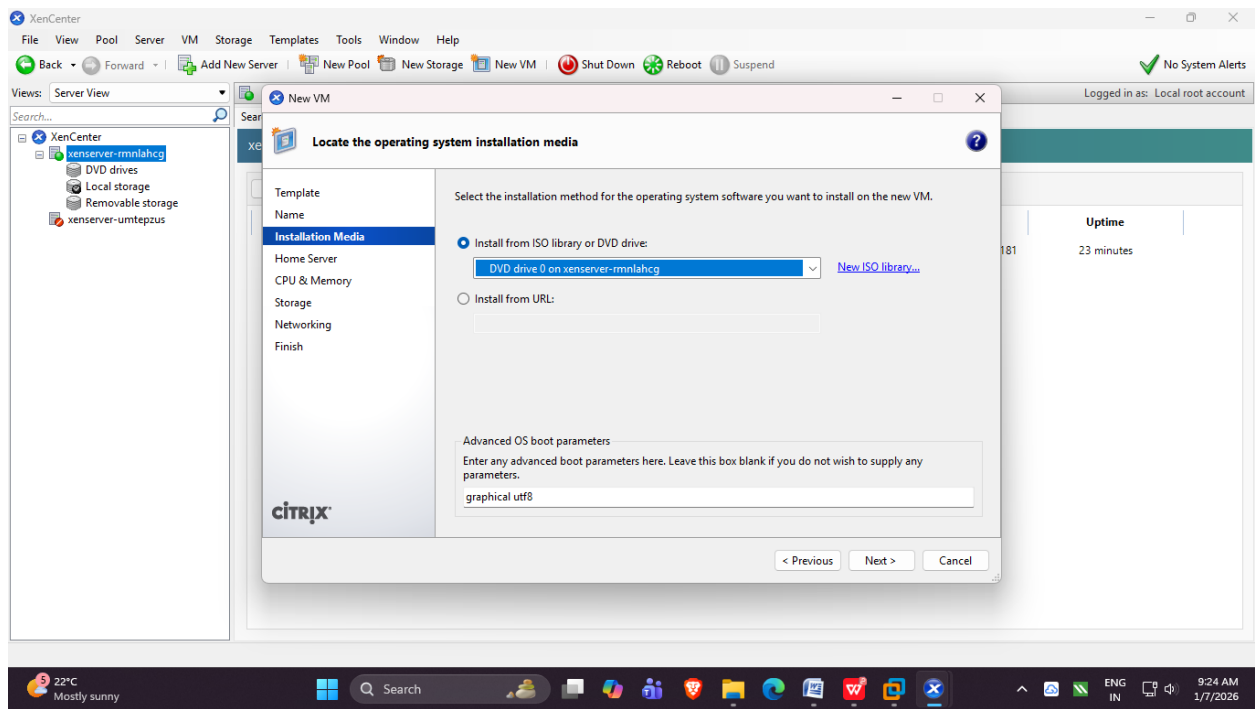
STEPS

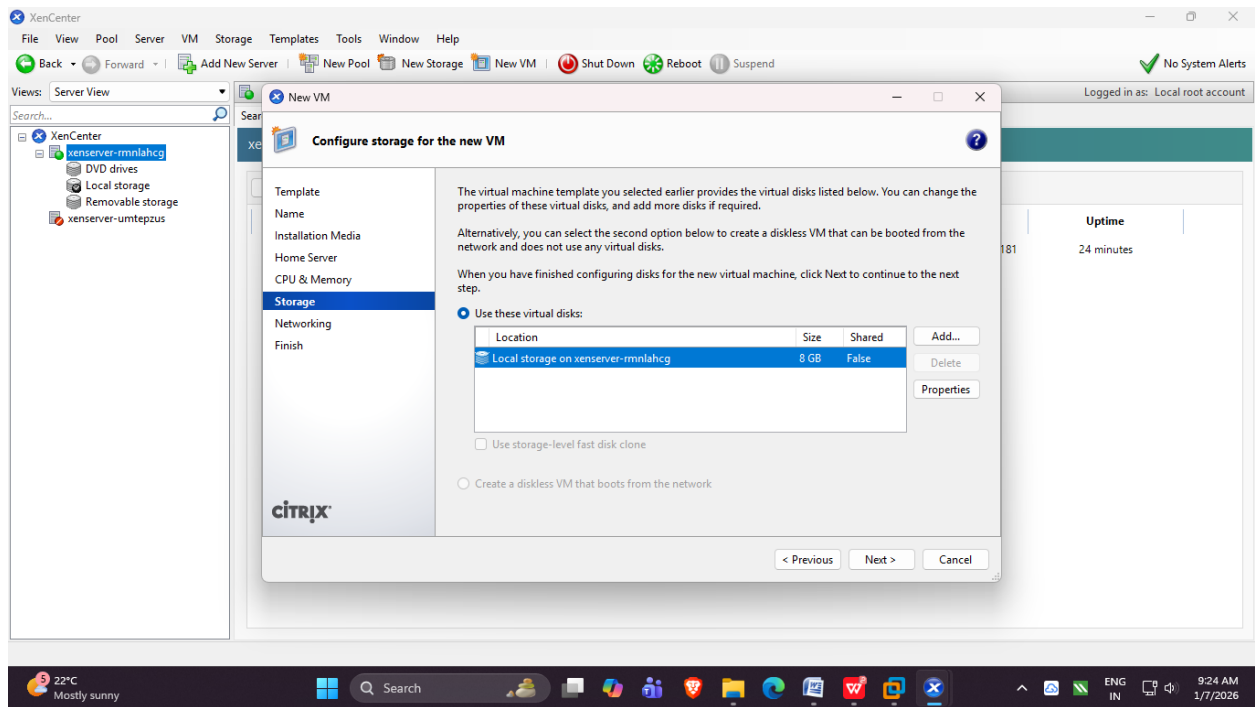
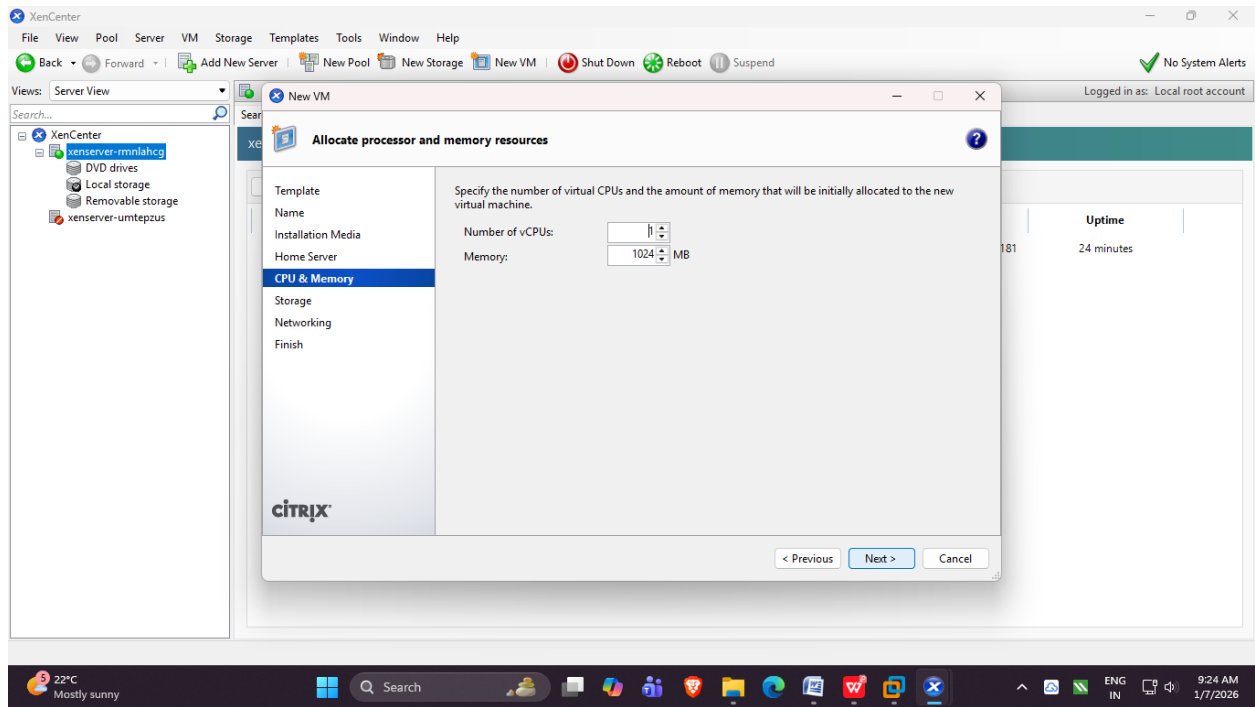
Install xenCenter.msi

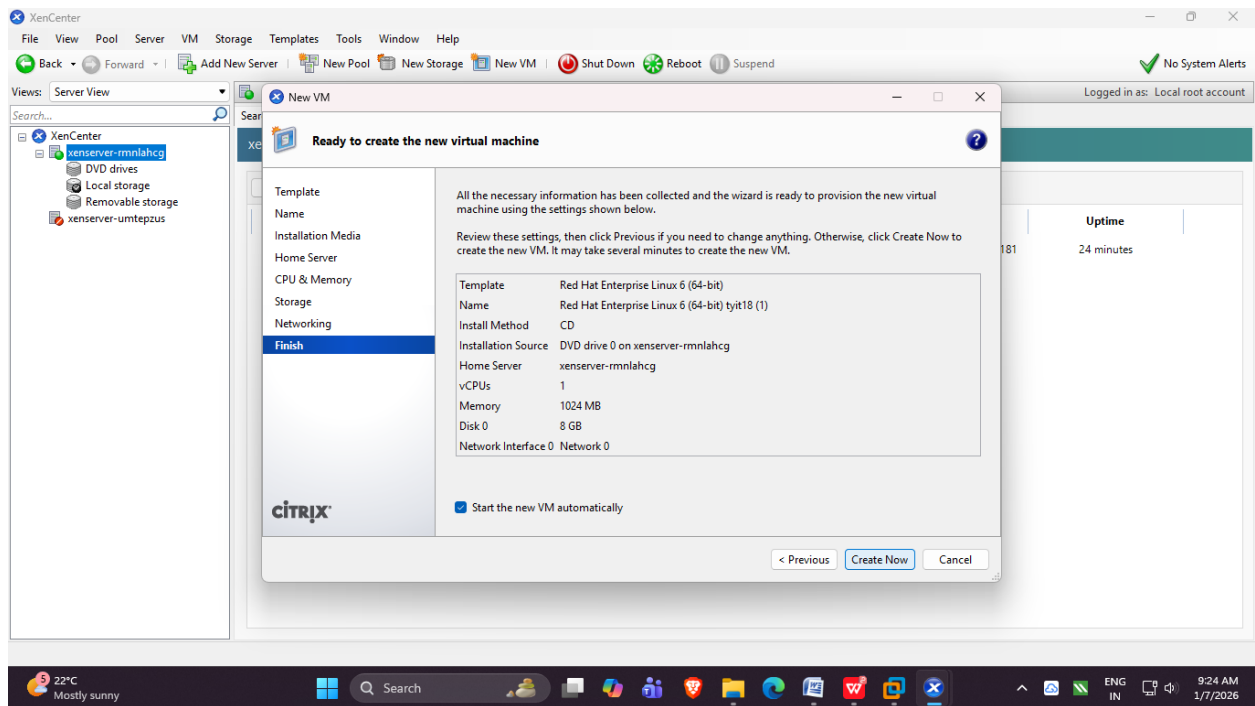
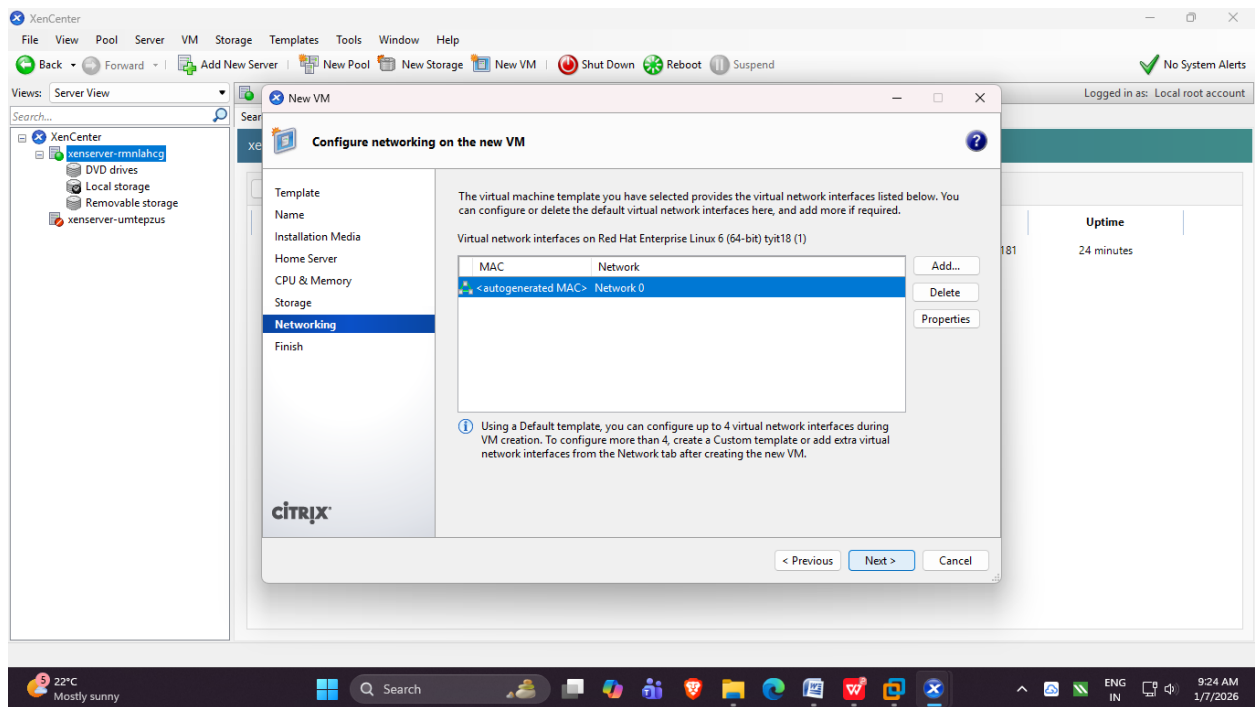
Open the app

Add server









XenCenter

FileViewPoolServerVMStorageTemplatesToolsWindowHelp

BackForwardAdd New ServerNew PoolNew StorageNew VMShut DownRebootSuspend

Views: Server View

Search...

TYIT Pool

TYIT Pool Overview

Search Options

Name	CPU Usage	Used Memory	Disks (avg / max KBs)	Network (avg / max KBs)	Address	Uptime
TYIT Pool	-	-	-	-	-	-
xenserver-rmnlahcg			-	0/0	192.168.174.181	42 minutes
Windows 7 (32-bit) (1)	-	-	-	-	-	-

22°C
Mostly sunny

Search

ENG
IN

9:42 AM
1/7/2026

PRACTICAL NO: - 04

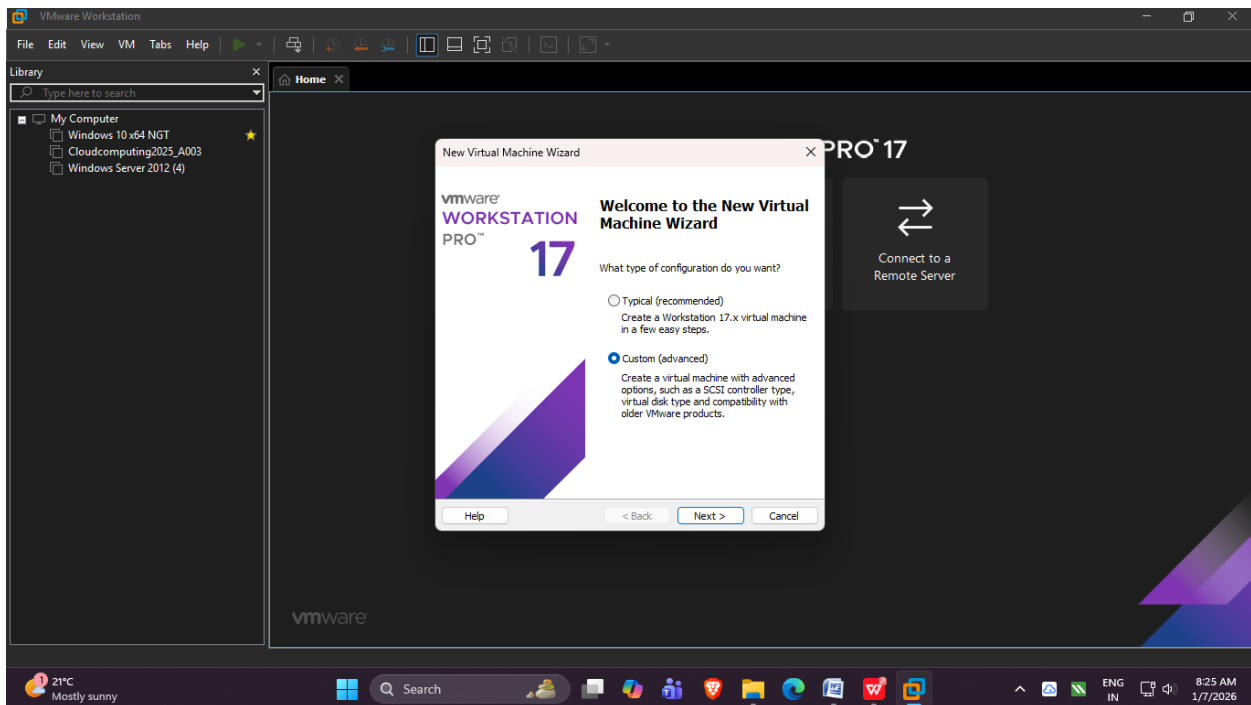
Date: - 07/01/2025

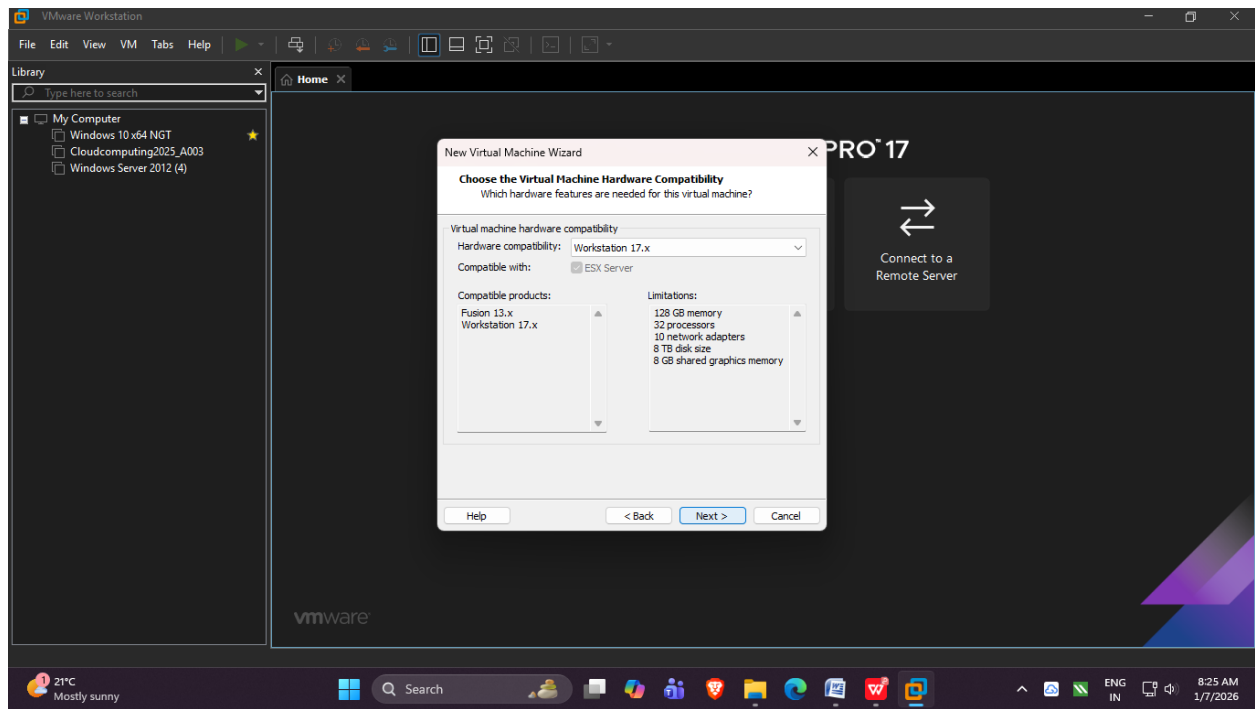
4. Implementing operating system on XenServer.

Implementing an operating system on VMware ESXi 7 involves creating virtual machines (VMs) on a physical server using a Type-1 bare-metal hypervisor that installs directly on hardware without a host OS. This architecture delivers superior performance, security, and stability compared to Type-2 hypervisors. After ESXi 7 installation, administrators can deploy multiple independent VMs running various operating systems like Windows Server or Linux, each with dedicated CPU, RAM, storage, and network resources for efficient hardware utilization and server consolidation.

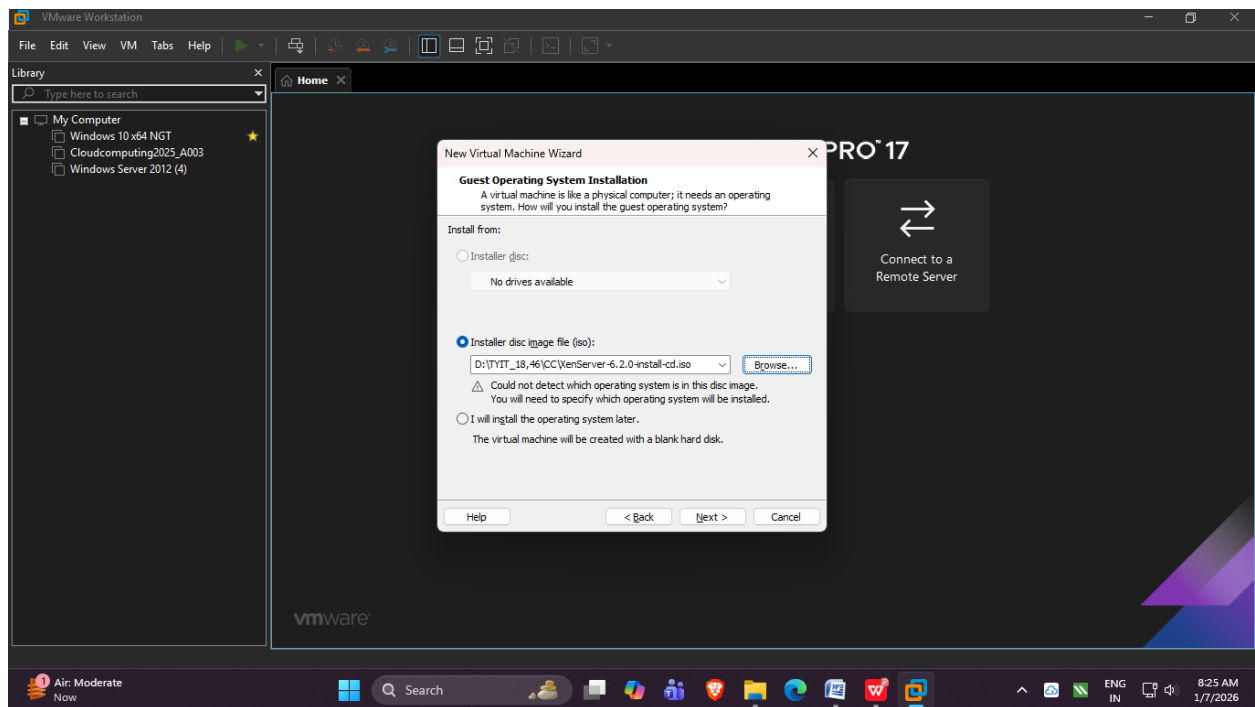
STEPS

Create virtual machine

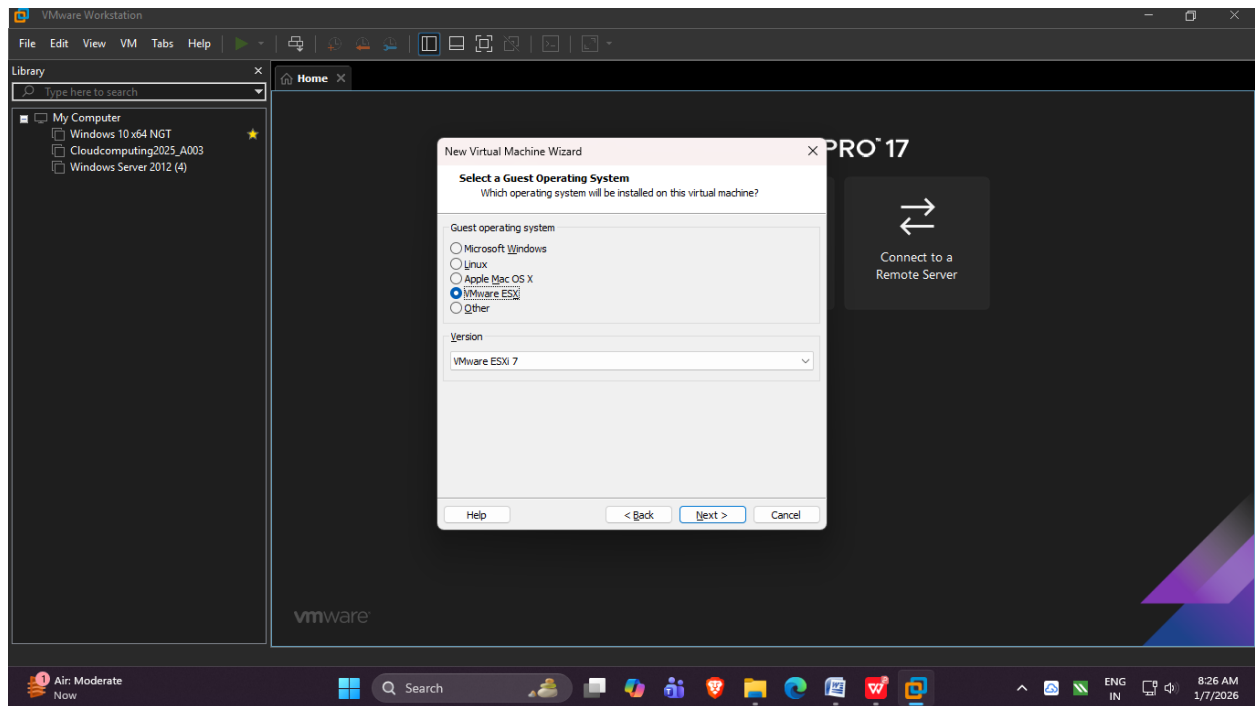




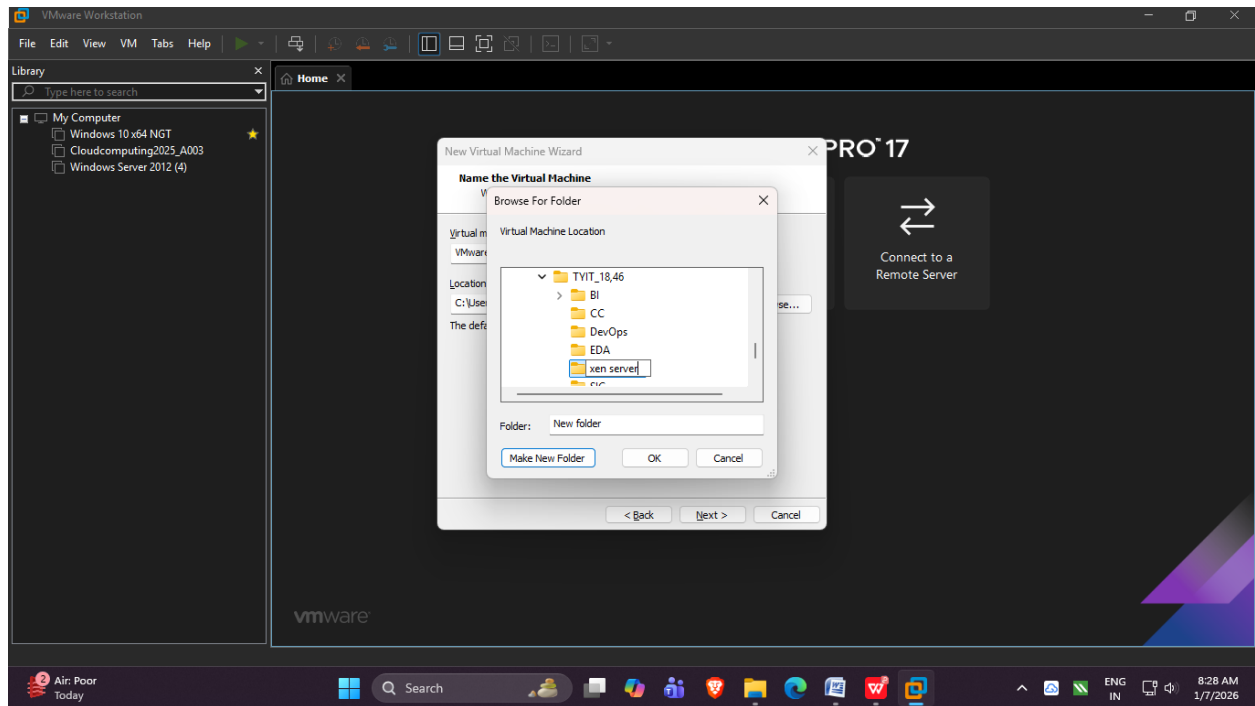
Browse XenServer.iso file



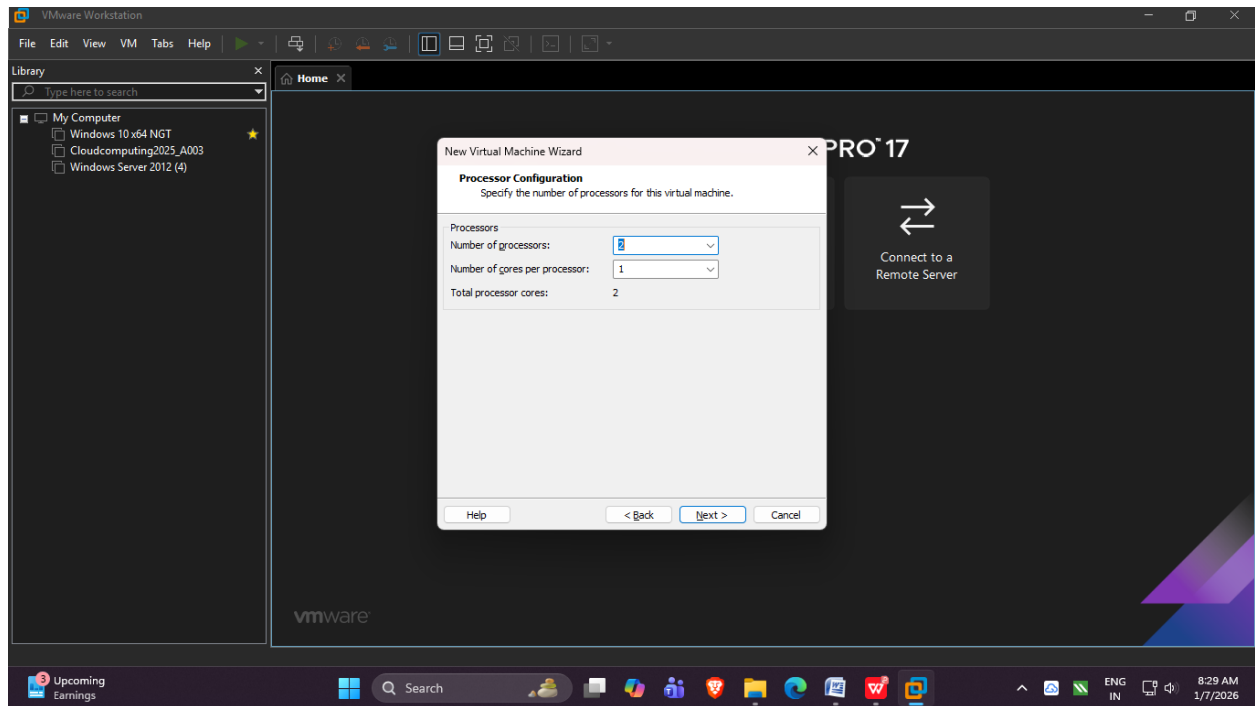
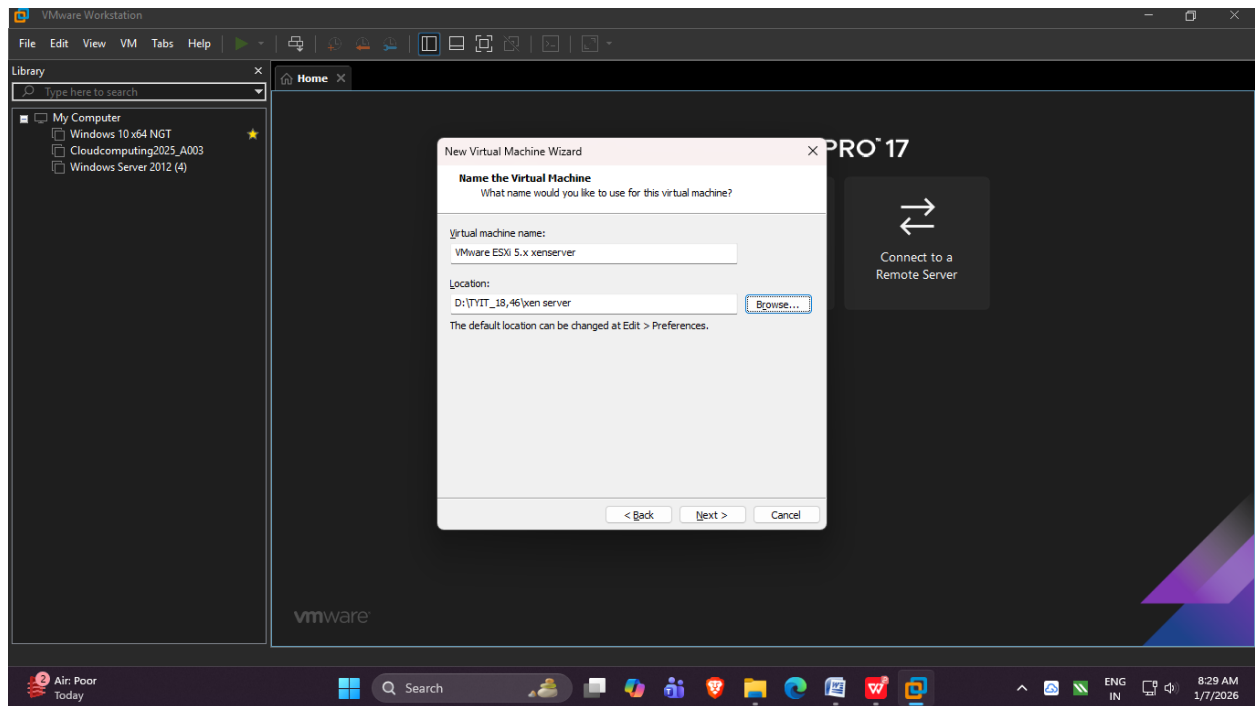
Select OS: VMware ESX, Version: VMware ESXi 7

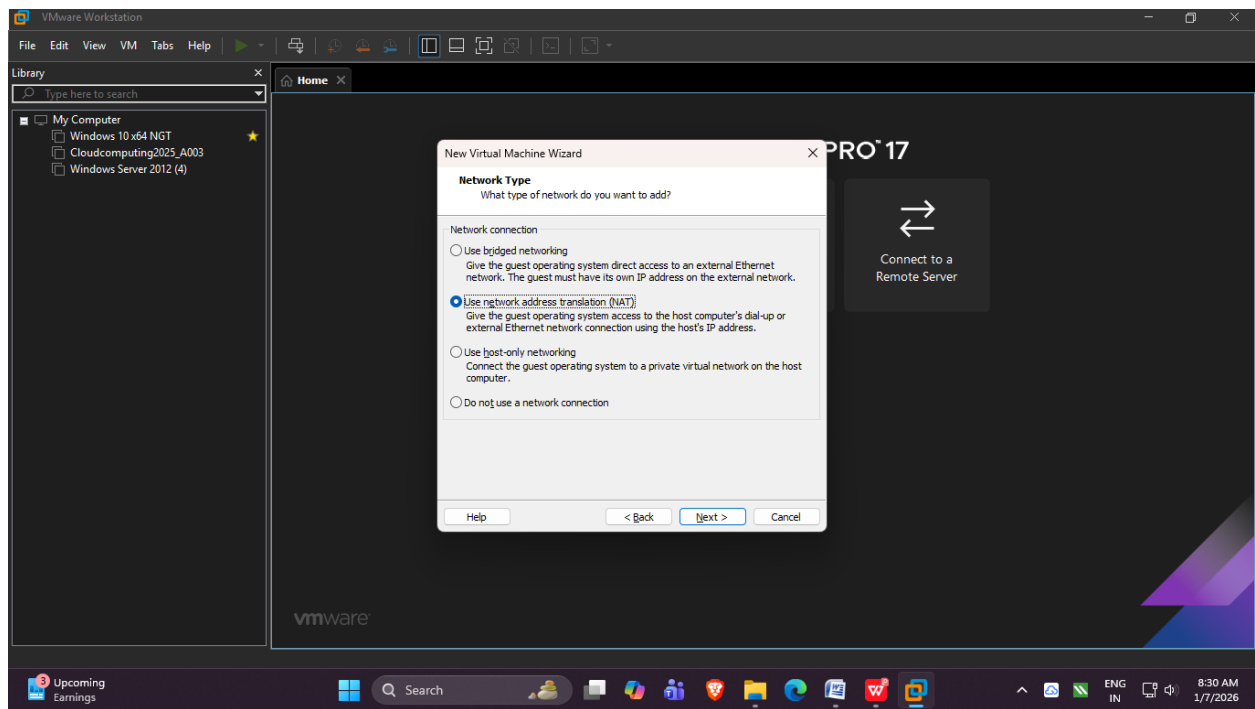
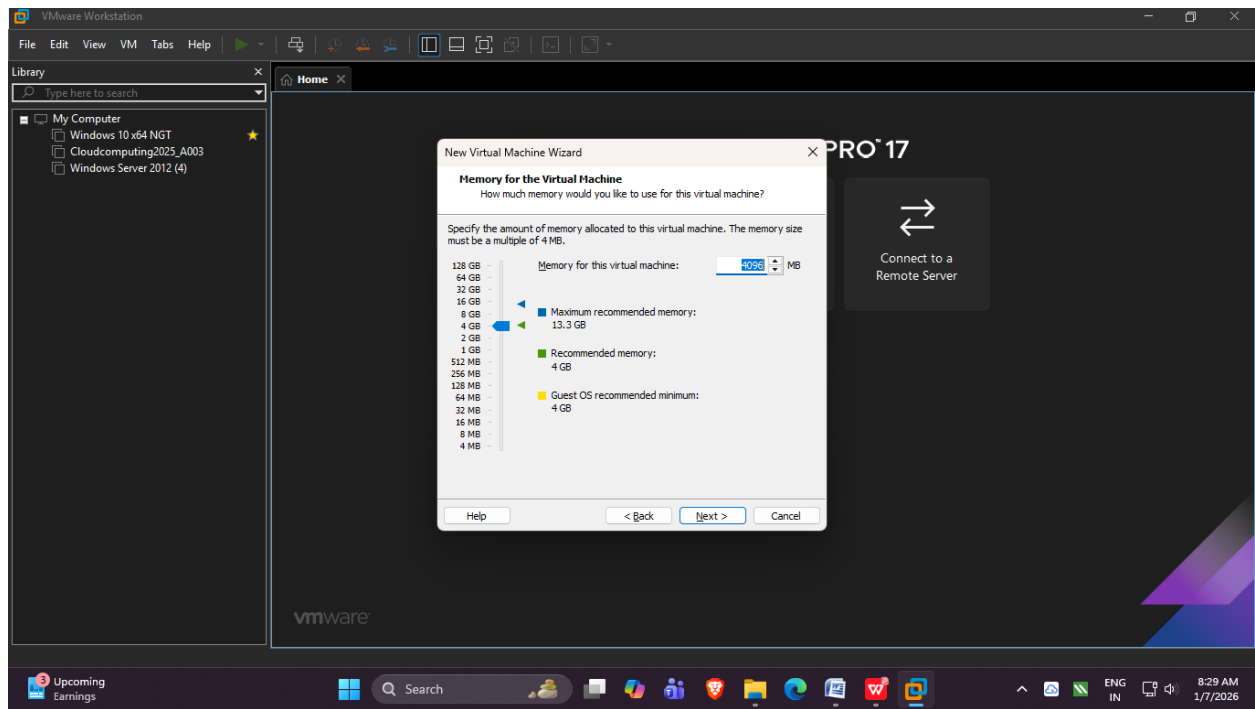


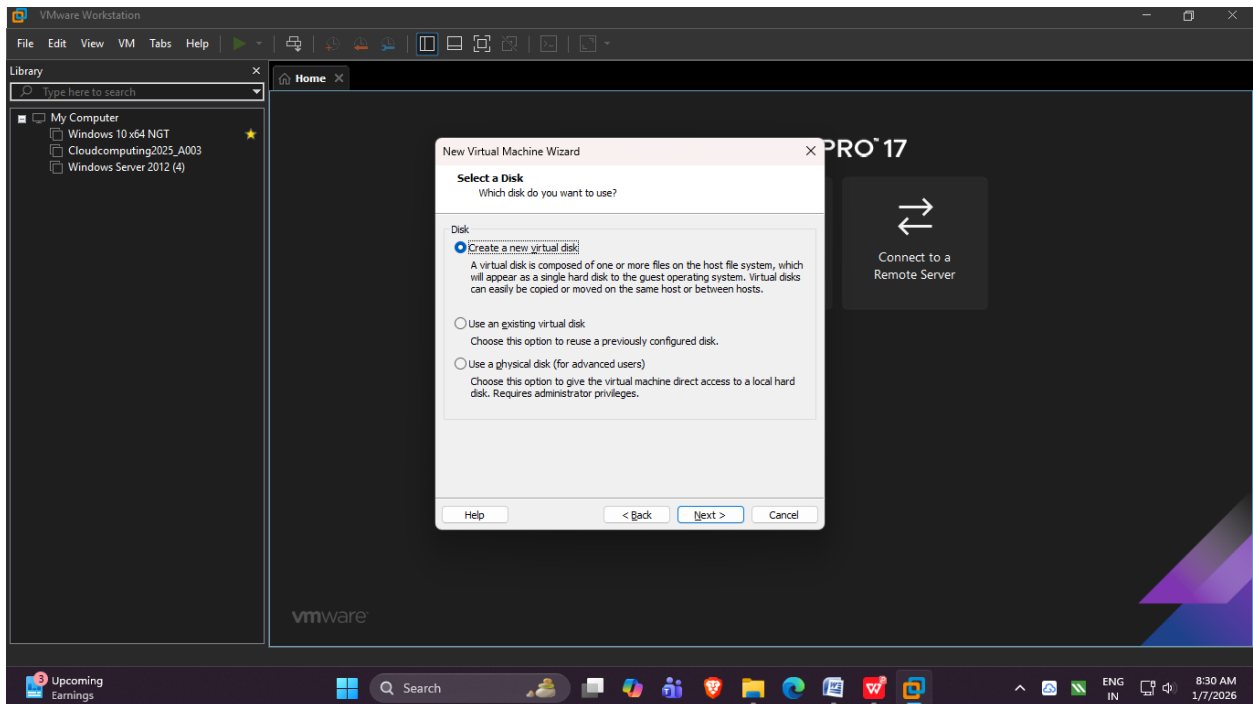
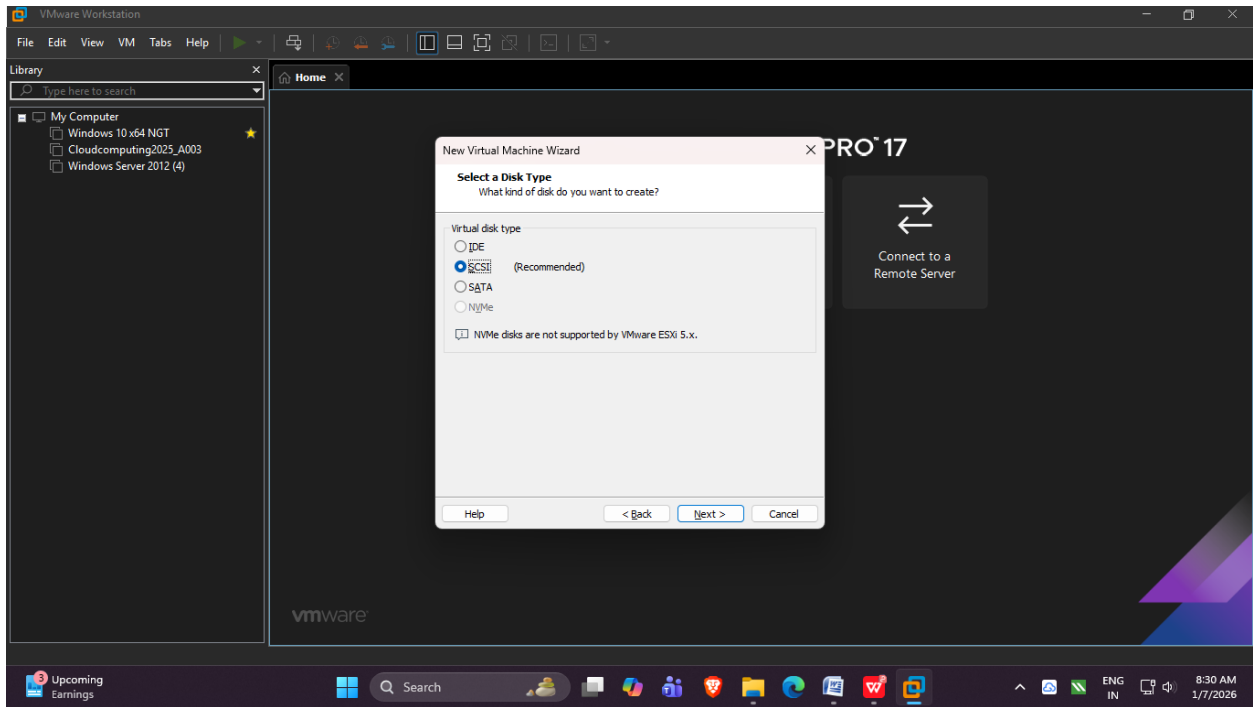
Create new folder named as xen server in D drive

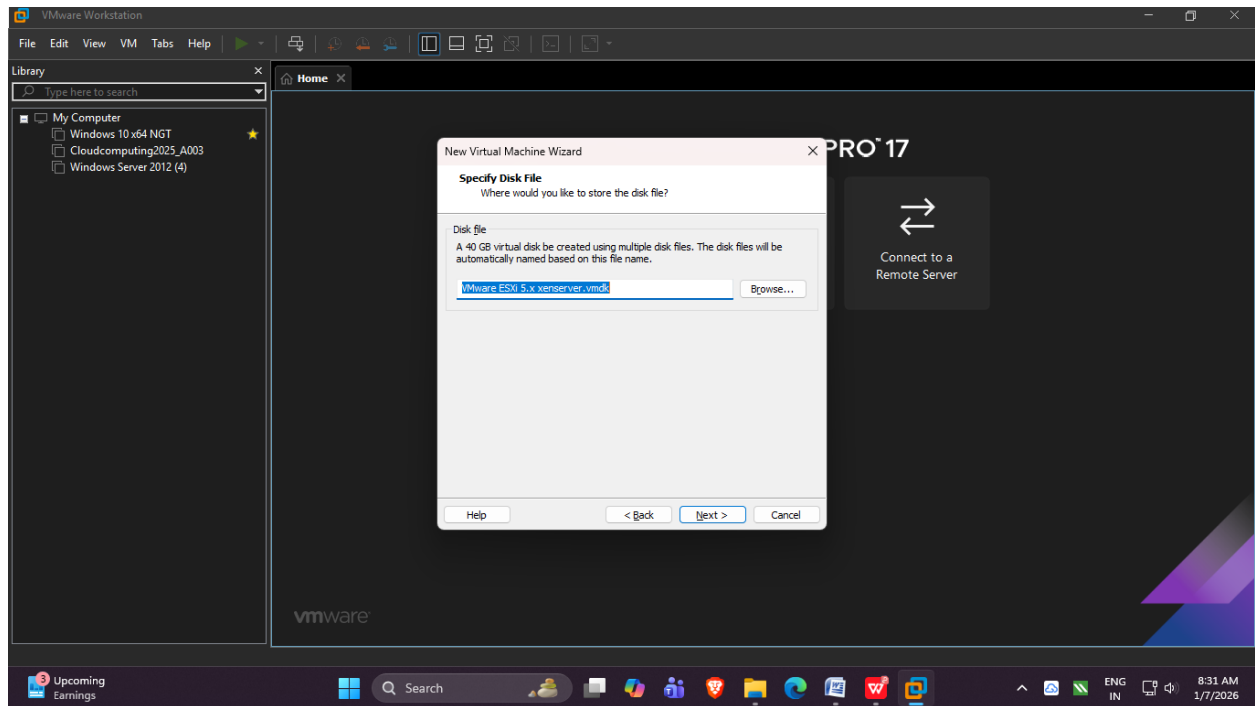
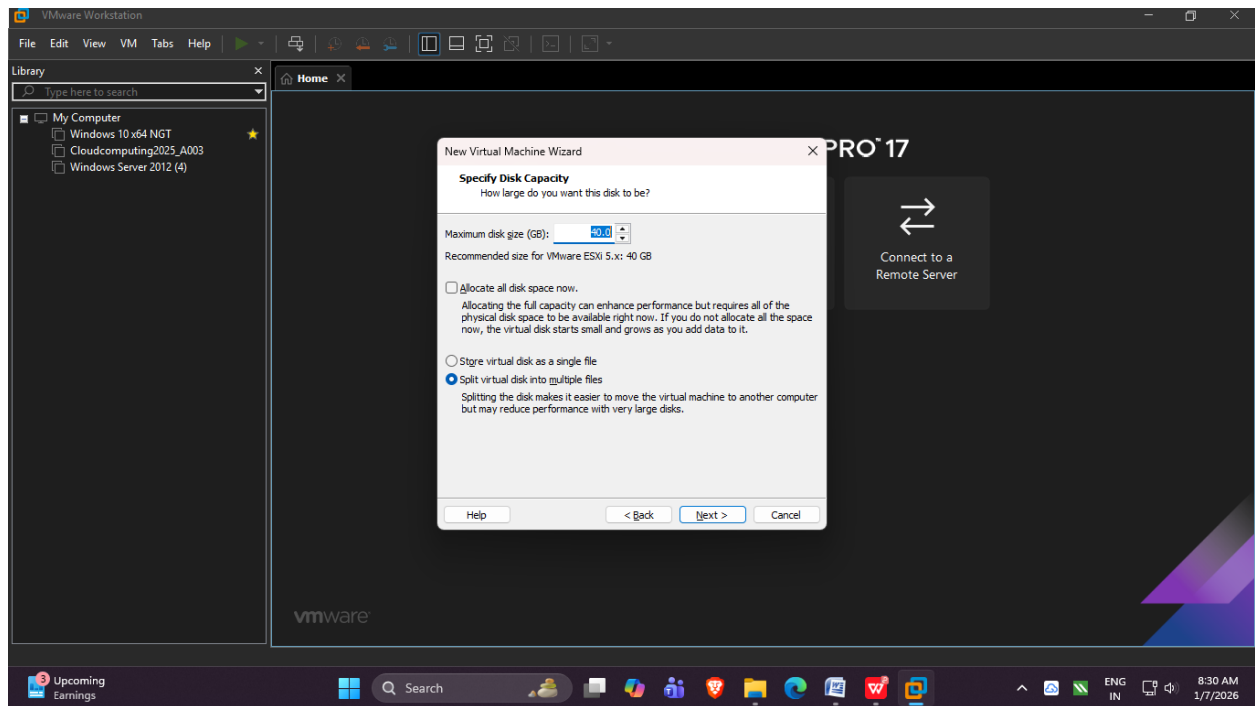


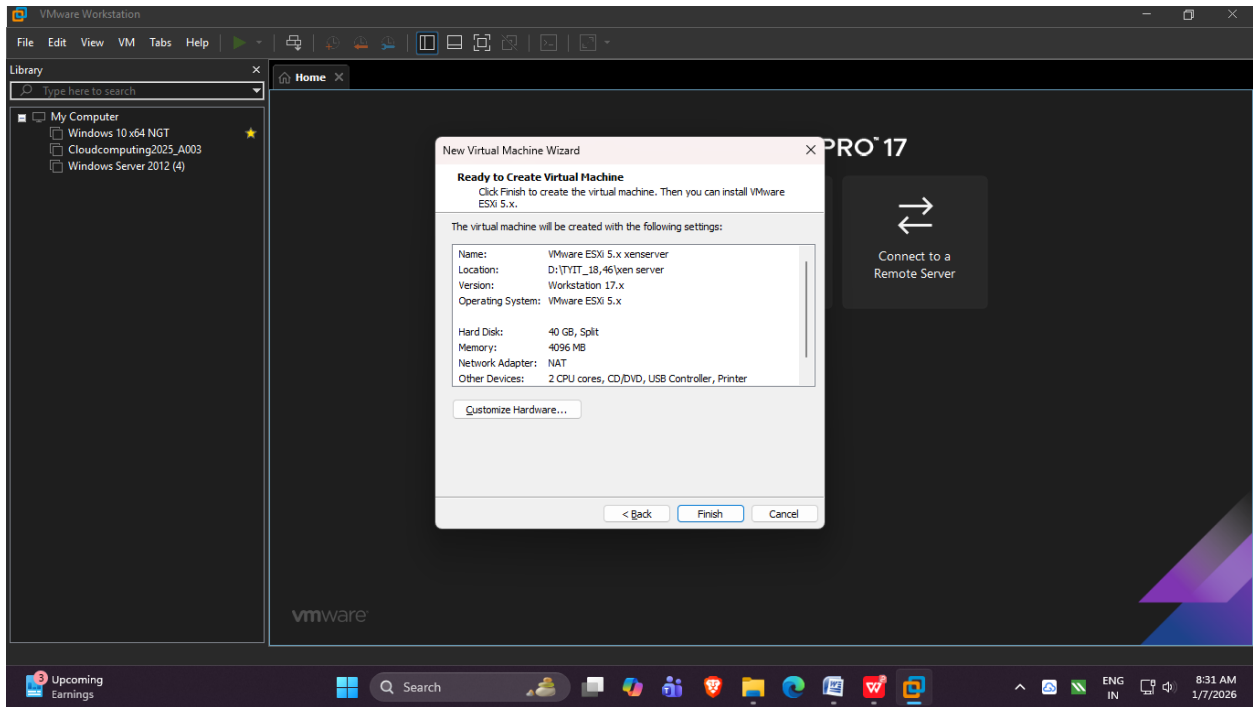
Rename vm as VMware ESXi 5.x xenserver



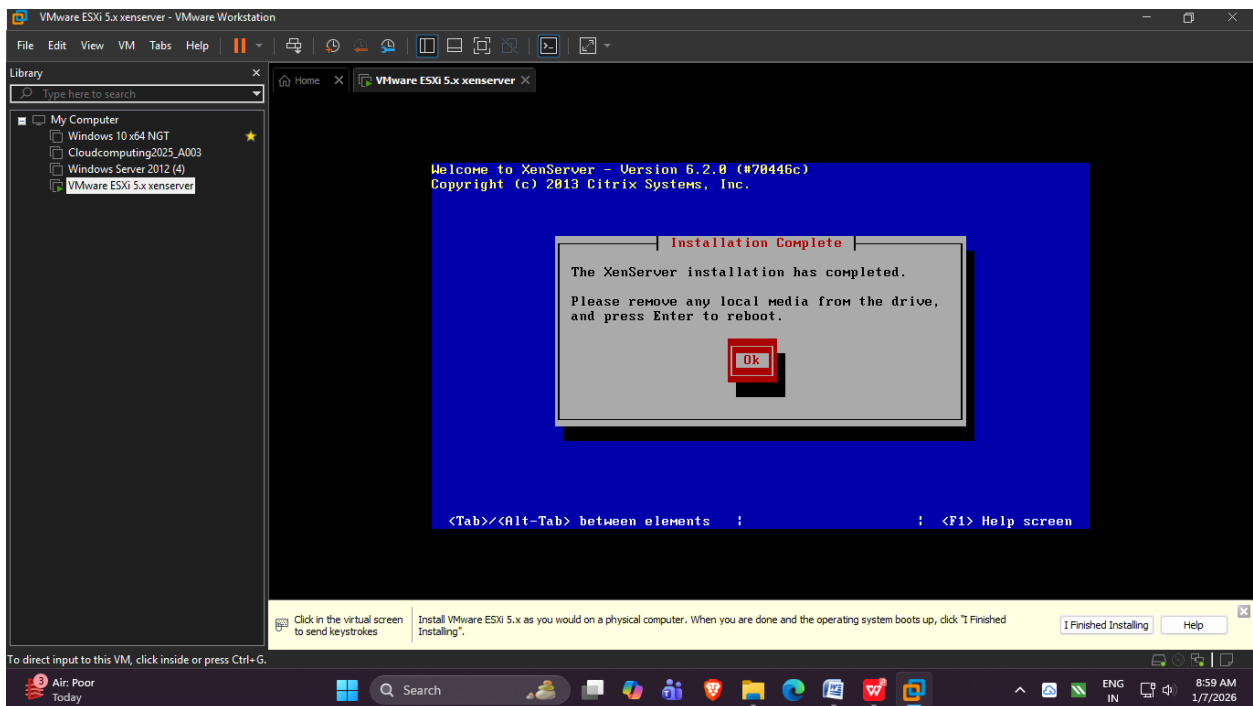


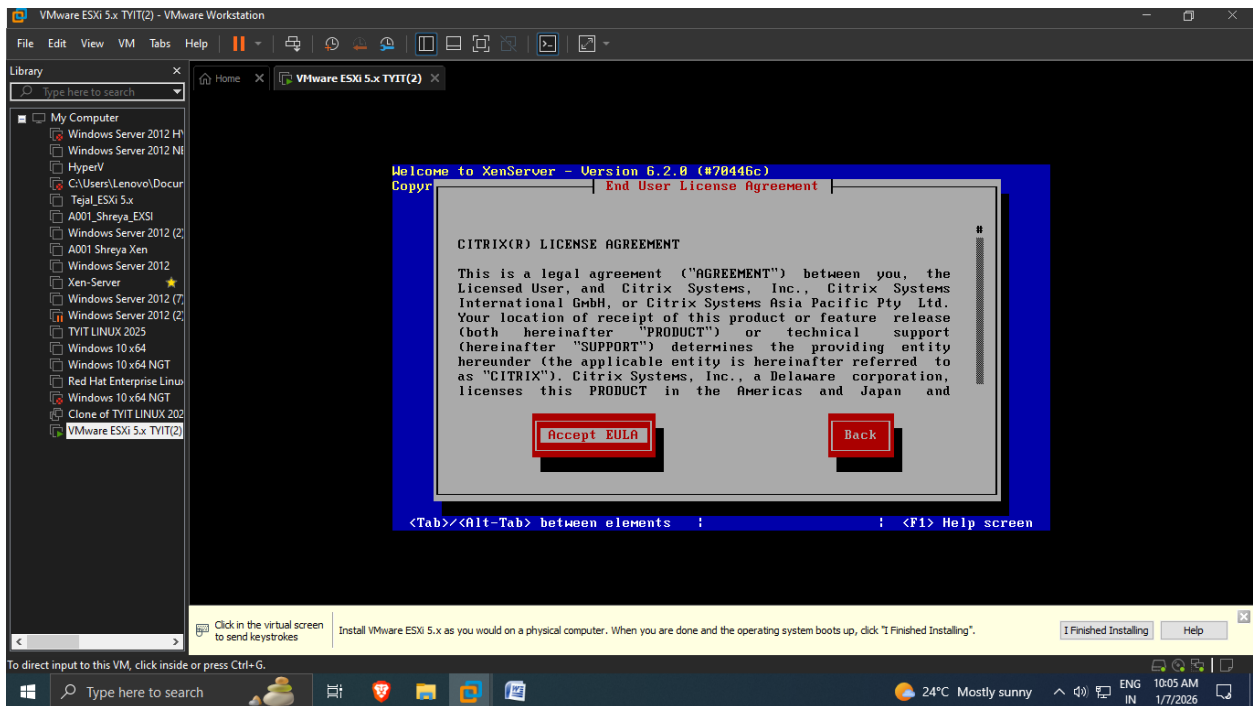
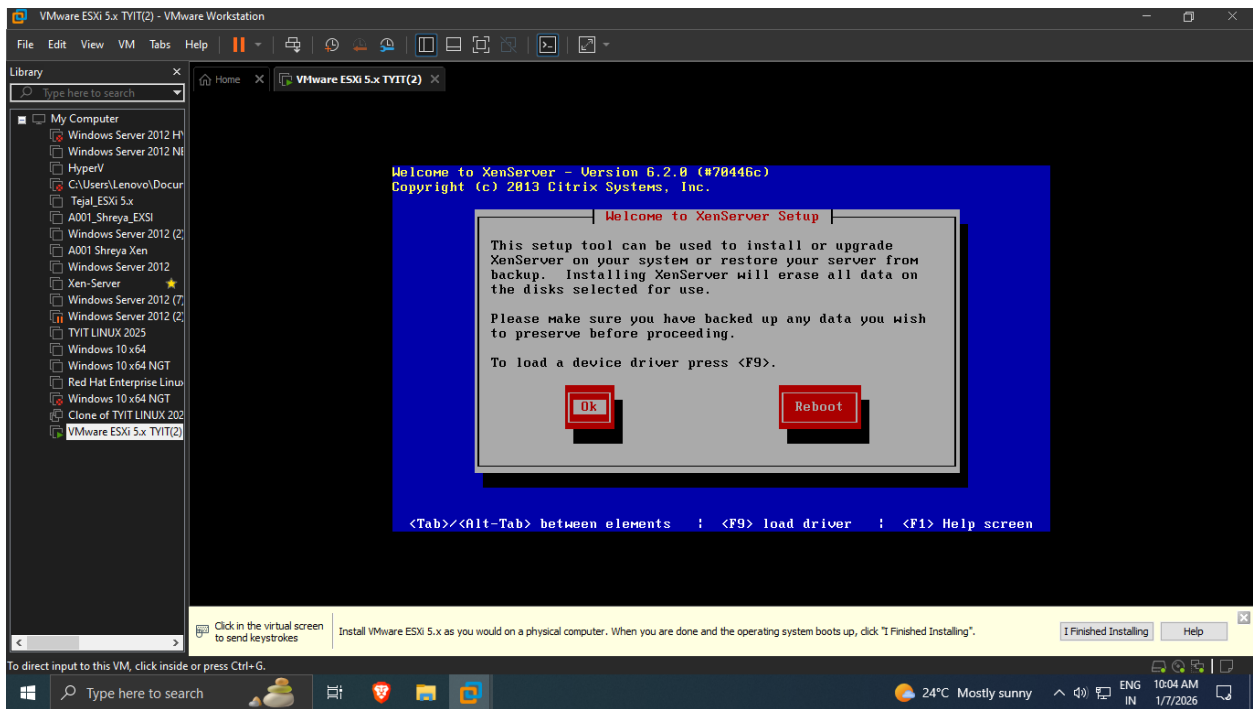


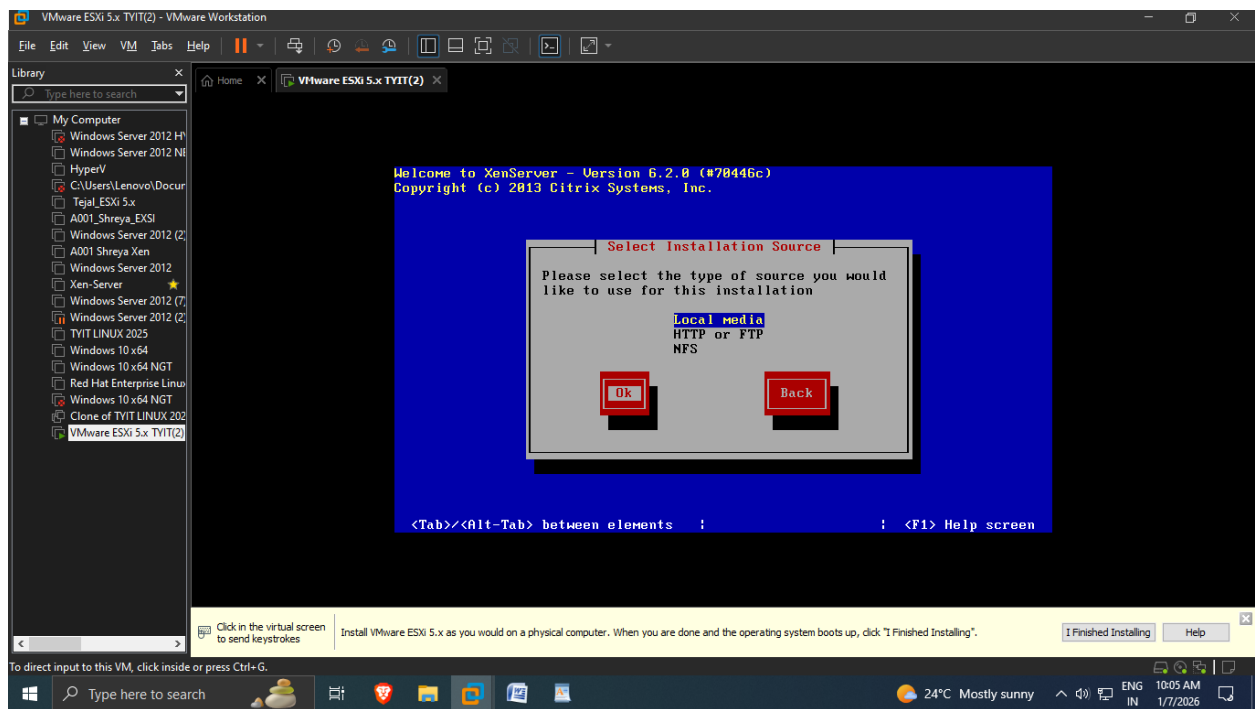
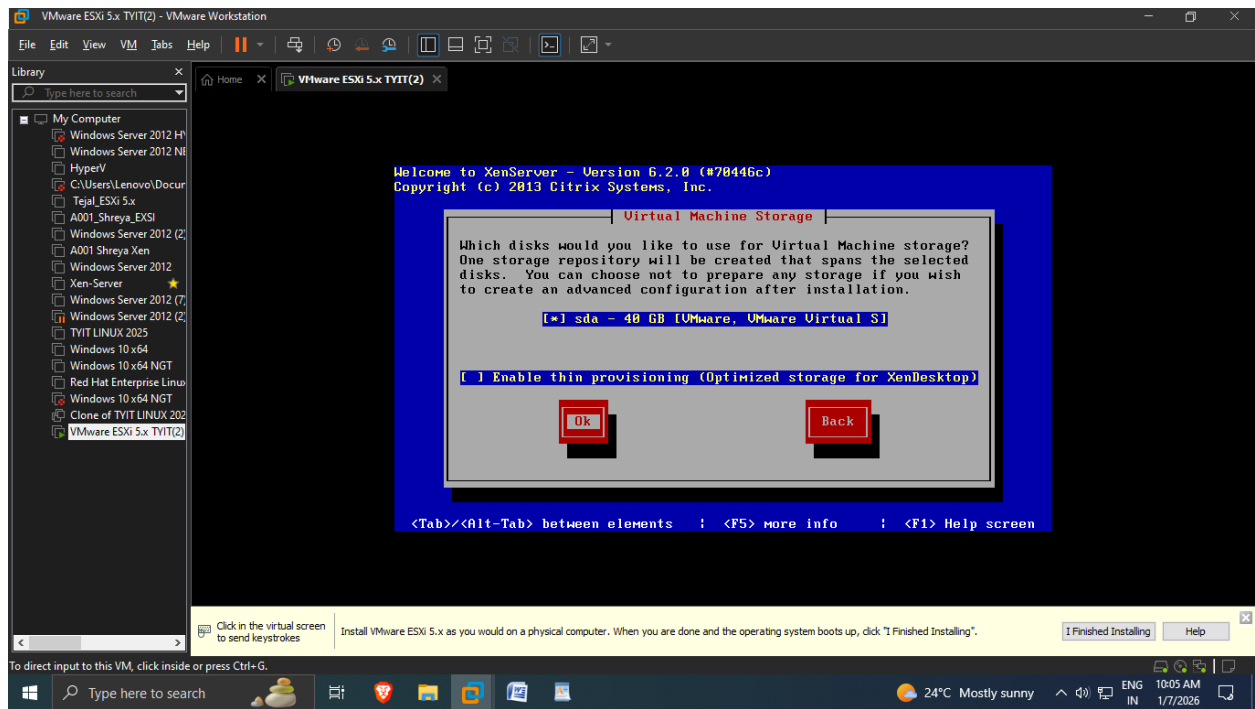


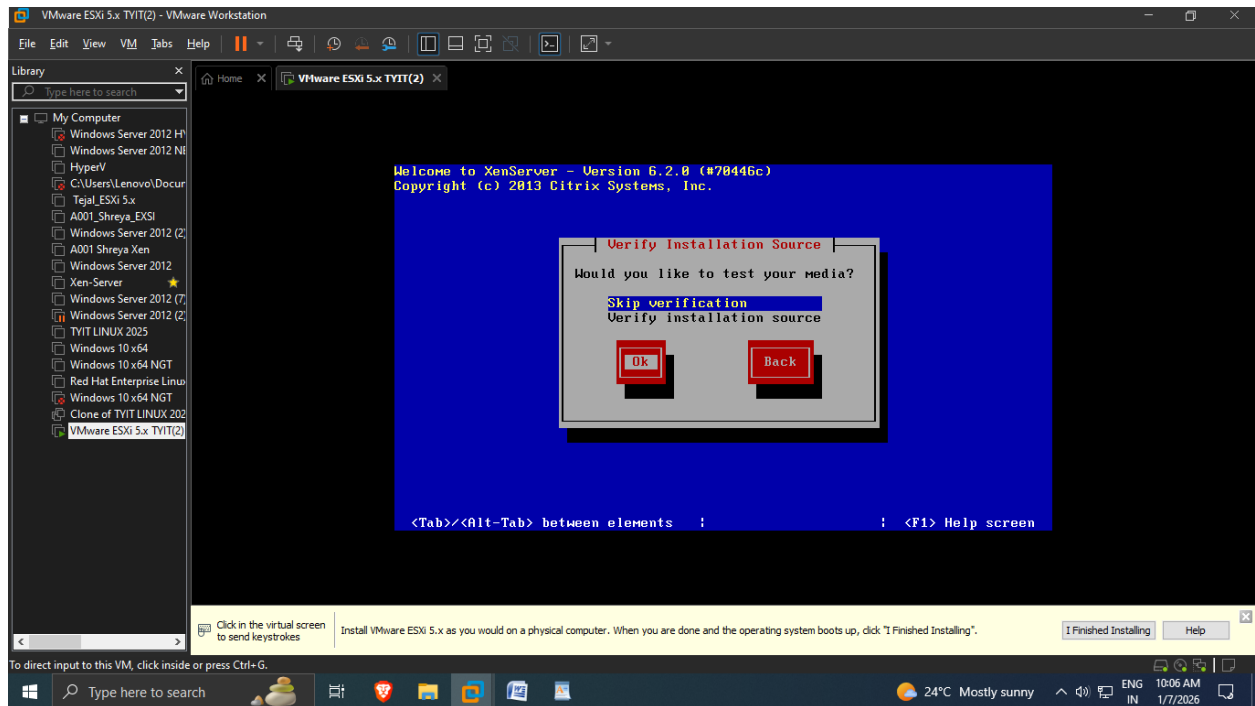
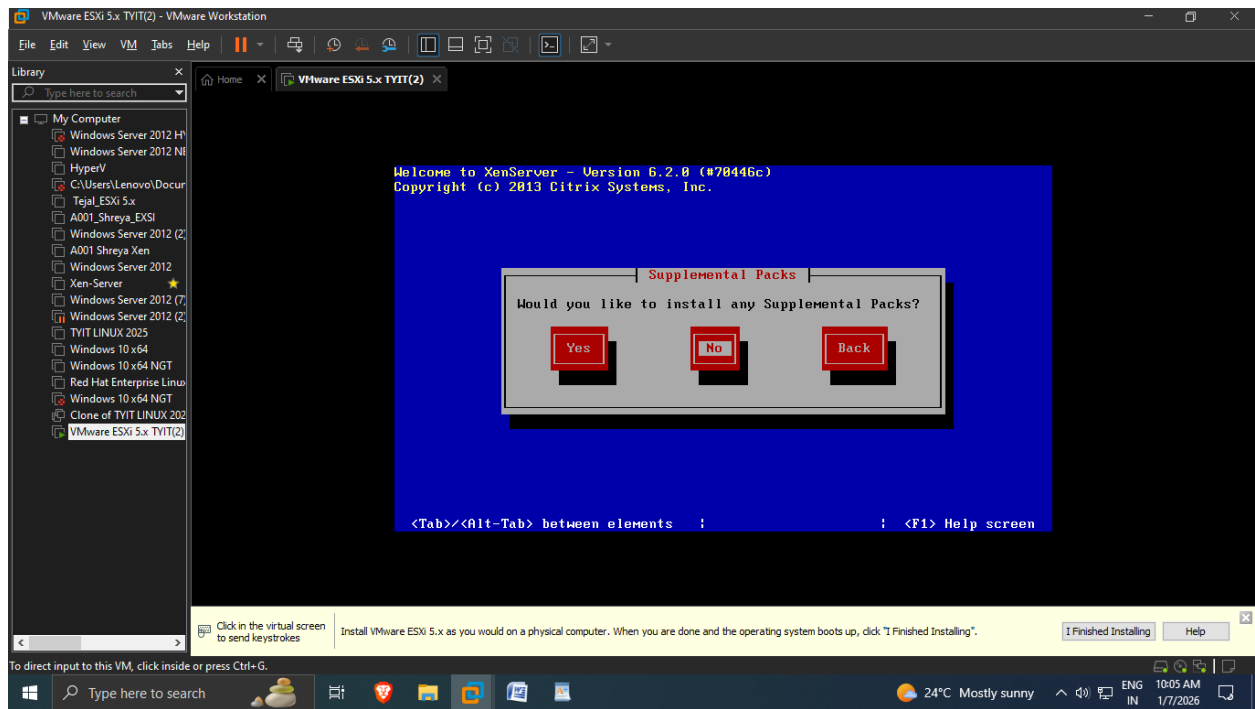


Installation completed.

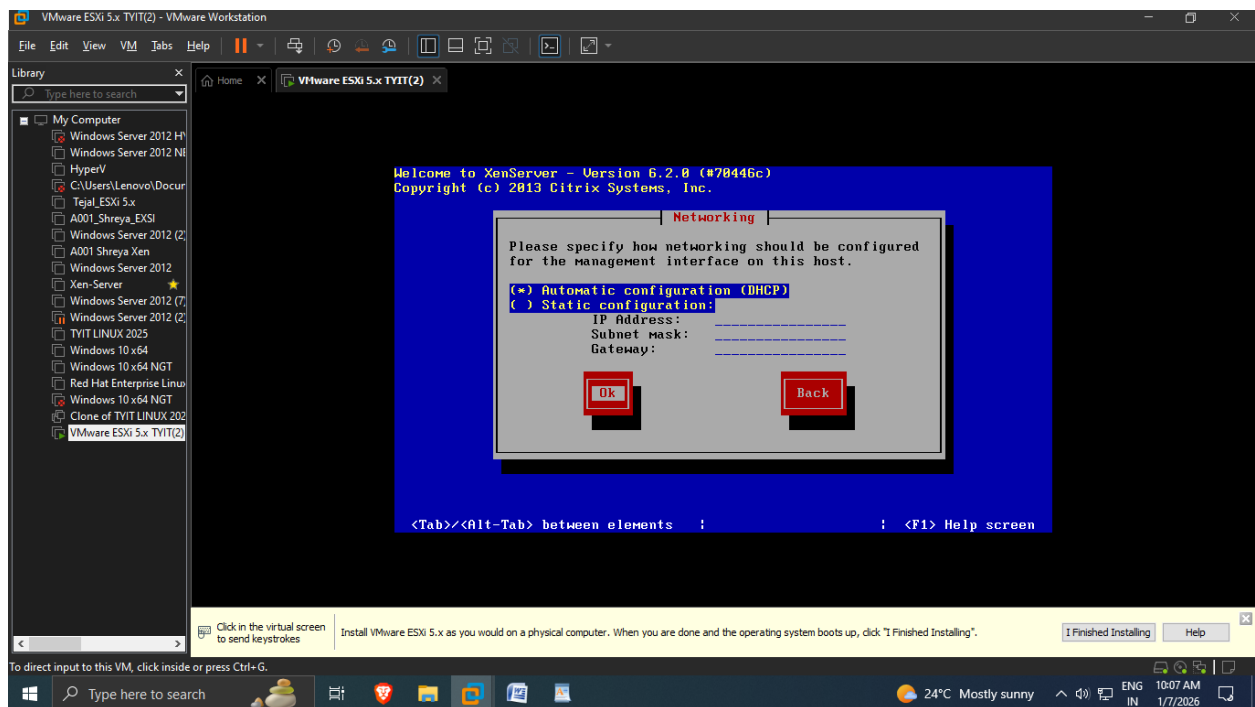
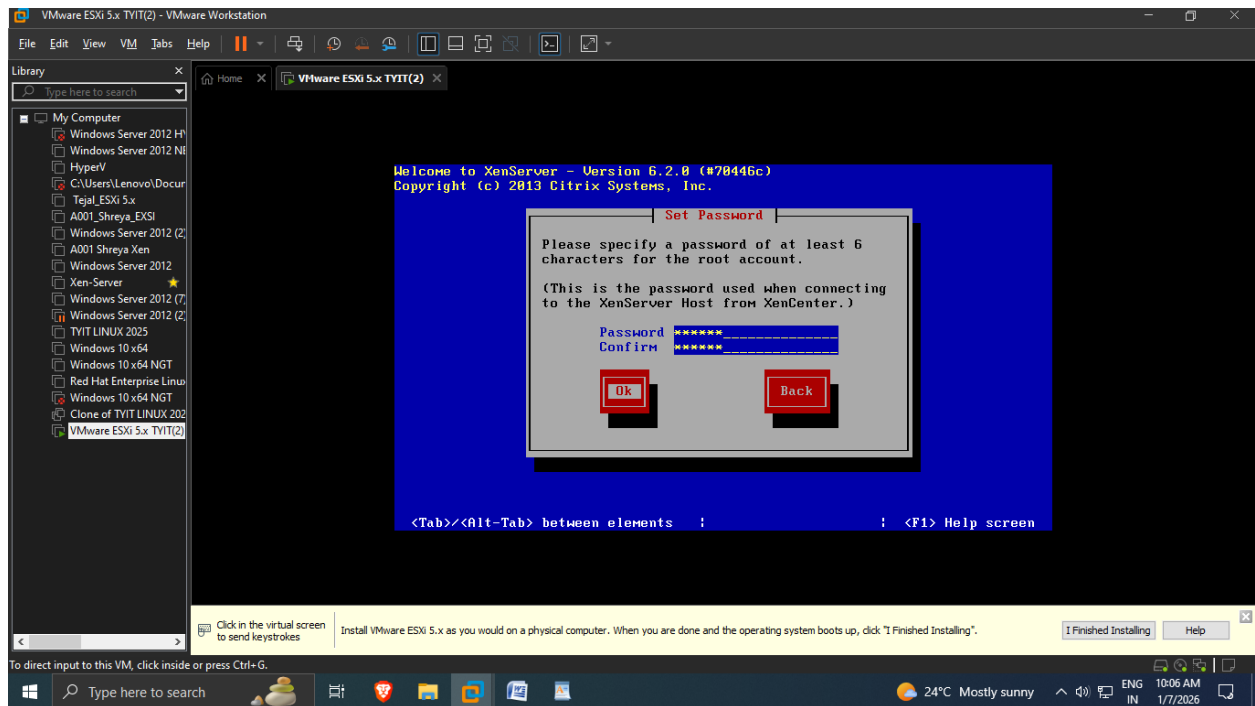


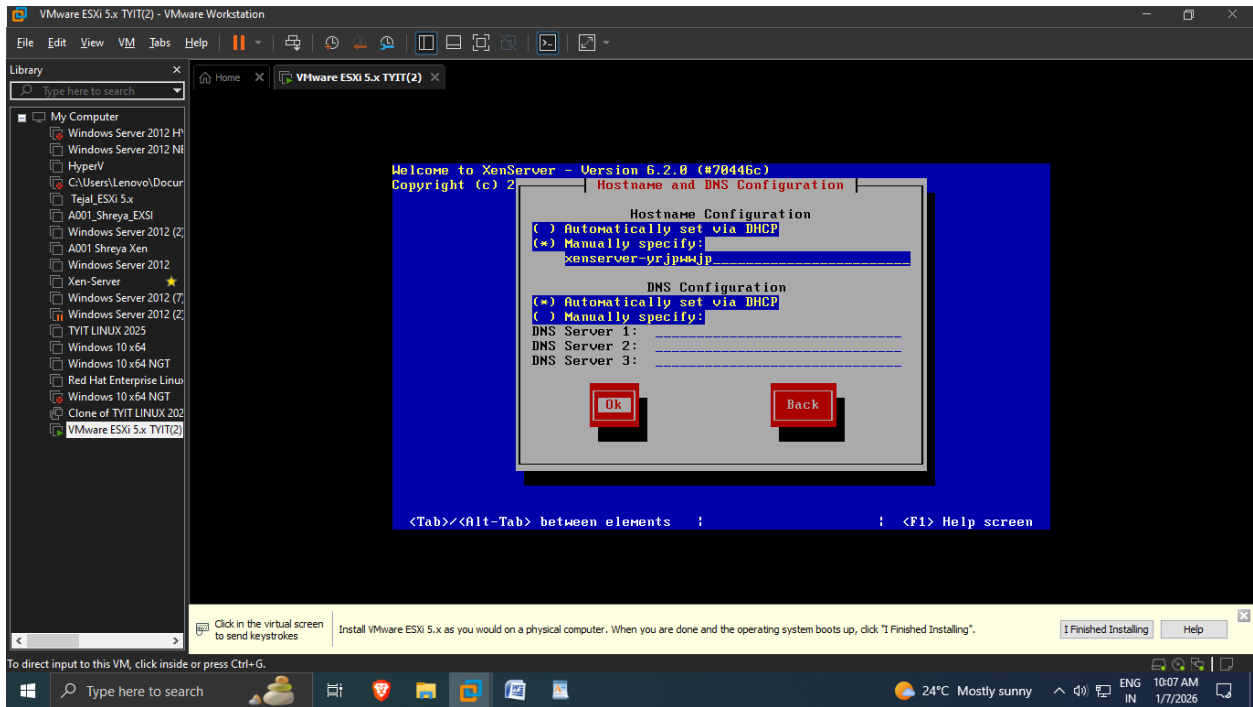




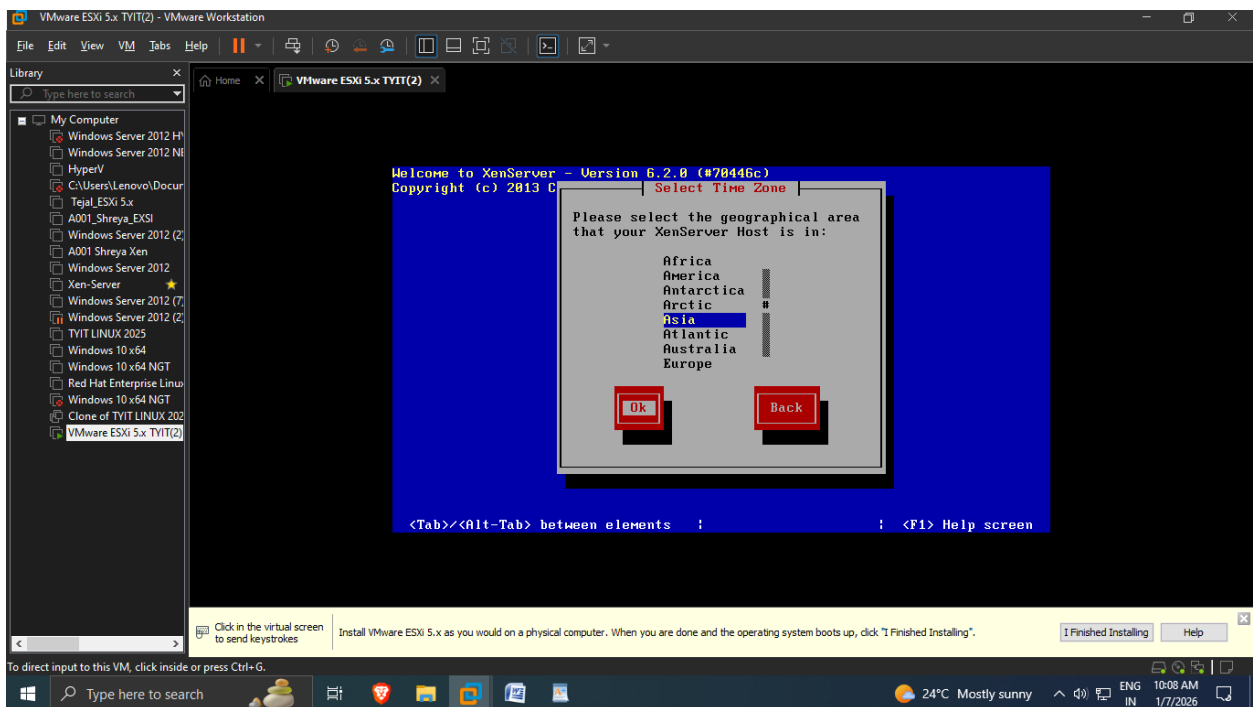


Set Password 123456.

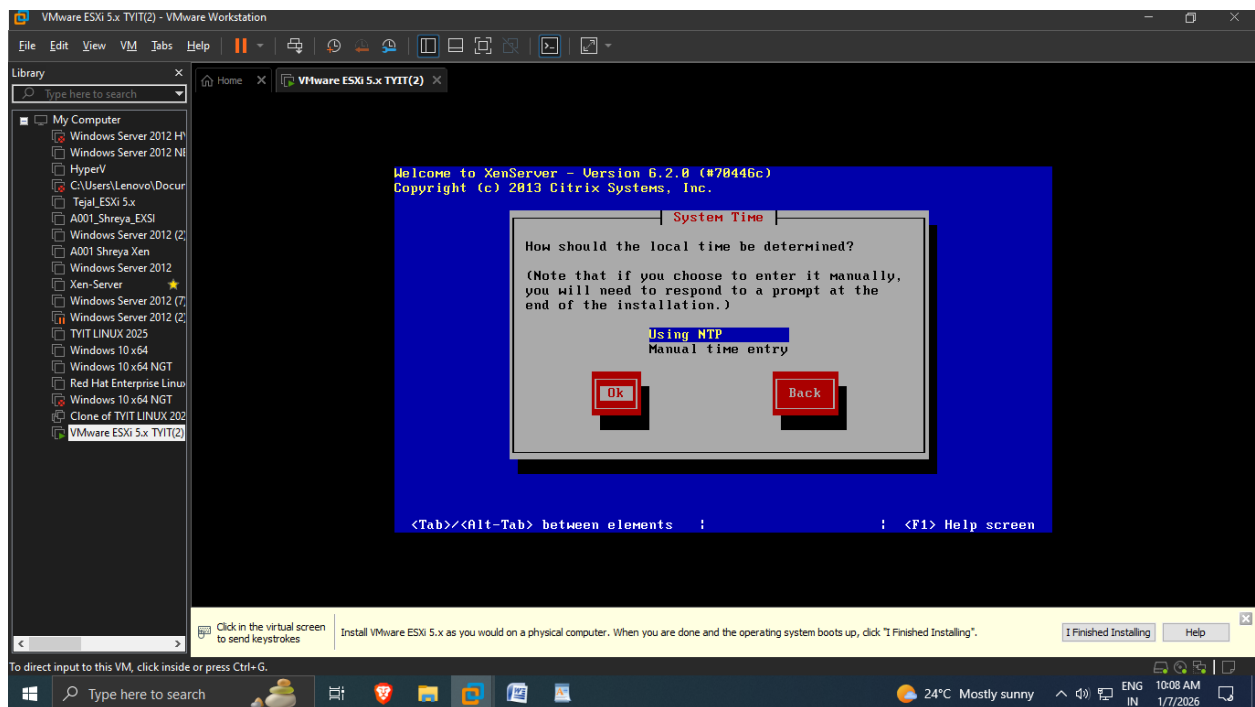
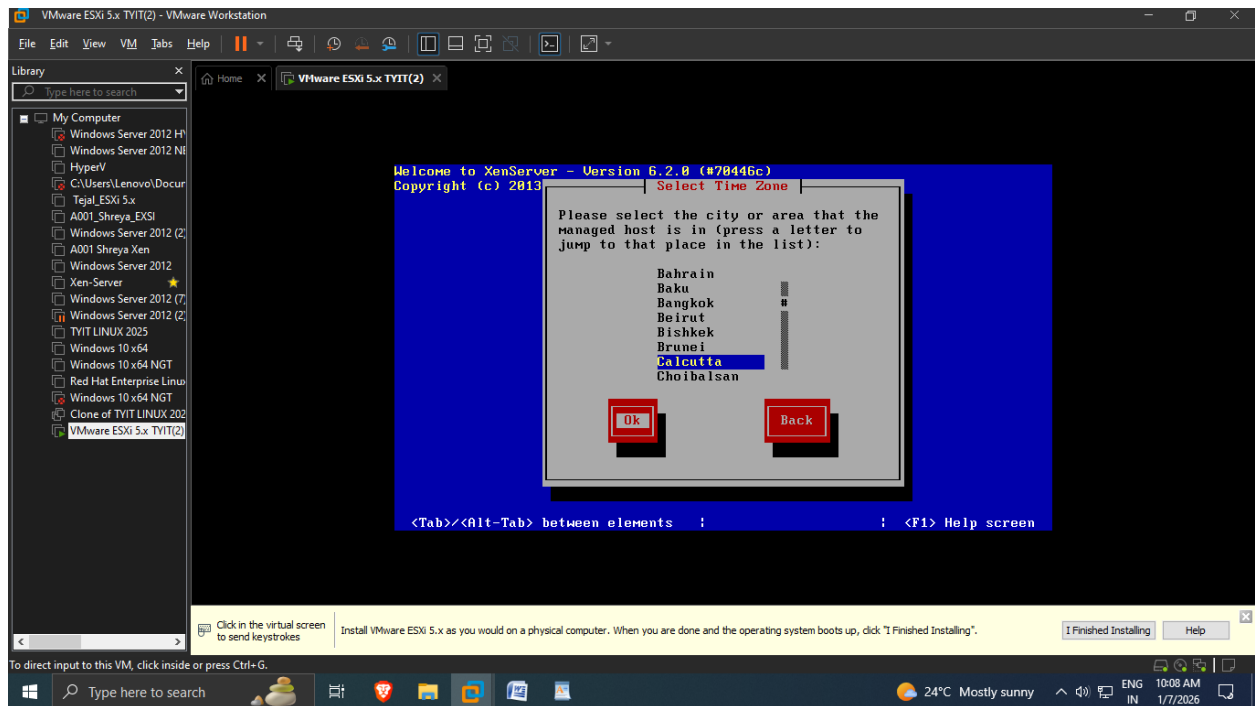


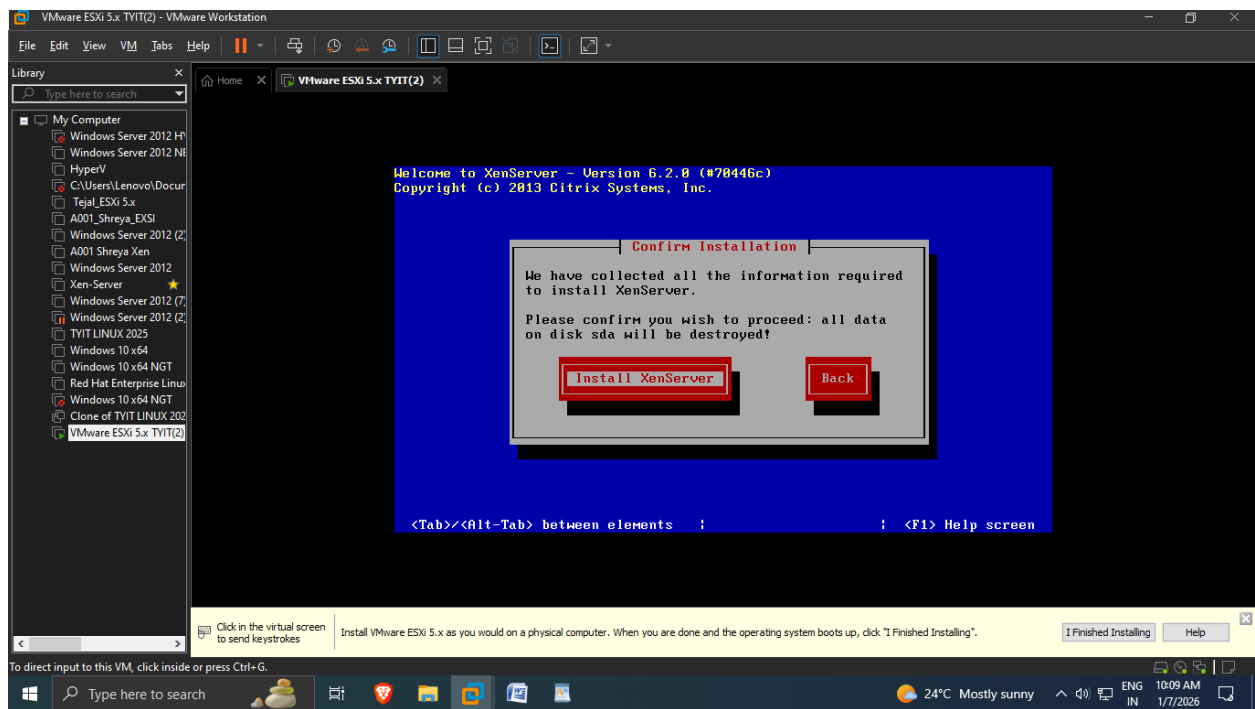
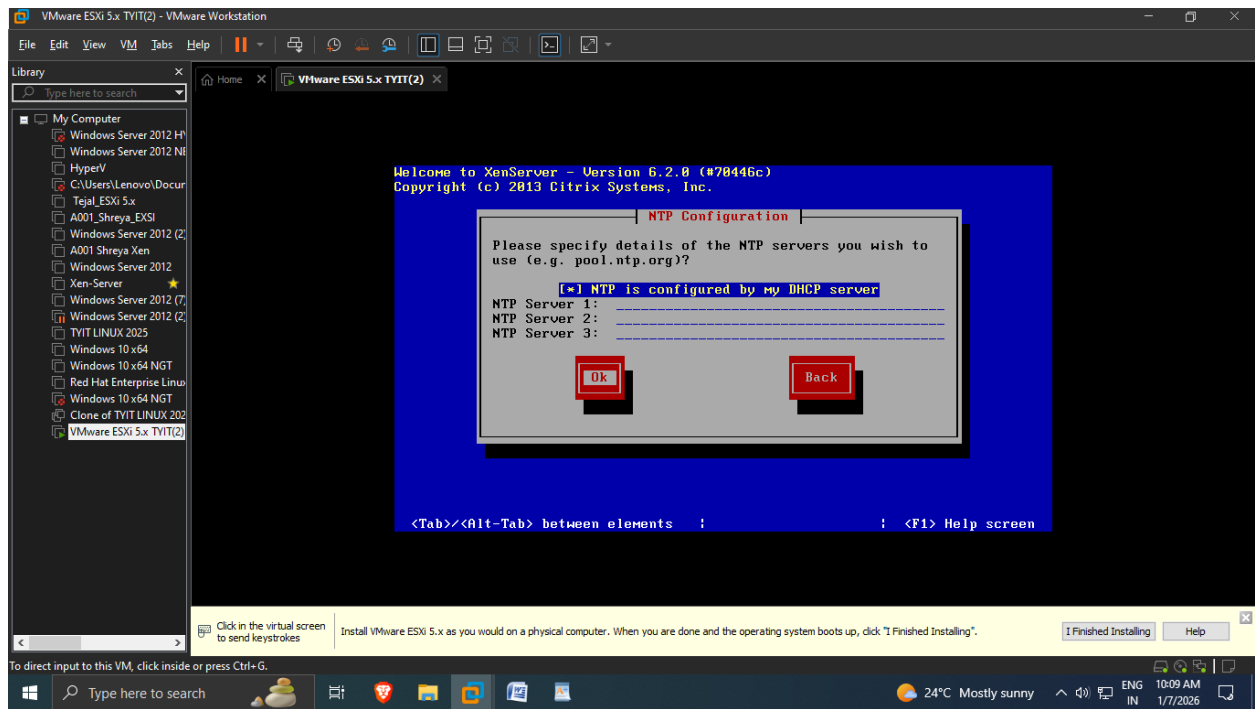


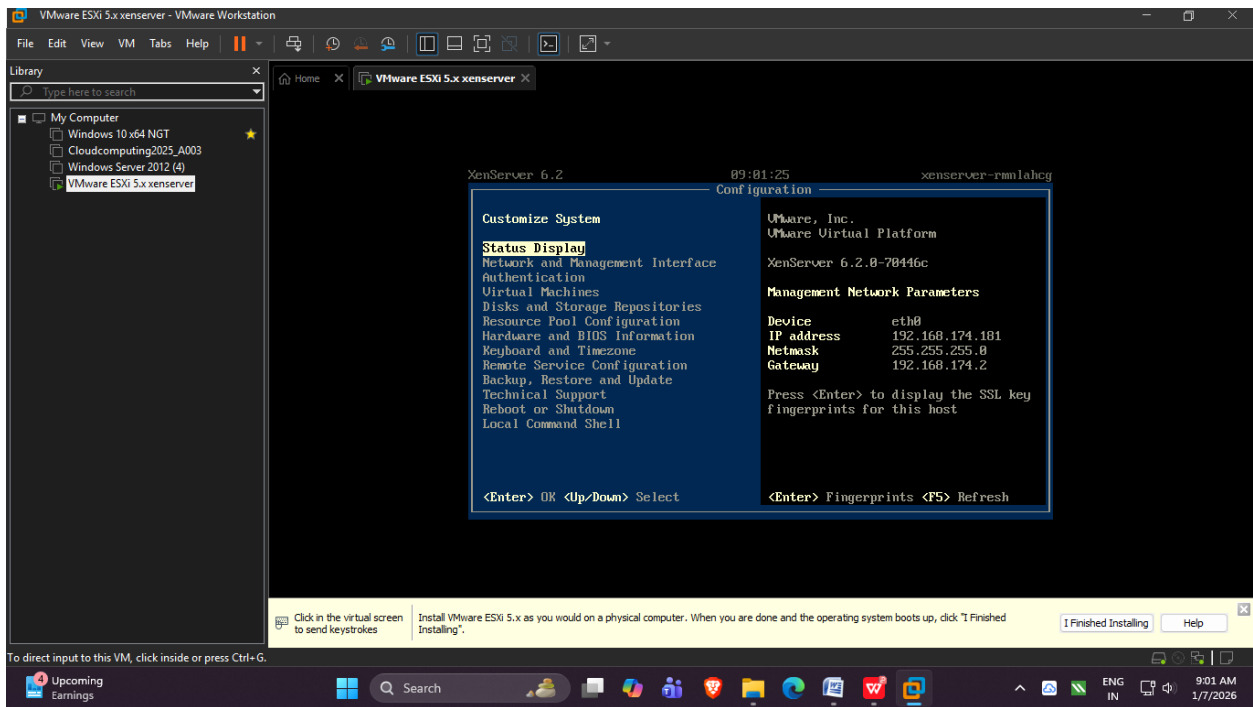
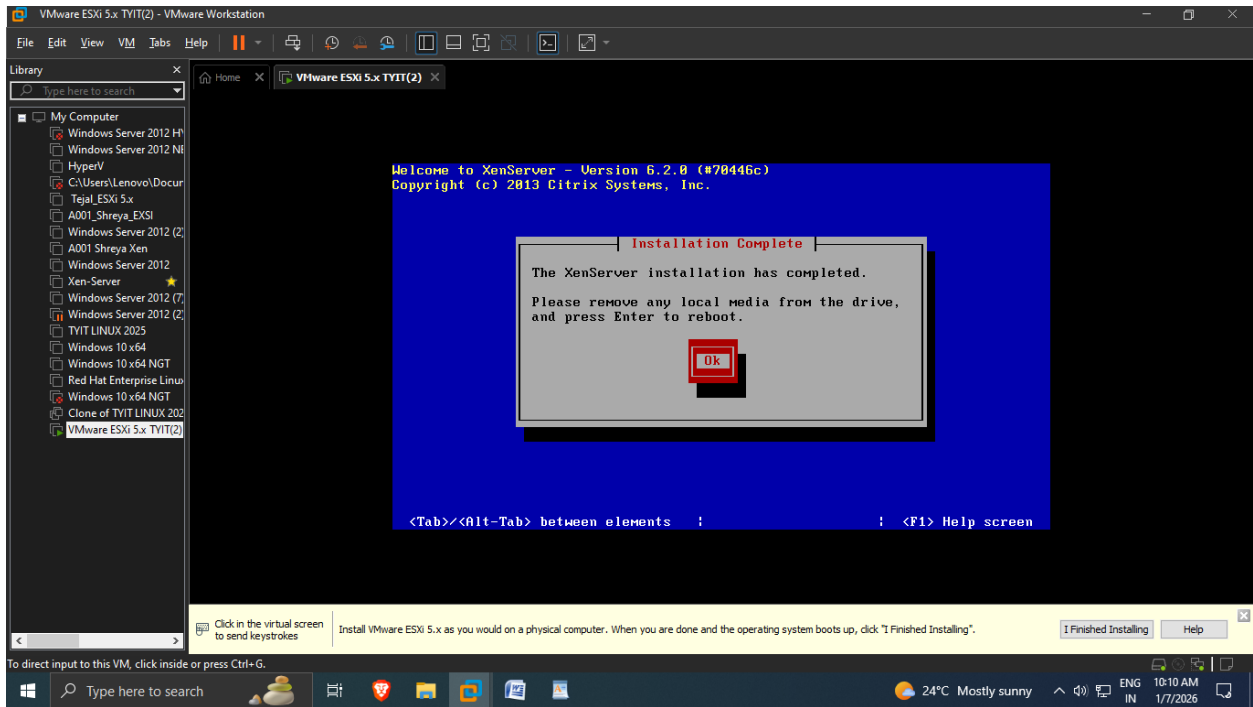
Select time zone Asia.



Select city as Calcutta.







PRACTICAL NO: - 07

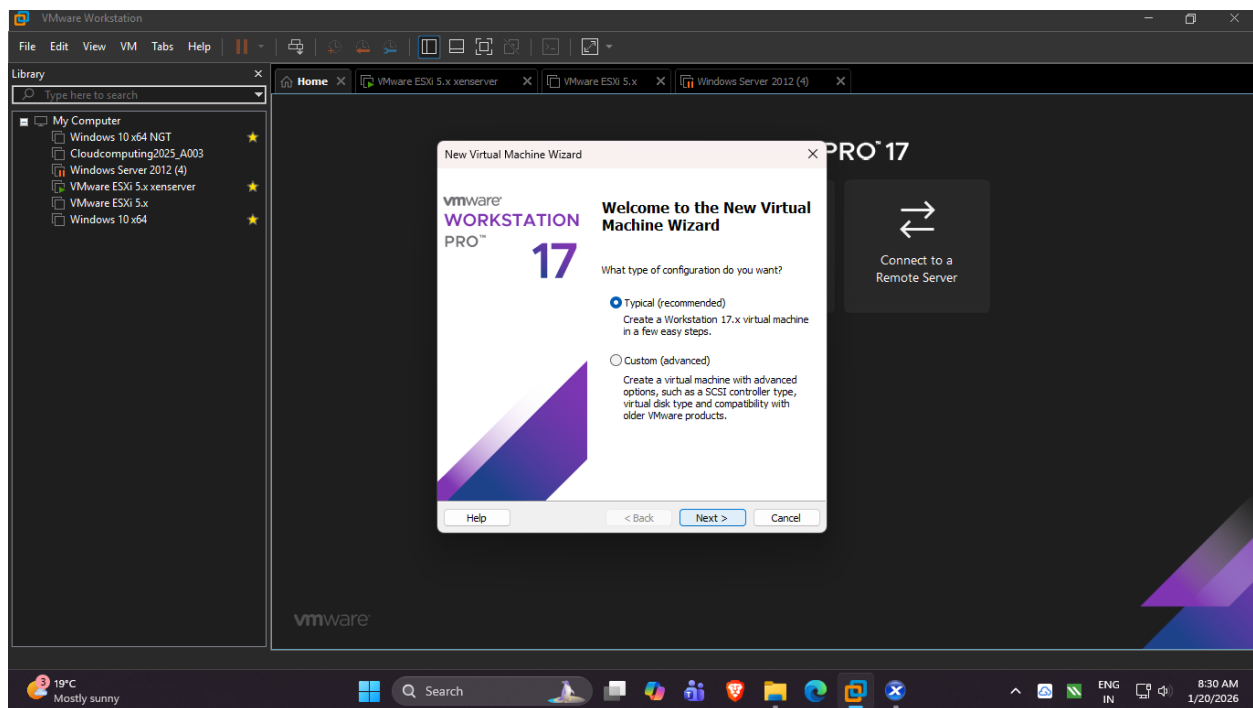
Date: - 20/01/2026

7. Implement VMWAREESXI server.

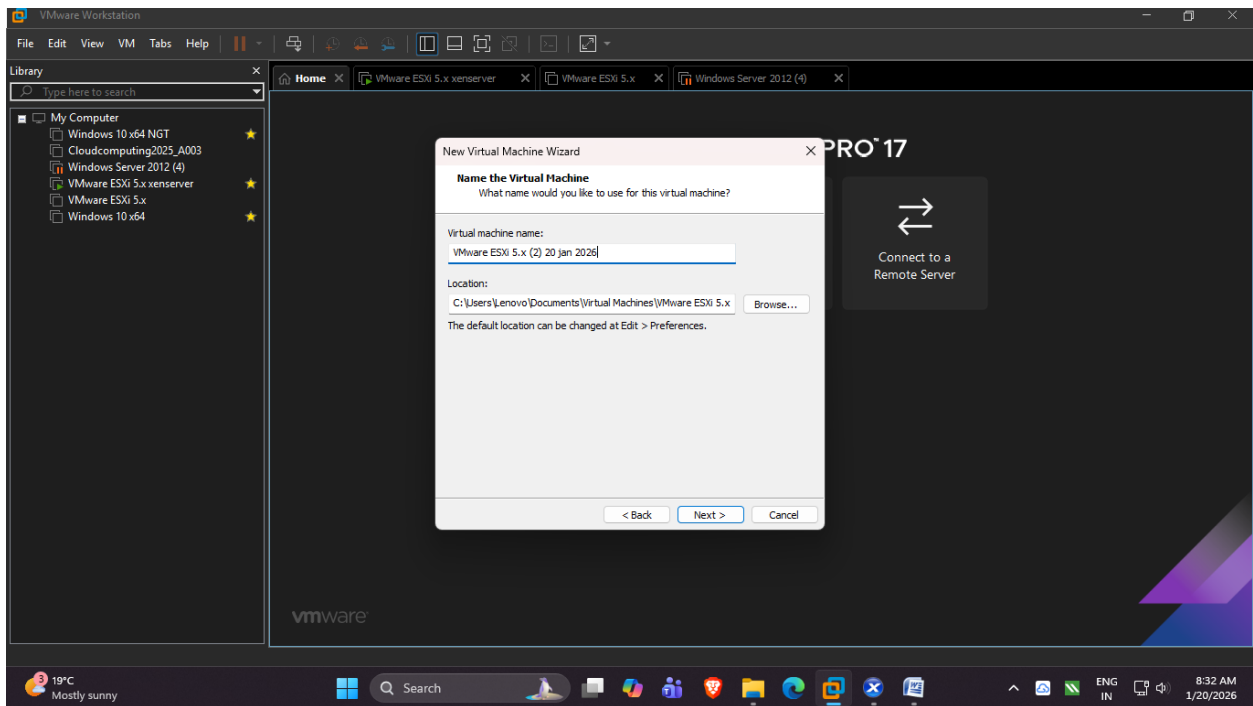
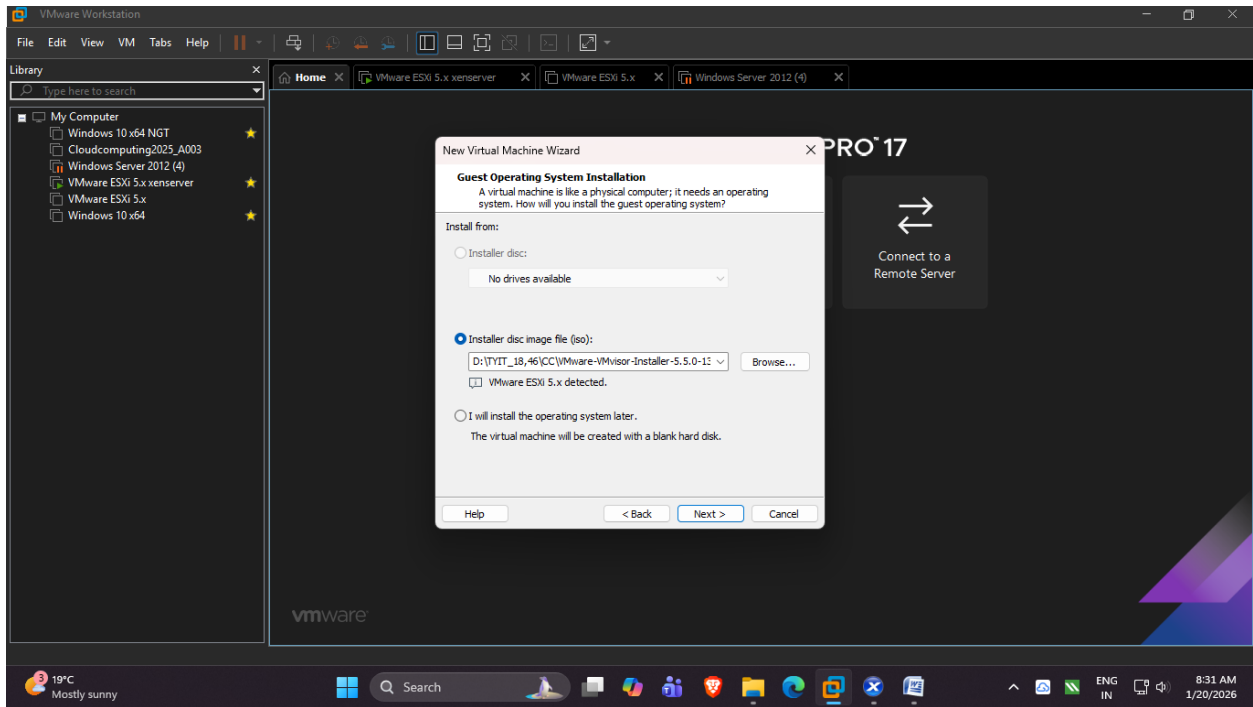
ESXi Server is a VMware virtualization platform that allows multiple virtual machines to run on a single physical server. It is a Type-1 (bare-metal) hypervisor that runs directly on hardware, providing high performance and strong security. ESXi is widely used in data centers to reduce hardware costs and improve efficiency. It supports features such as live migration (vMotion), high availability, snapshots, backups, and centralized management using vCenter.

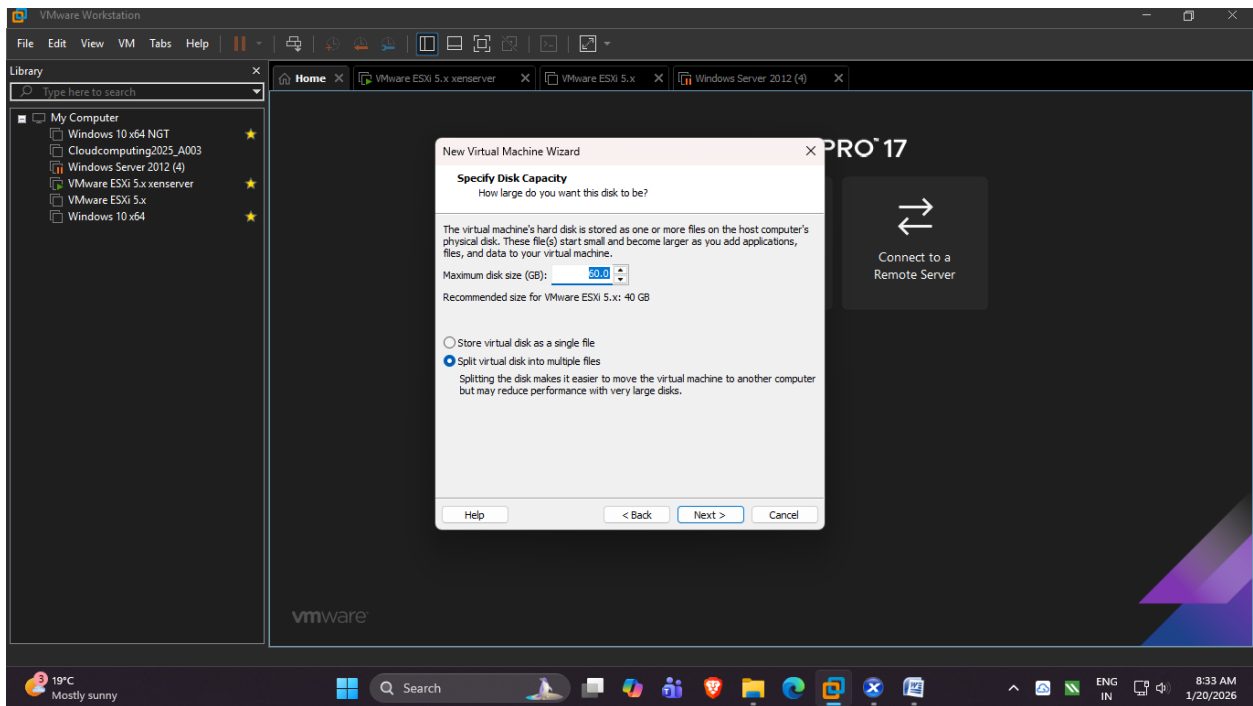
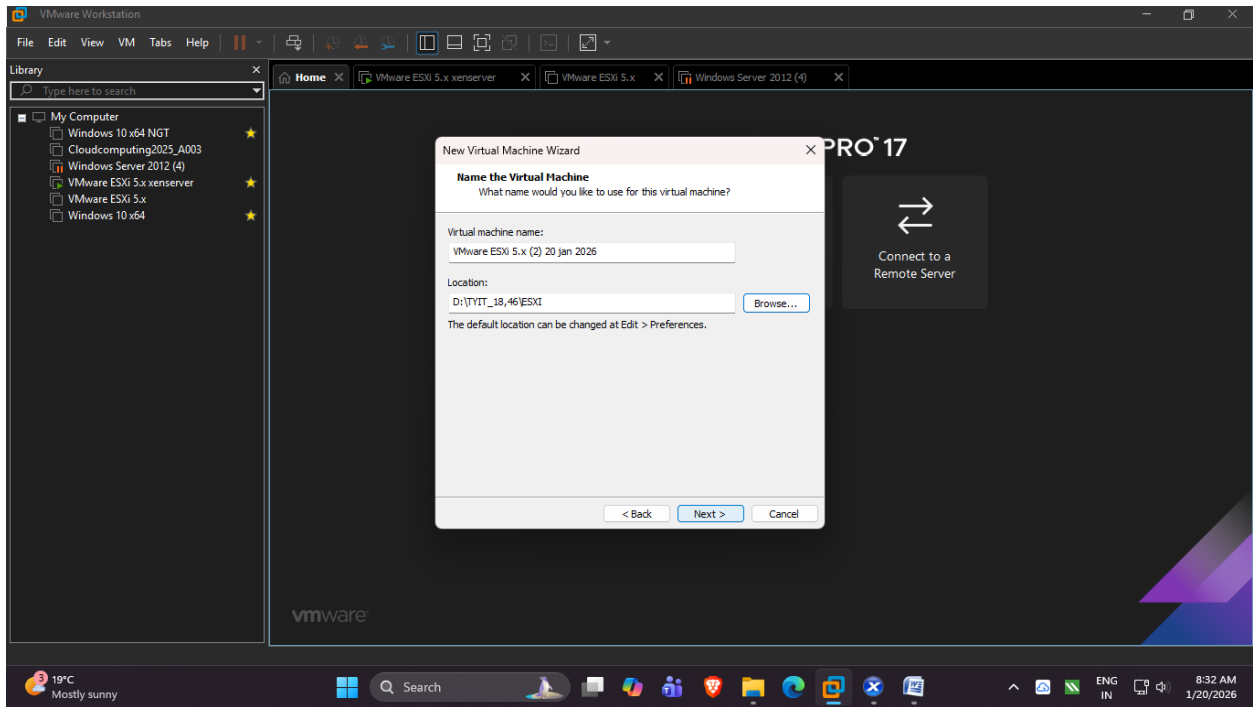
STEPS

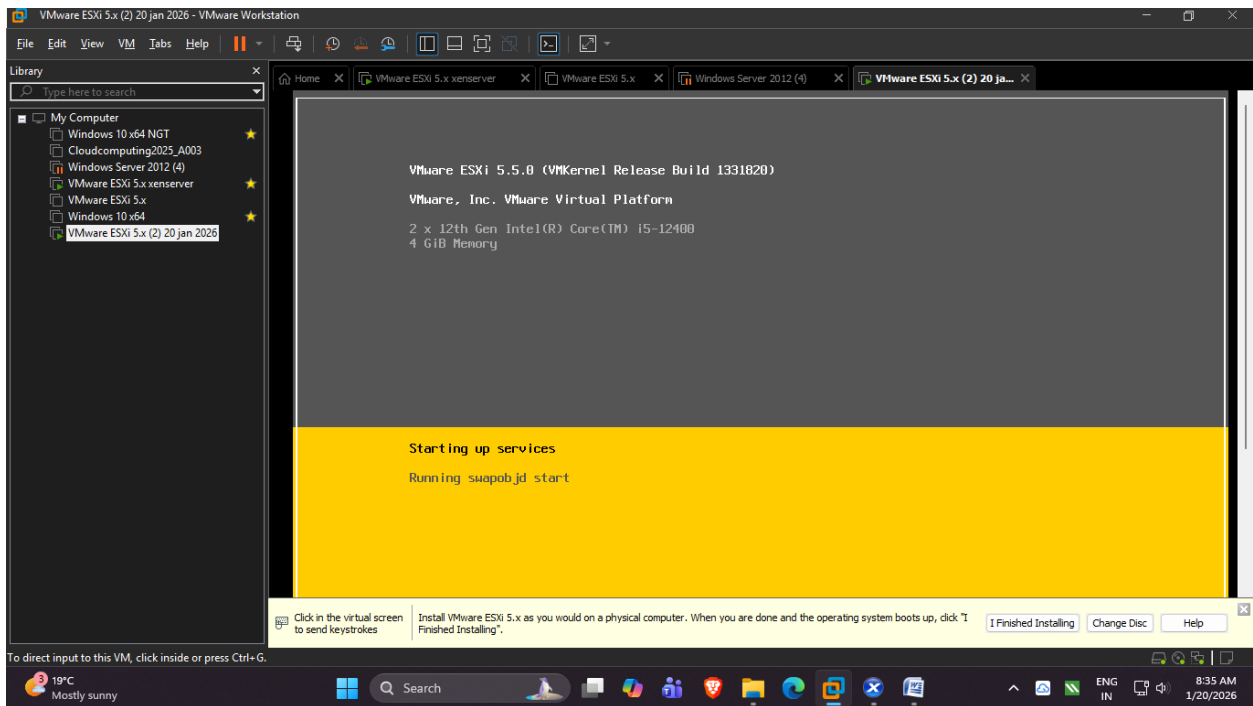
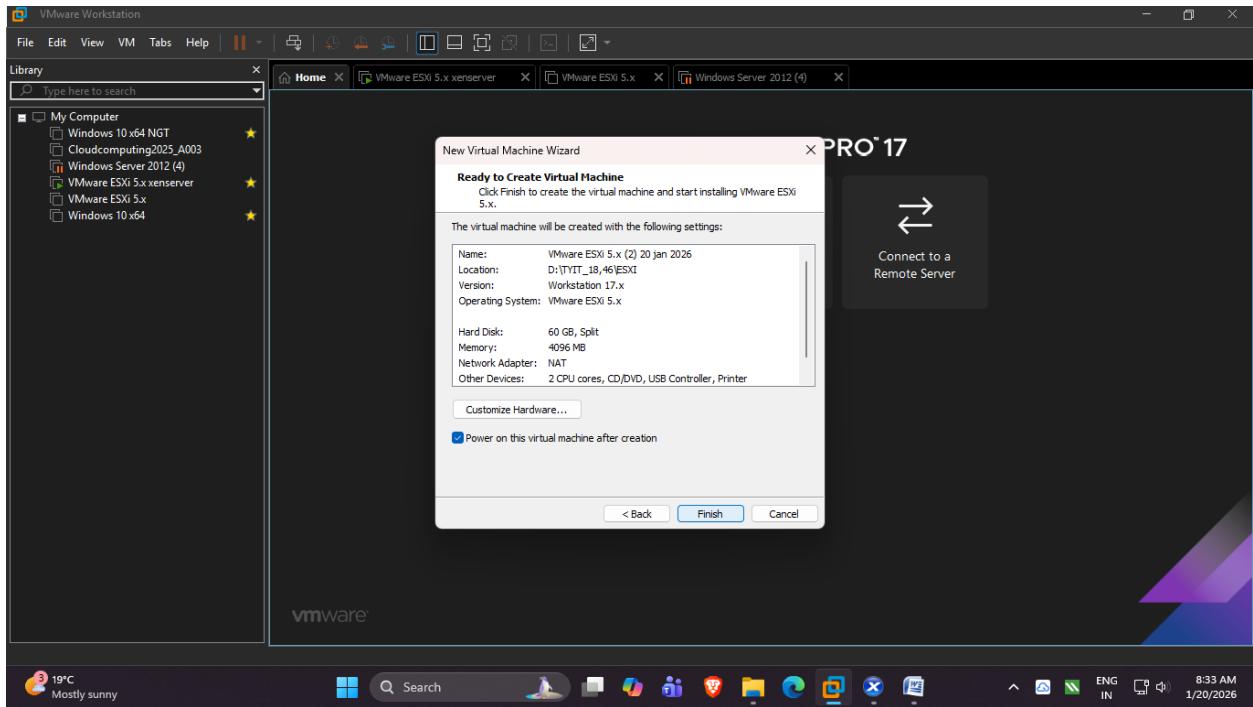
Create new virtual machine

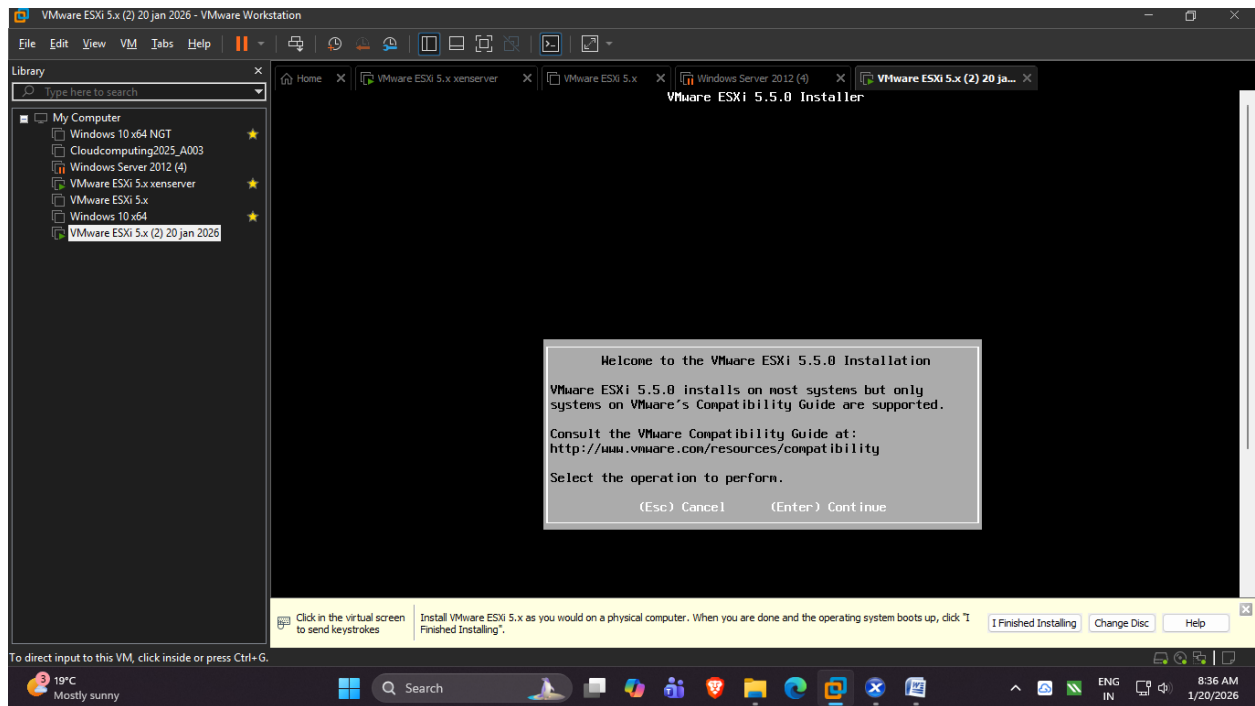
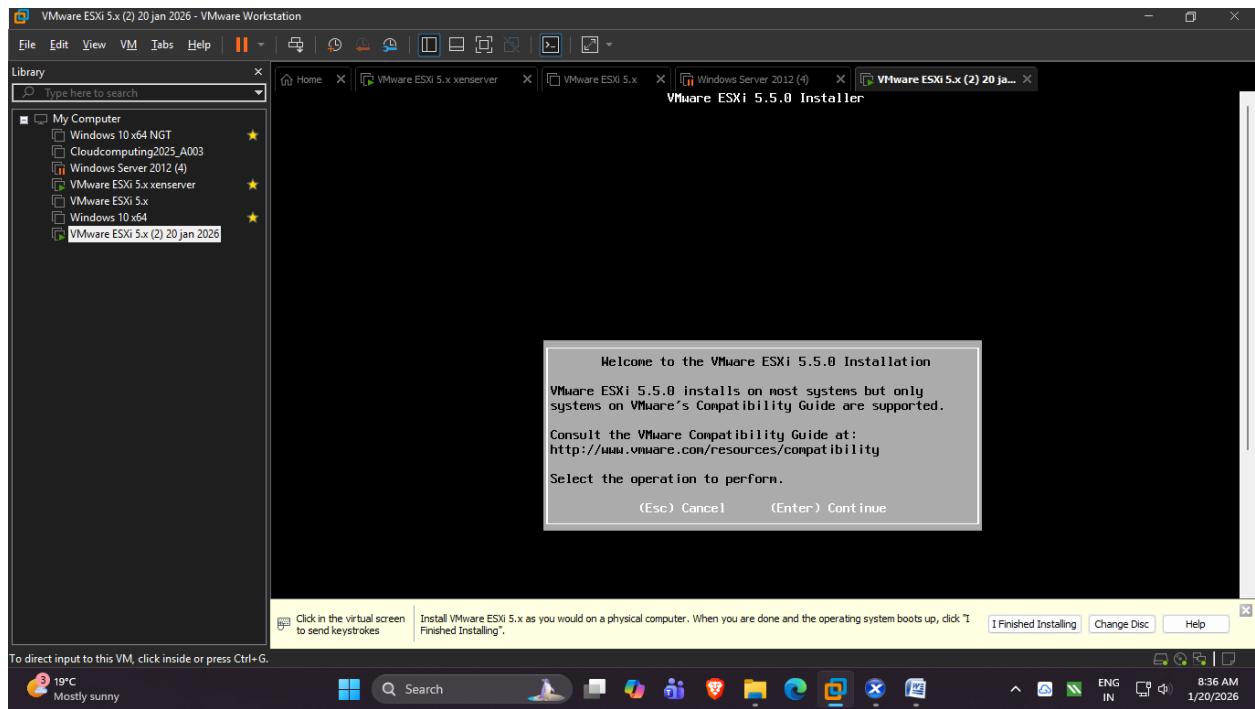


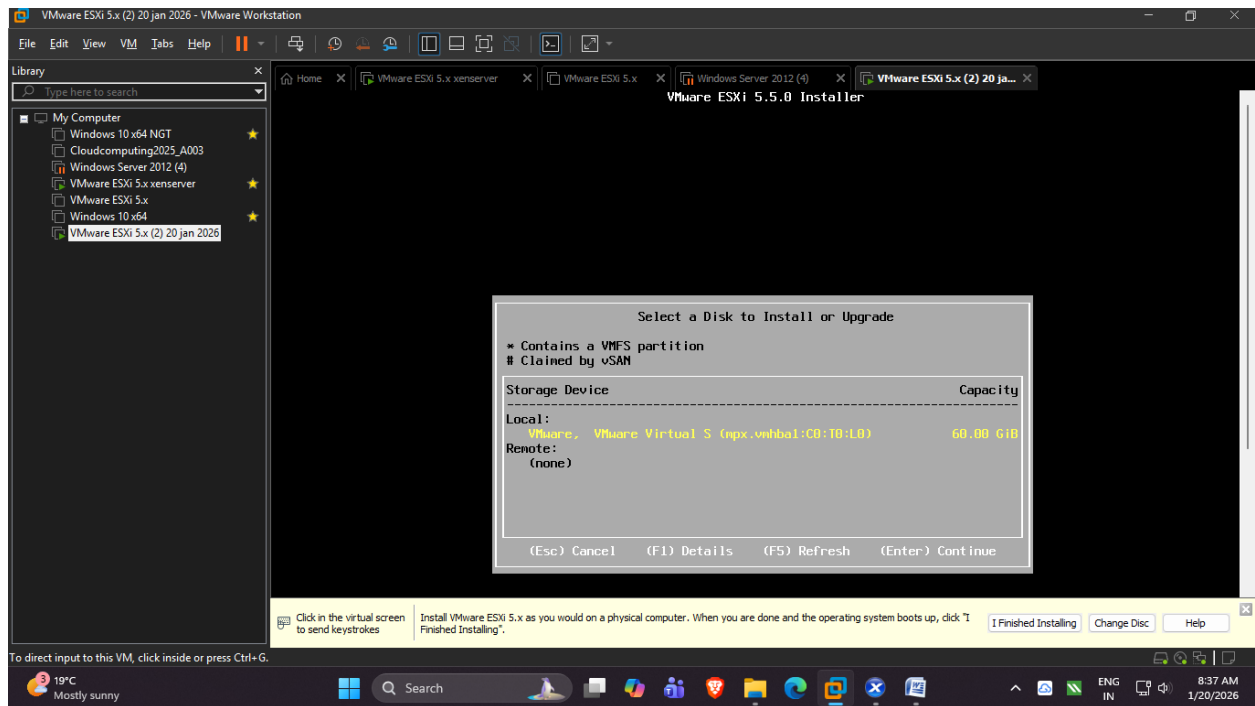
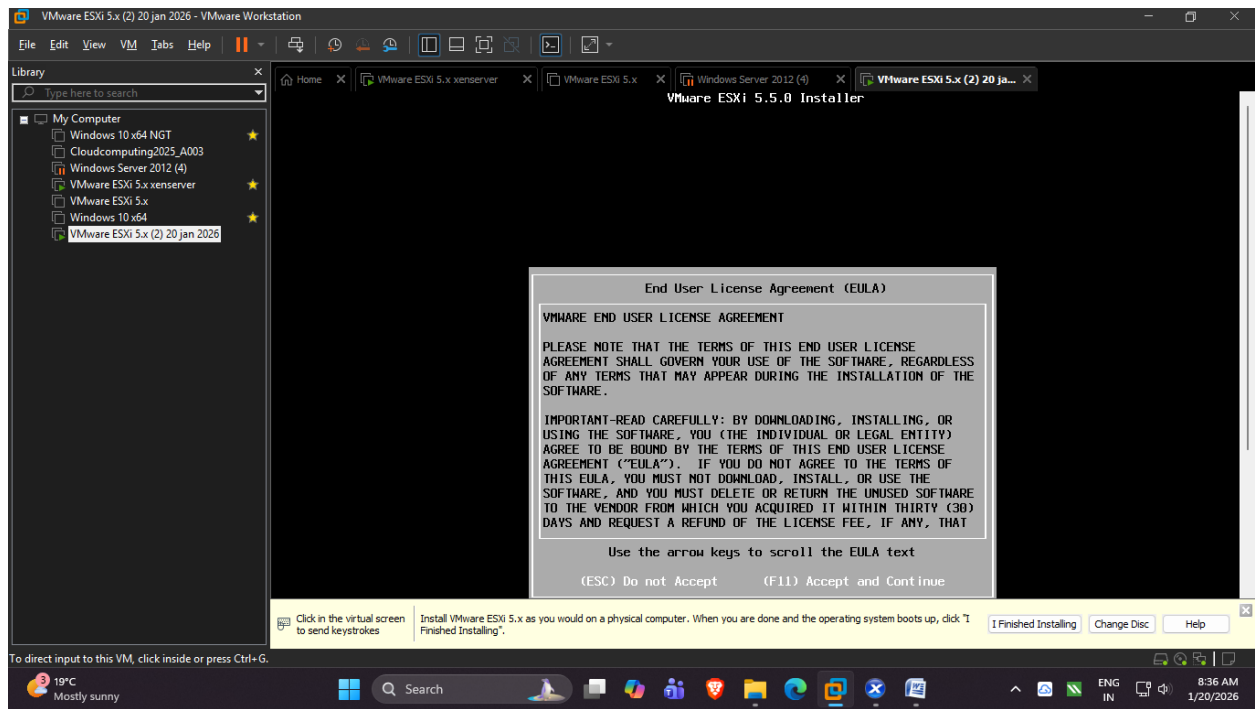
Browse VMware iso file



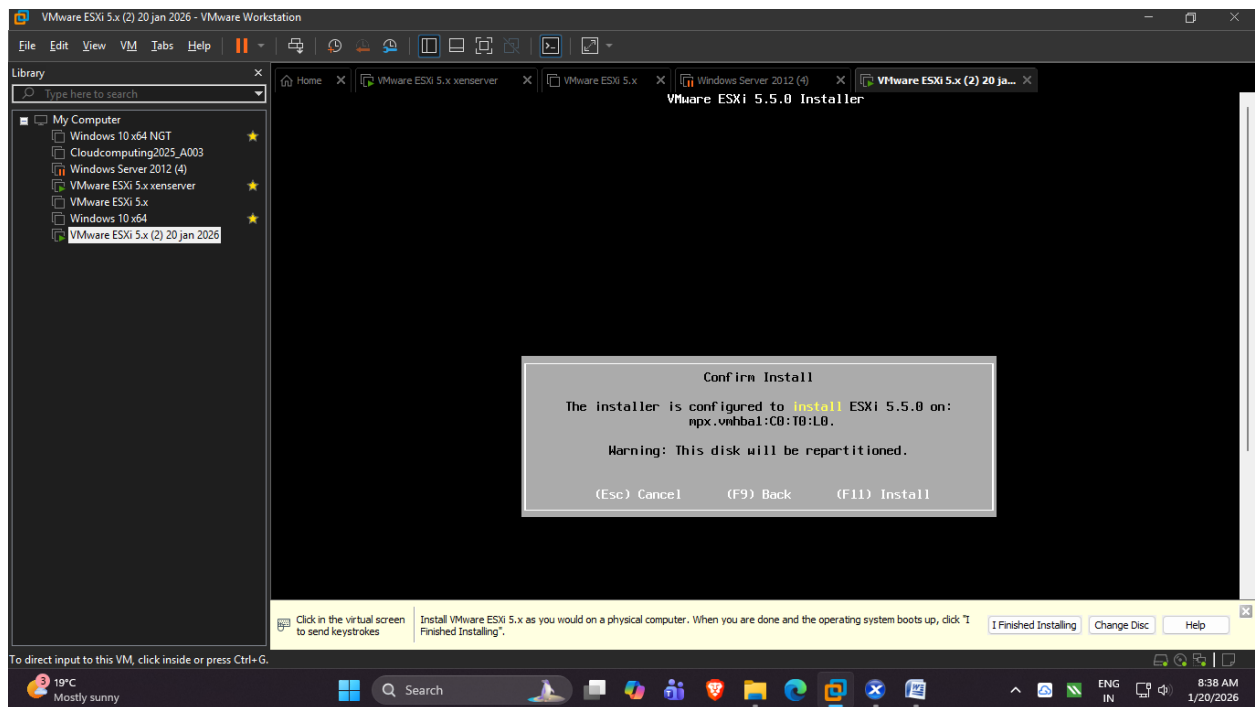
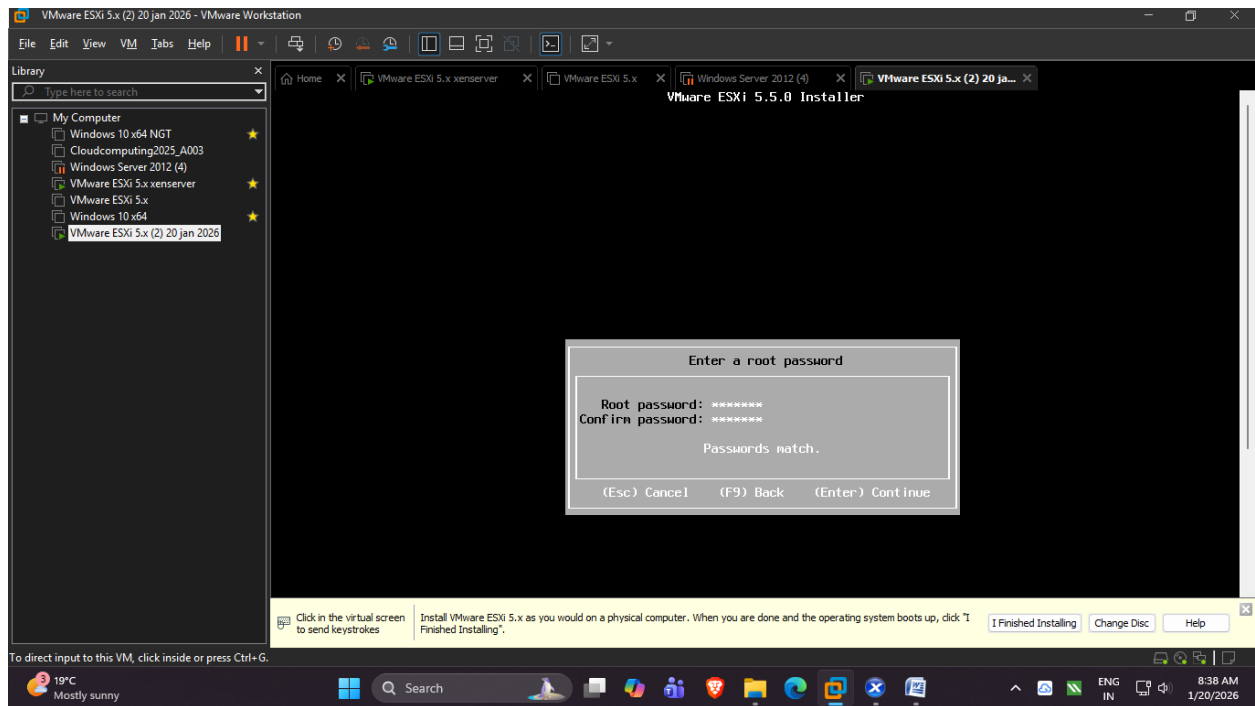


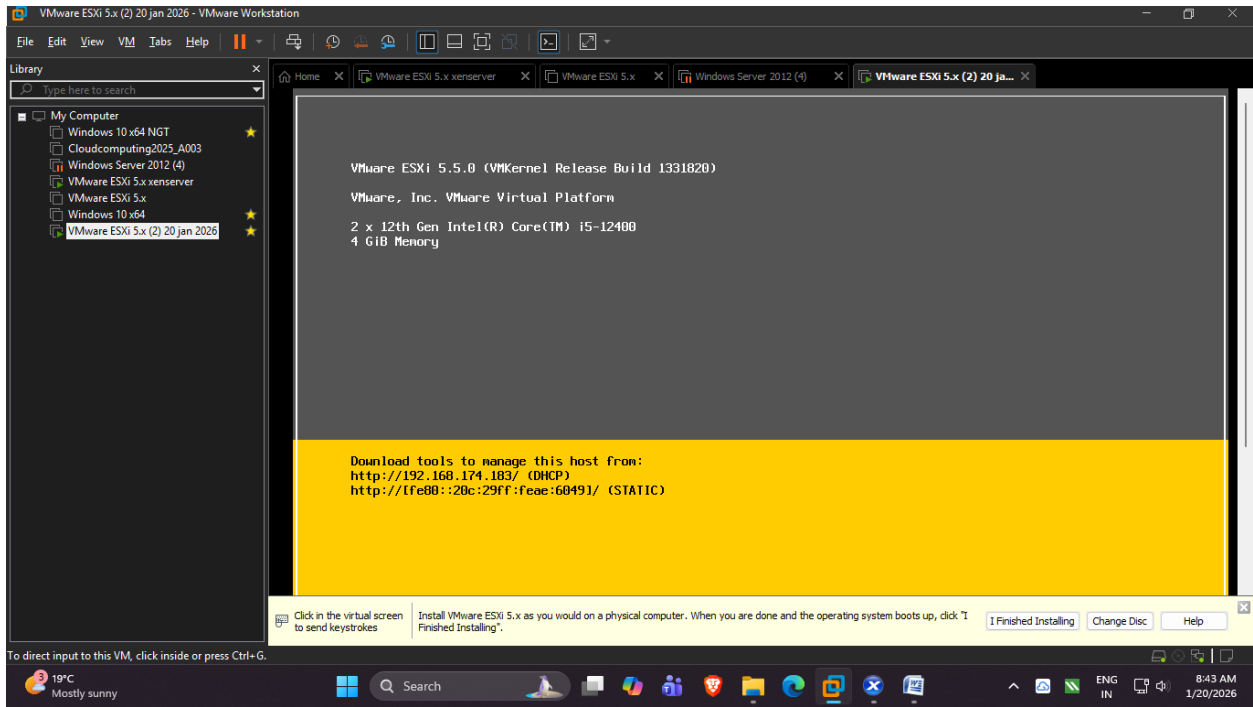
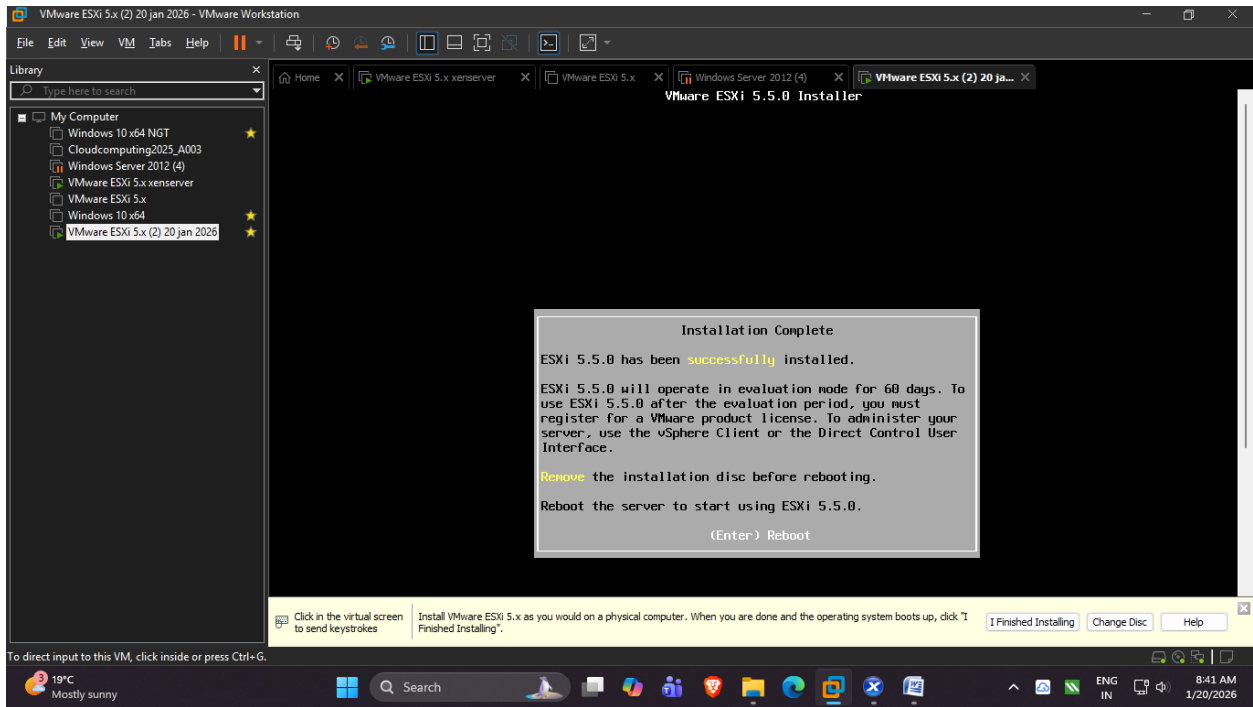




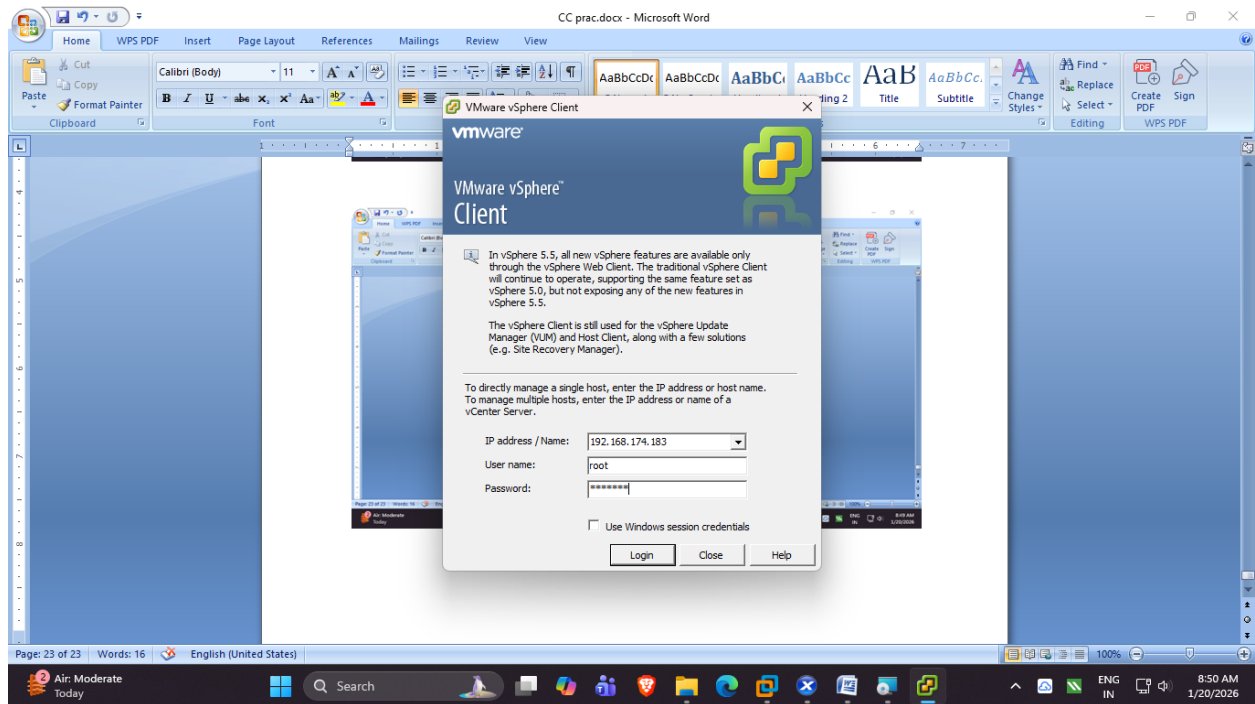
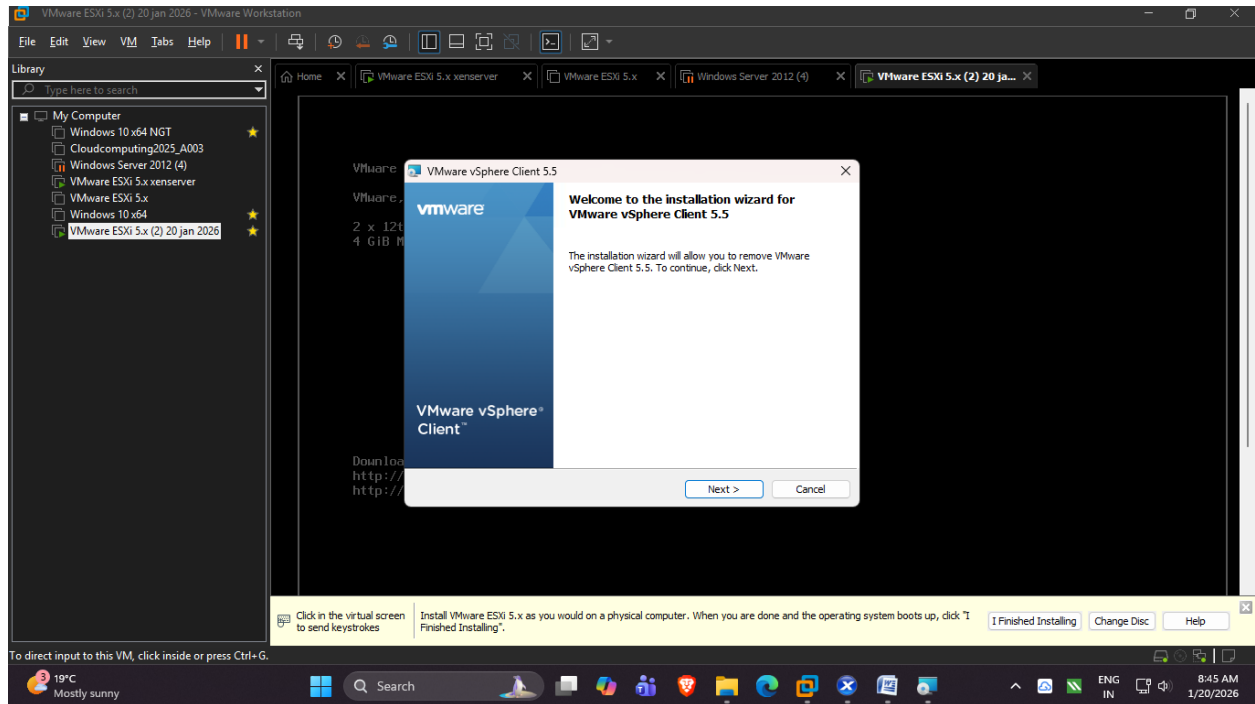


Set Password 1234567.





Open Vsphere Client



192.168.174.183 - vSphere Client

File Edit View Inventory Administration Plug-ins Help

Home Inventory

192.168.174.183

localhost.localdomain VMware ESXi, 5.5.0, 1331820 | Evaluation (60 days remaining)

Getting Started Summary Virtual Machines Resource Allocation Performance Configuration Local Users & Groups Events Permissions

What is a Host?

A host is a computer that uses virtualization software, such as ESX or ESXi, to run virtual machines. Hosts provide the CPU and memory resources that virtual machines use and give virtual machines access to storage and network connectivity.

You can add a virtual machine to a host by creating a new one or by deploying a virtual appliance.

The easiest way to add a virtual machine is to deploy a virtual appliance. A virtual appliance is a pre-built virtual machine with an operating system and software already installed. A new virtual machine will need an operating system installed on it, such as Windows or Linux.

Basic Tasks

- Deploy from VA Marketplace
- Create a new virtual machine

Explore Further

Recent Tasks

Name	Target	Status	Details	Initiated by	Requested Start Time	Start Time	Completed Time
Auto power On	192.168.174.1...	Completed		root	1/20/2026 8:43:29 AM	1/20/2026 8:43:29 AM	1/20/2026 8:43:29 AM

Tasks

Search

ENG IN 8:51 AM 1/20/2026

192.168.174.183 - vSphere Client

File Edit View Inventory Administration Plug-ins Help

Home Inventory

192.168.174.183

localhost.localdomain VMware ESXi, 5.5.0, 1331820 | Evaluation (60 days remaining)

Getting Started Summary Virtual Machines Resource Allocation Performance Configuration Local Users & Groups Events Permissions

Name, State or Guest OS contains: Clear

Name	State	Provisioned Space	Used Space	Host CPU - MHz	Host Mem - MB	Guest Mem - %	Notes
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Recent Tasks

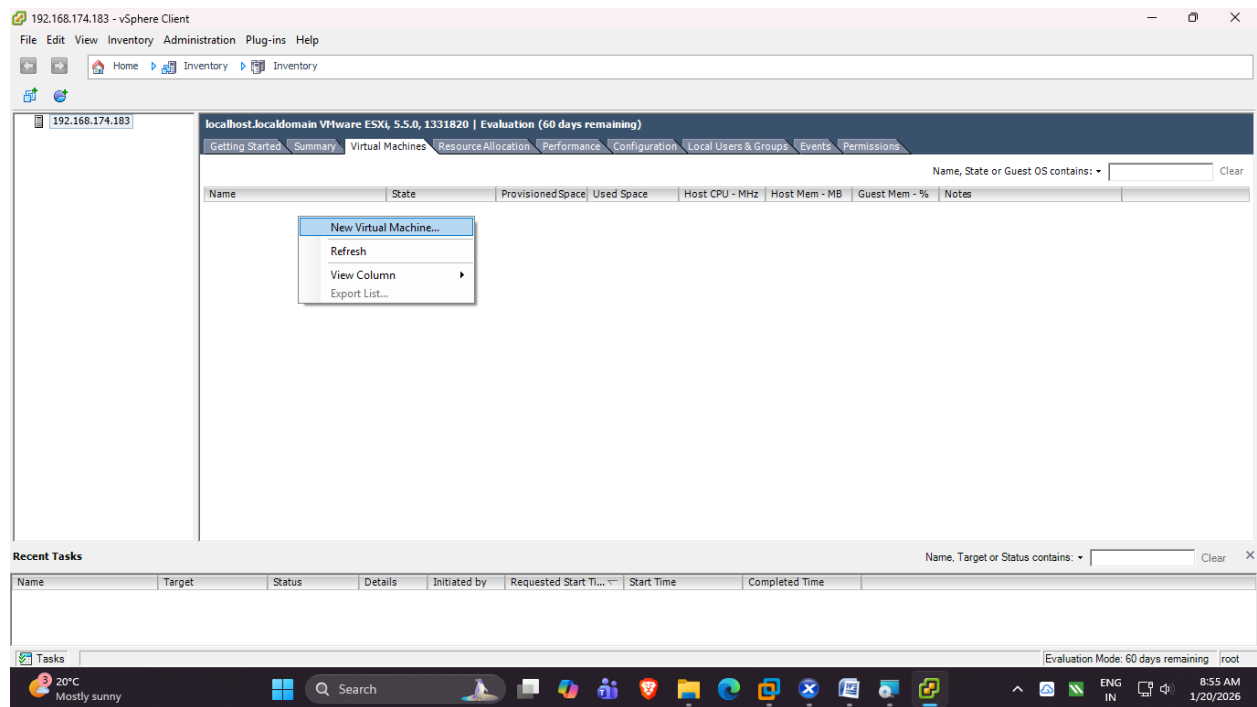
Name	Target	Status	Details	Initiated by	Requested Start Time	Start Time	Completed Time
------	--------	--------	---------	--------------	----------------------	------------	----------------

Tasks

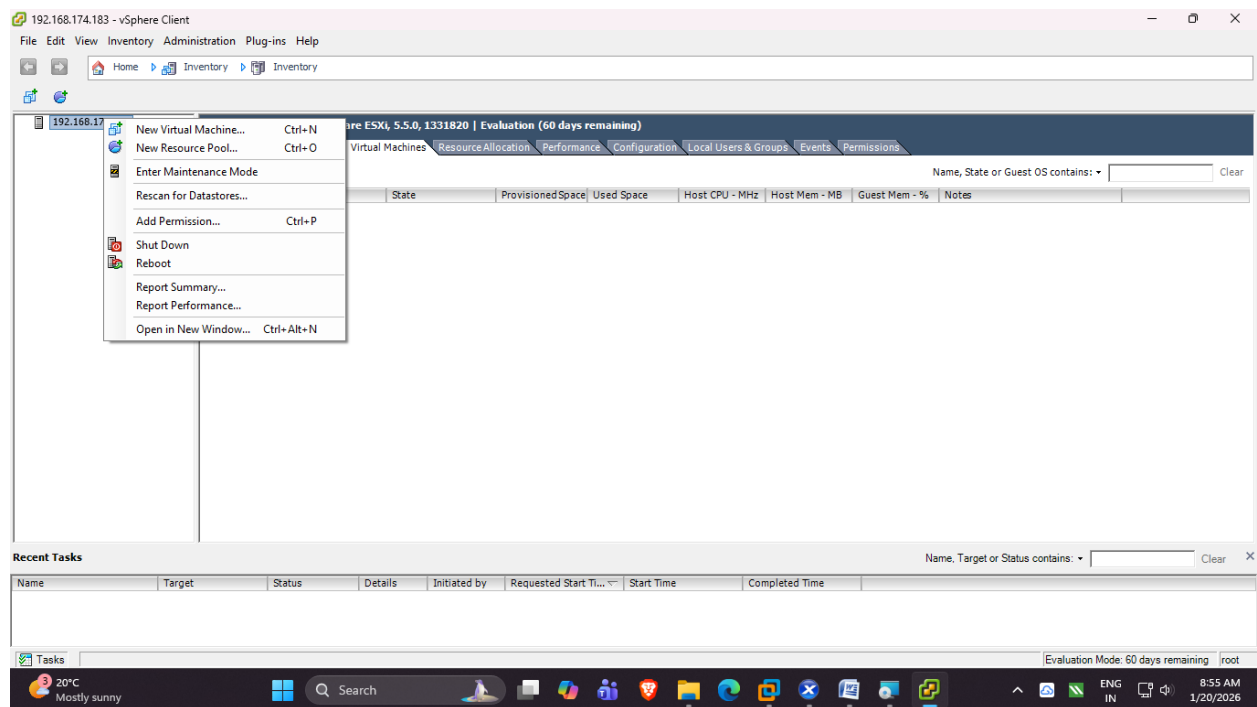
Search

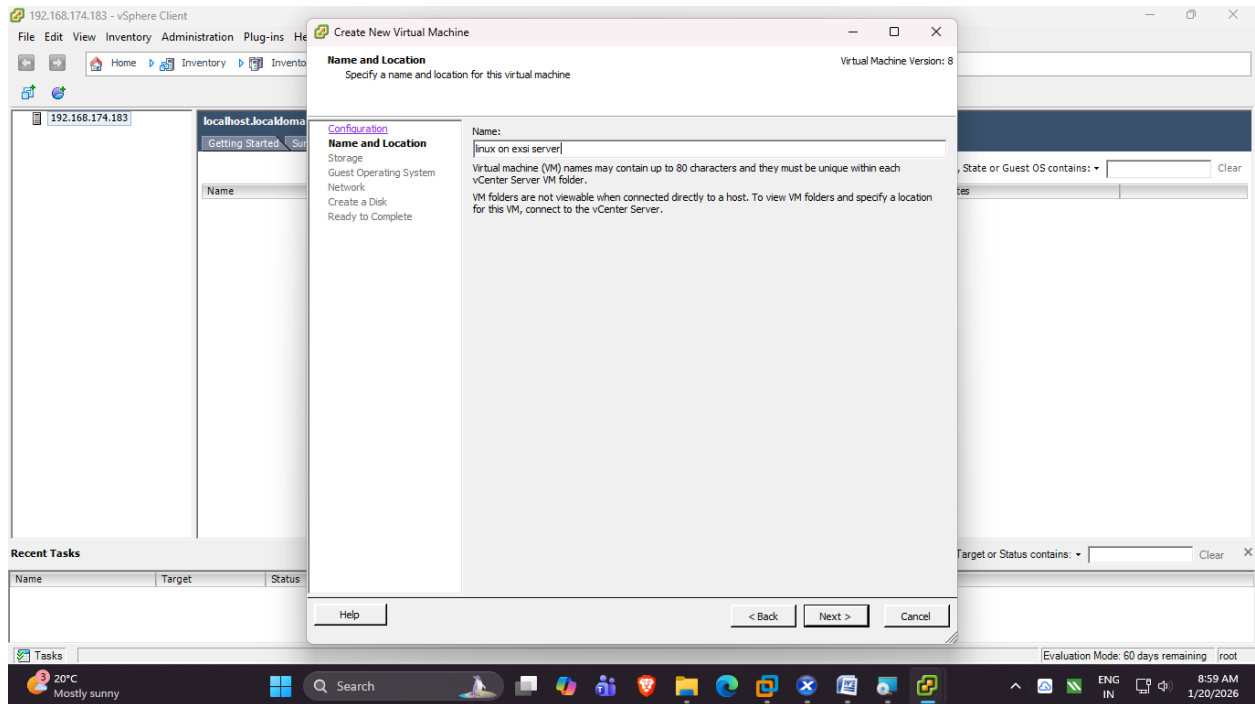
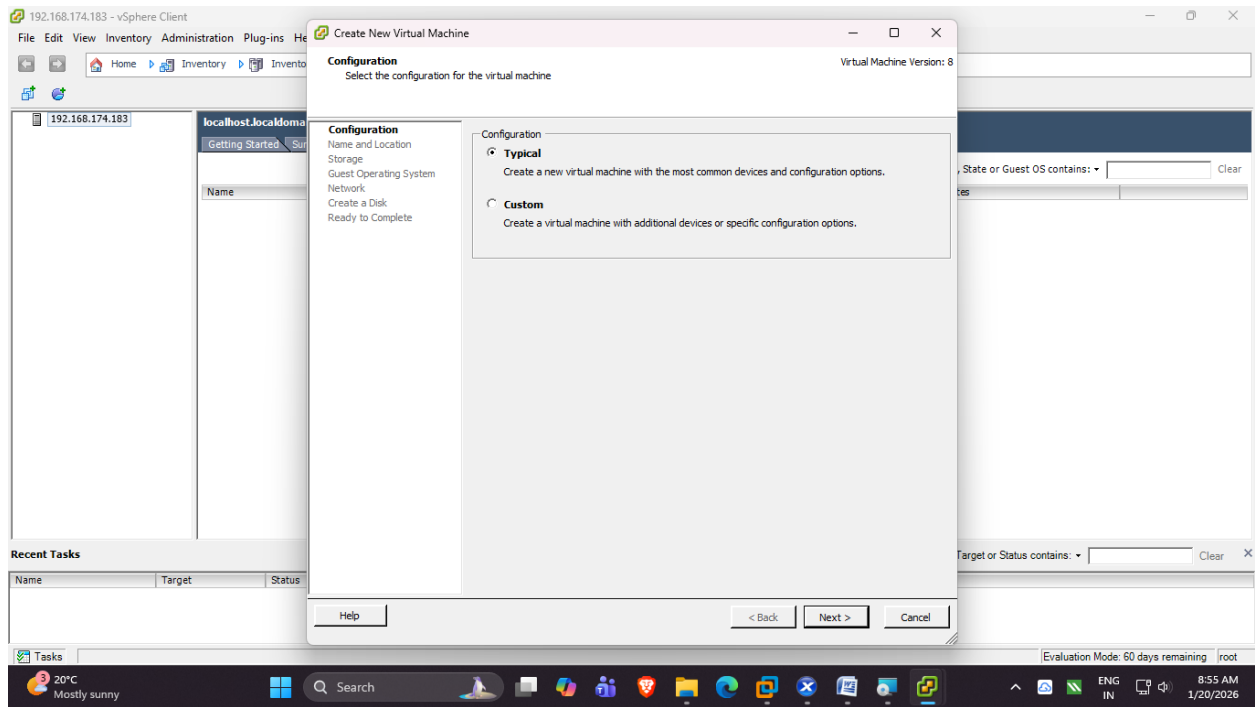
ENG IN 8:55 AM 1/20/2026

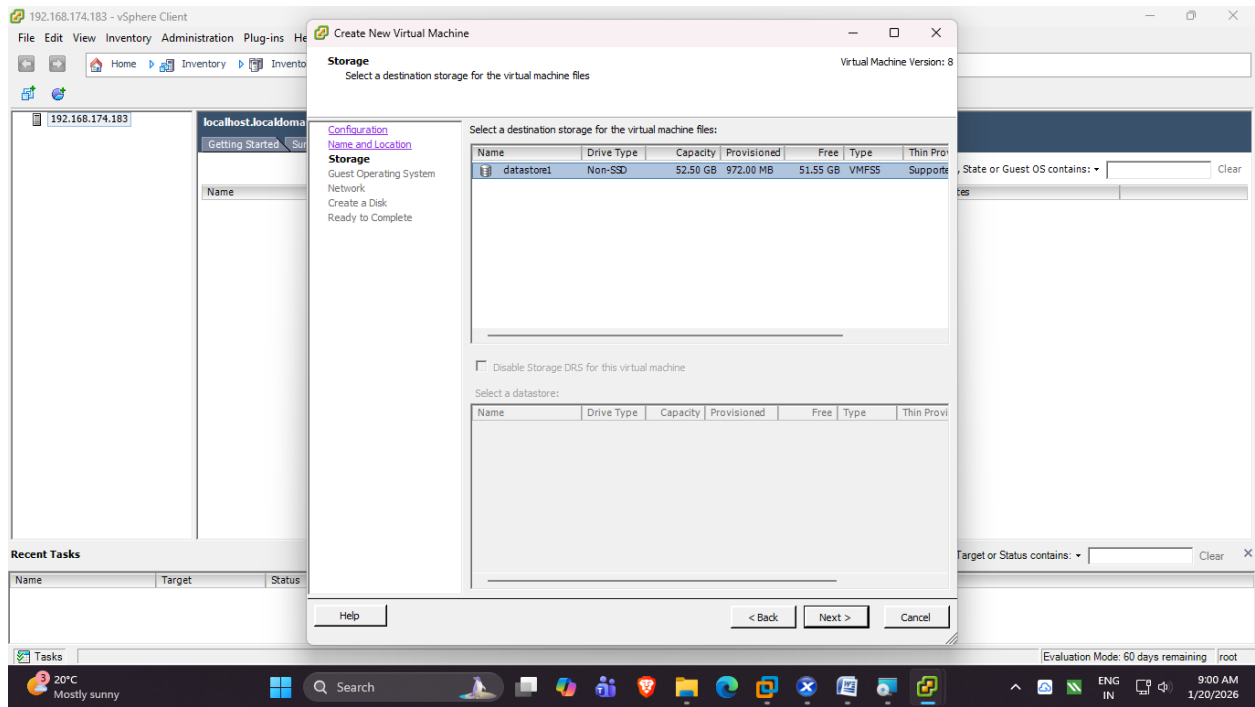
Two ways to create vm: 1. Right click on empty space on virtual machine tab



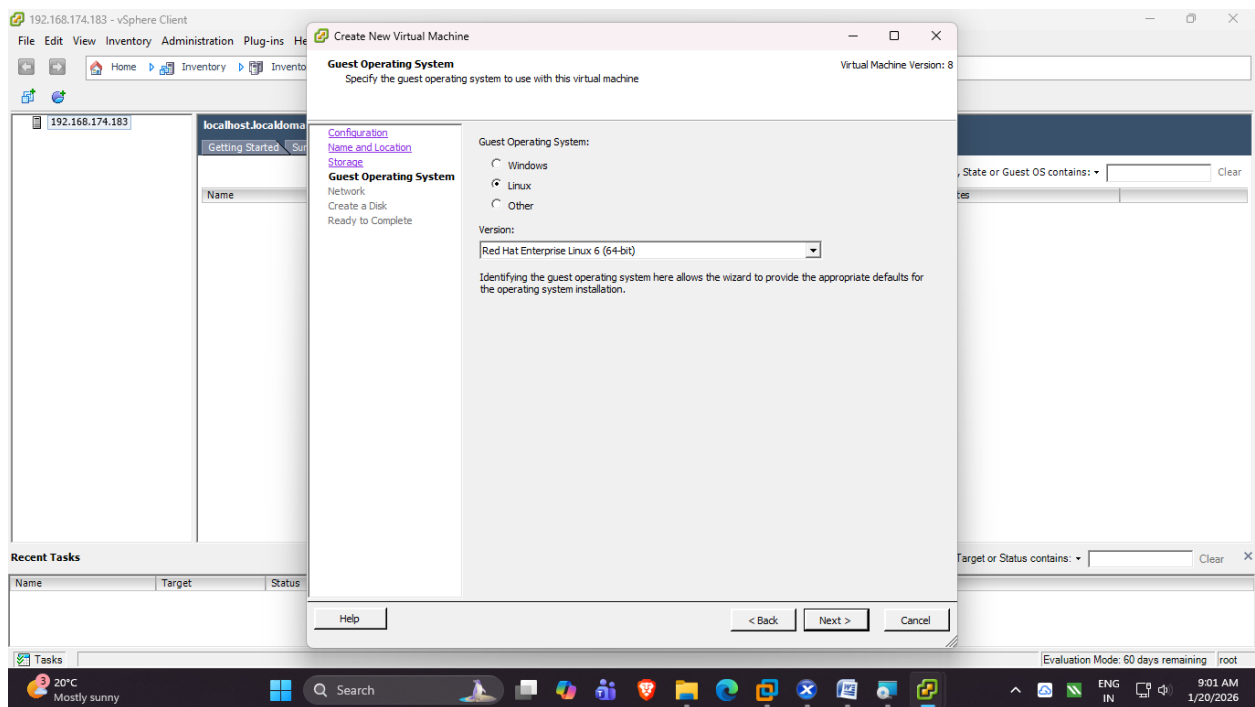
2. Right click on left side ip address

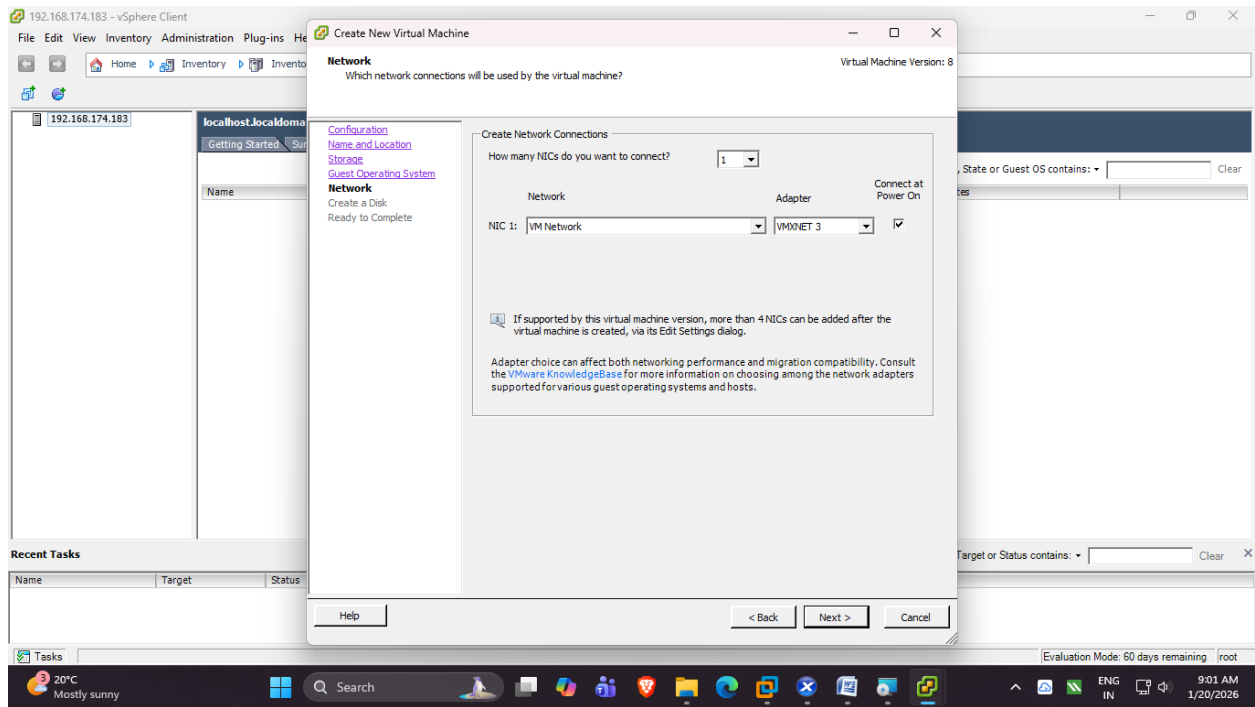




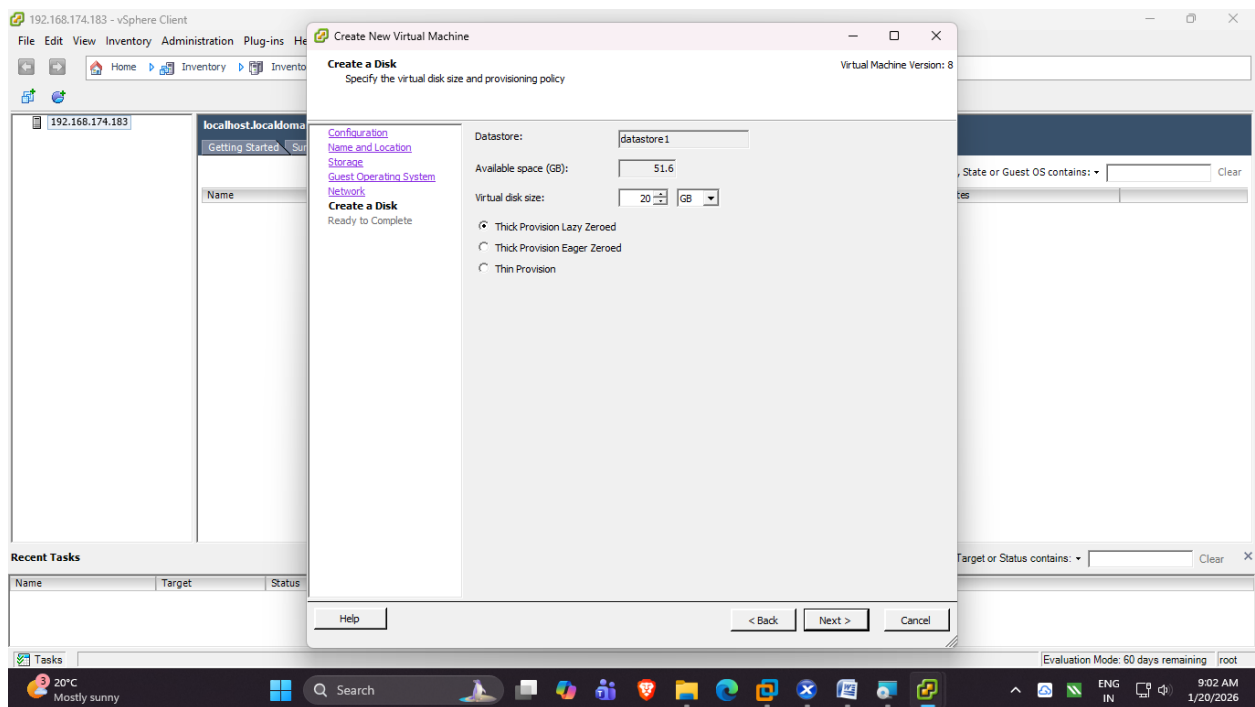


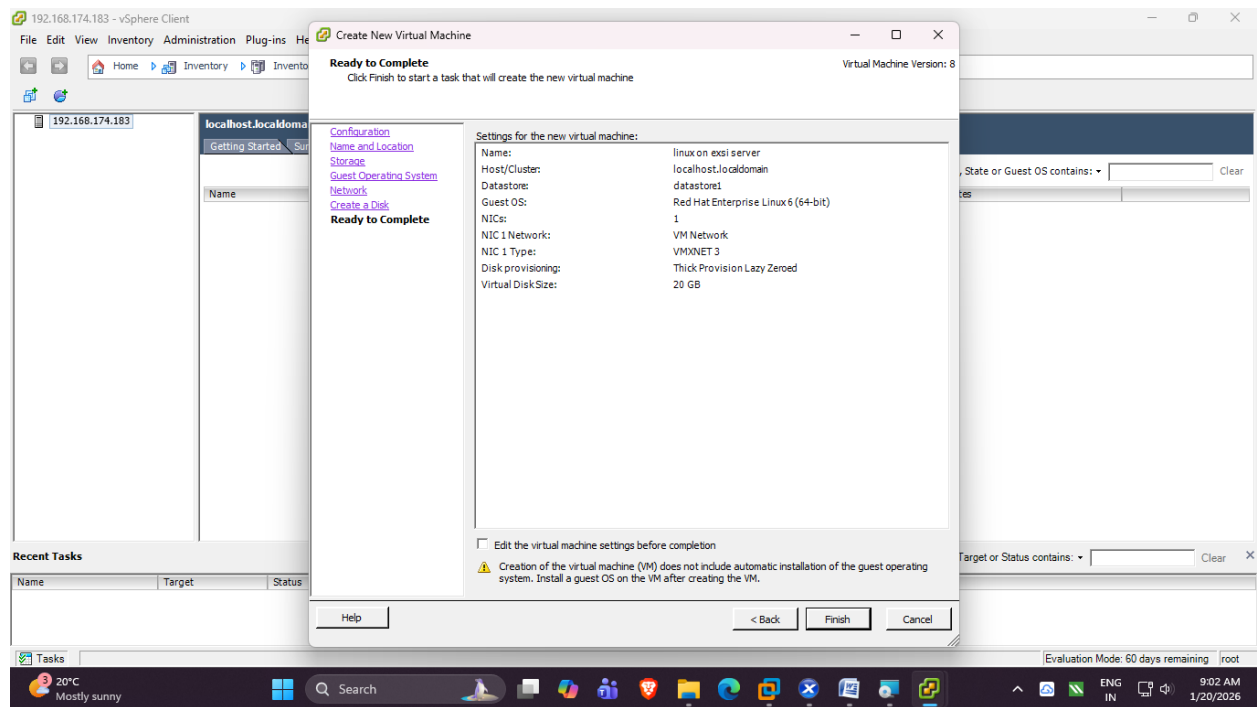
Select Linux and version Red hat enterprise Linux 6 (64 bit)



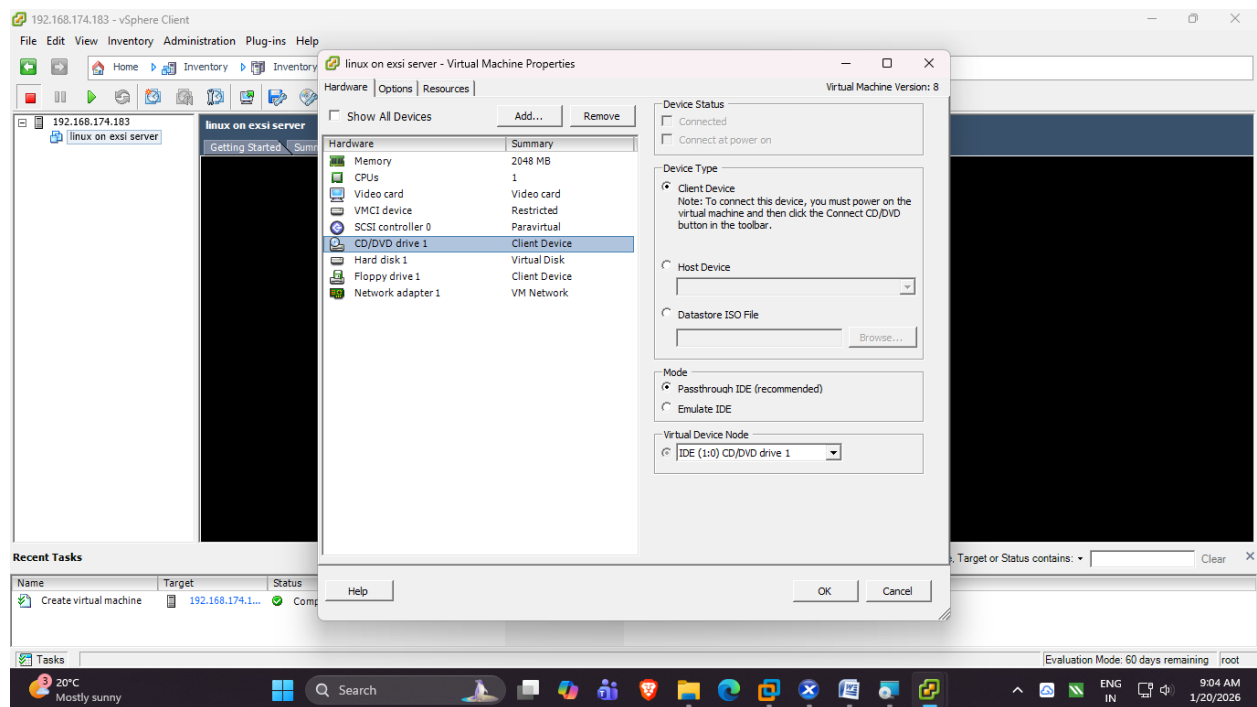


Allot any virtual disk size



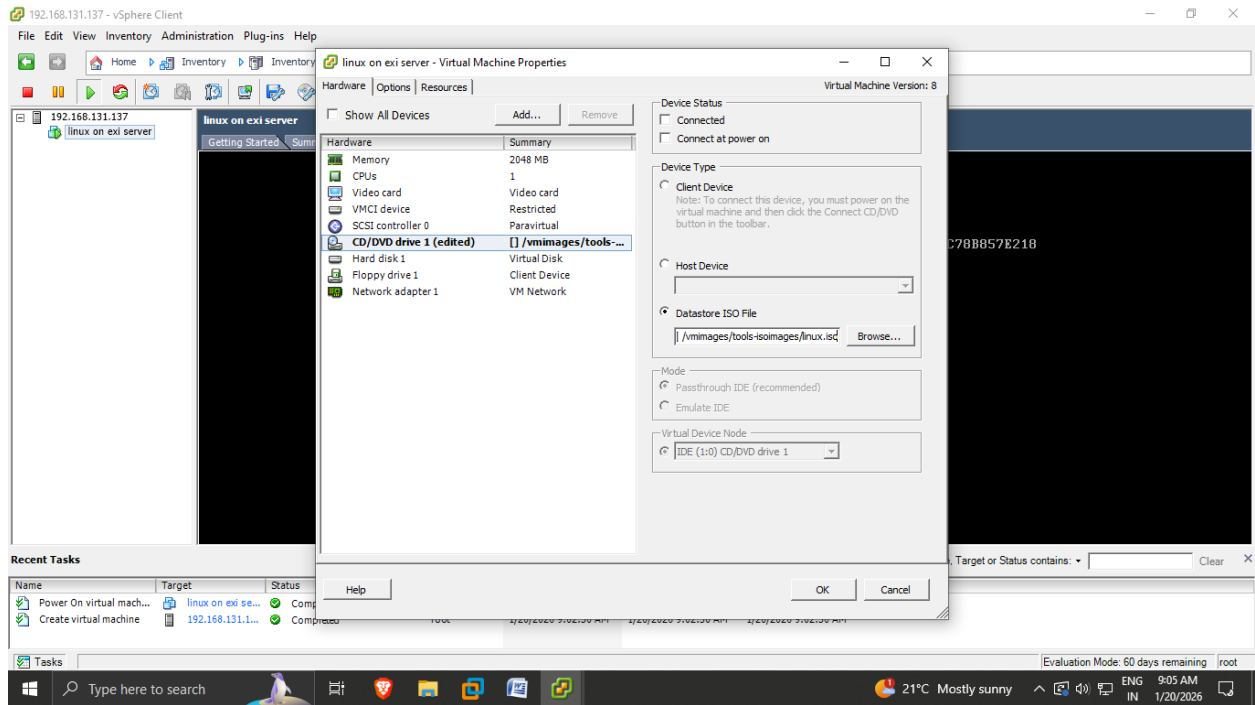


Right click on machine (left side) then click on edit setting

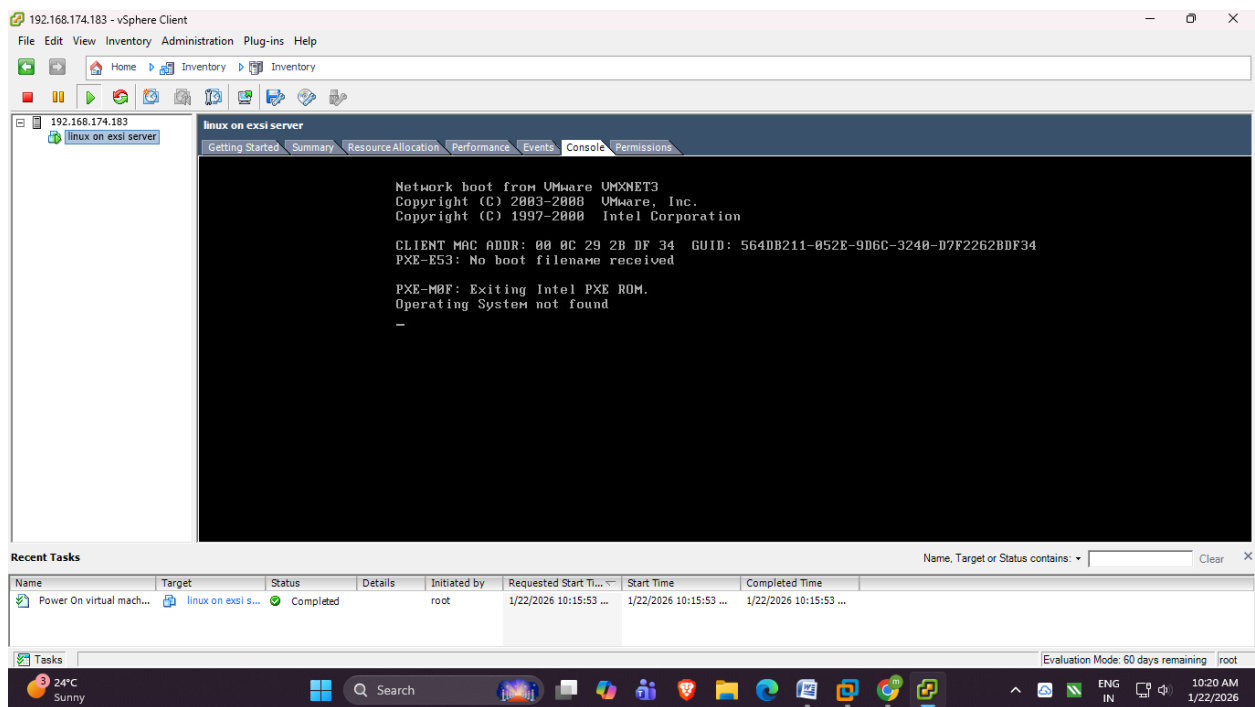


Select database iso file option on right side.

Click on vmimages then tools images then linux.iso



Click on console tab.



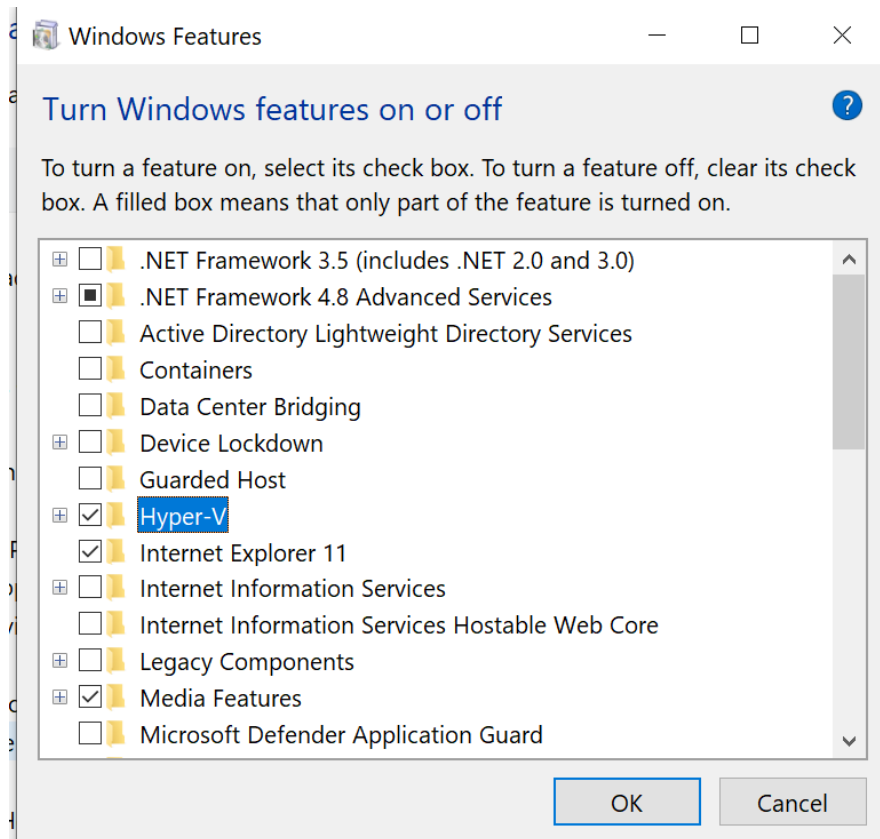
8. Native virtualization using Hyper-V.

Native virtualization, in the context of Hyper-V, refers to its architecture as a Type 1 or "bare-metal" hypervisor. Unlike hosted virtualization solutions that run as an application on an existing operating system, Hyper-V's hypervisor runs directly on the physical hardware. This design allows for more direct control over system resources, reduced overhead, and significantly improved performance for virtual machines. It achieves this through a micro kernelized architecture, a strict partition isolation model, and high-speed communication channels, all while leveraging modern hardware-assisted virtualization technologies from CPU manufacturers like Intel

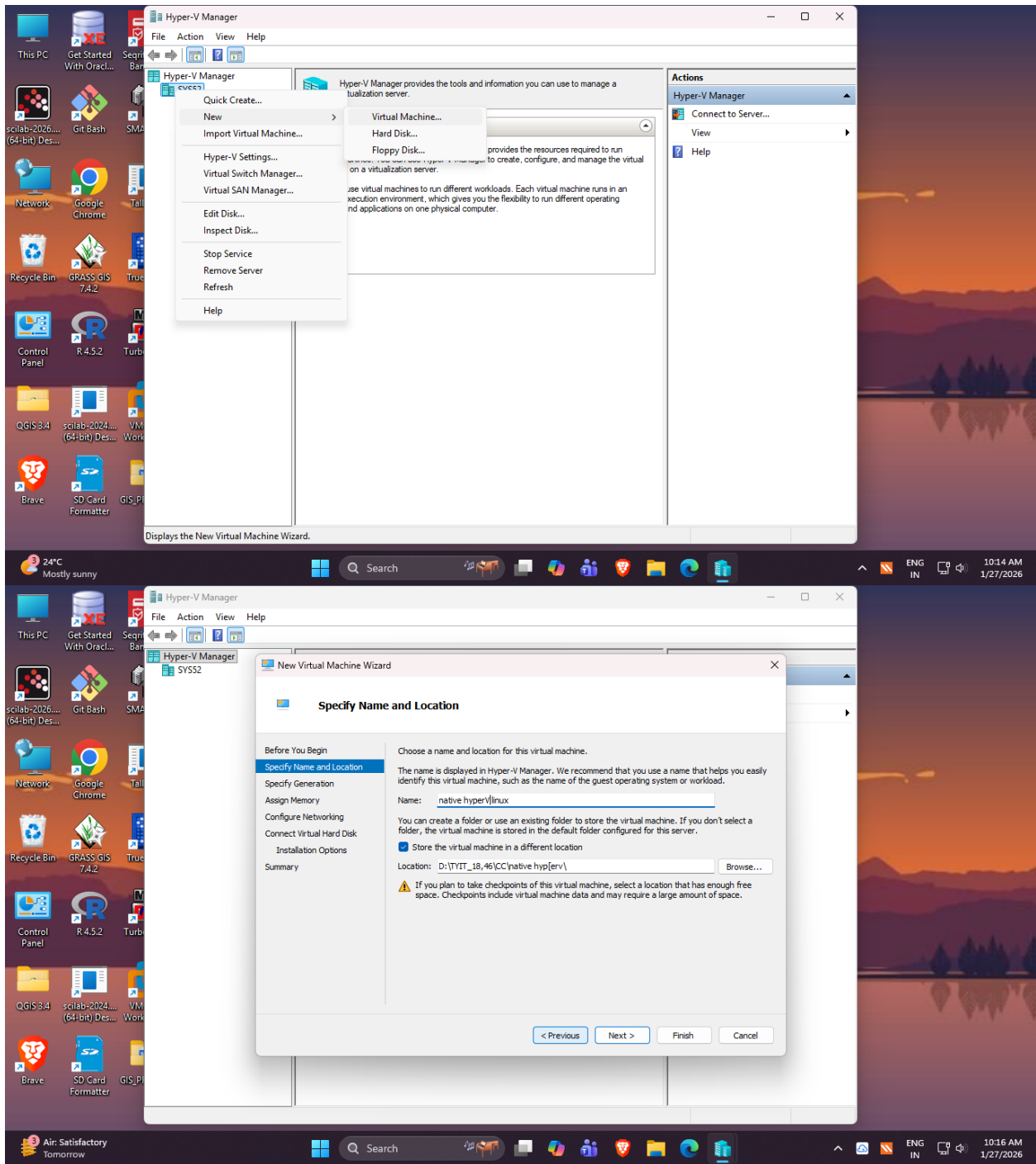
STEPS

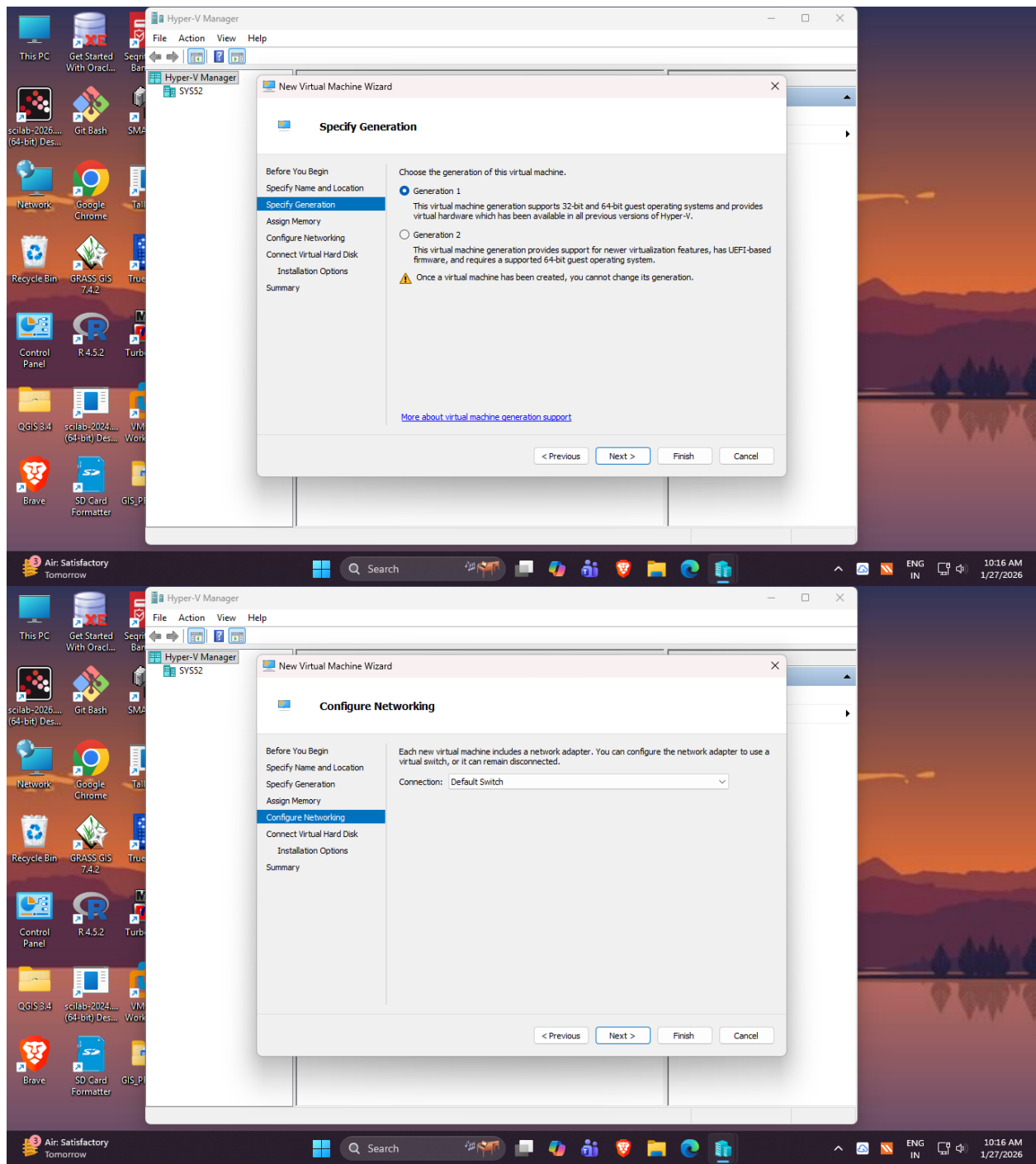
On your base machine.

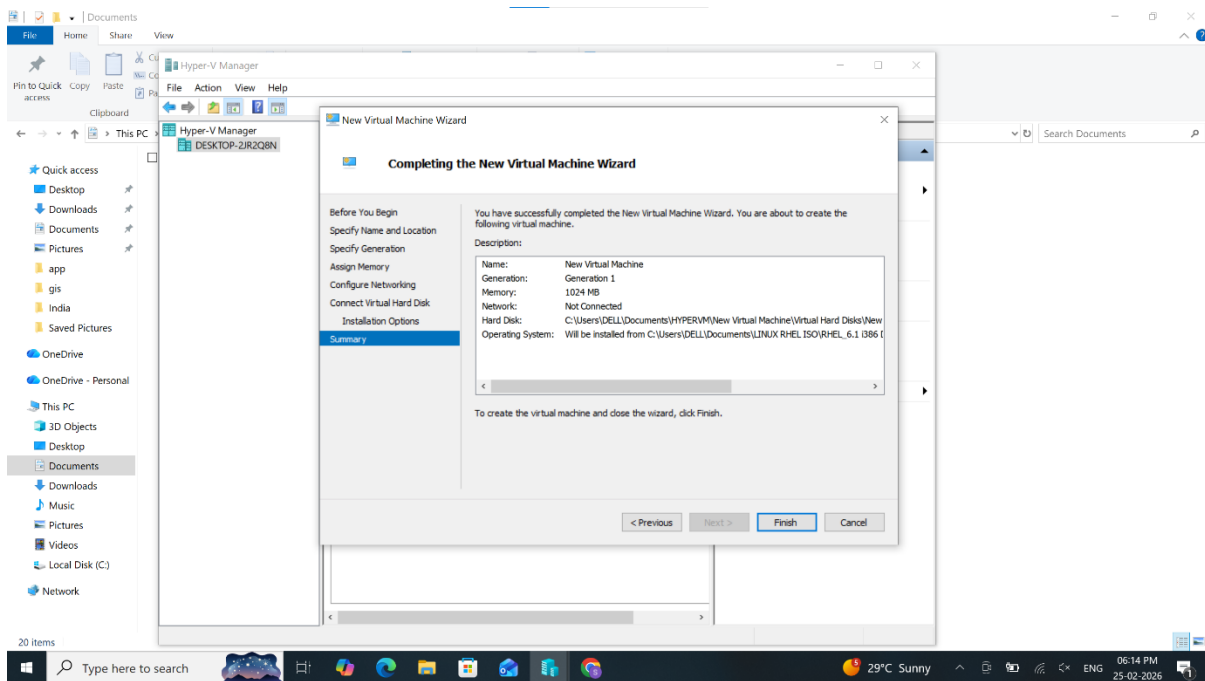
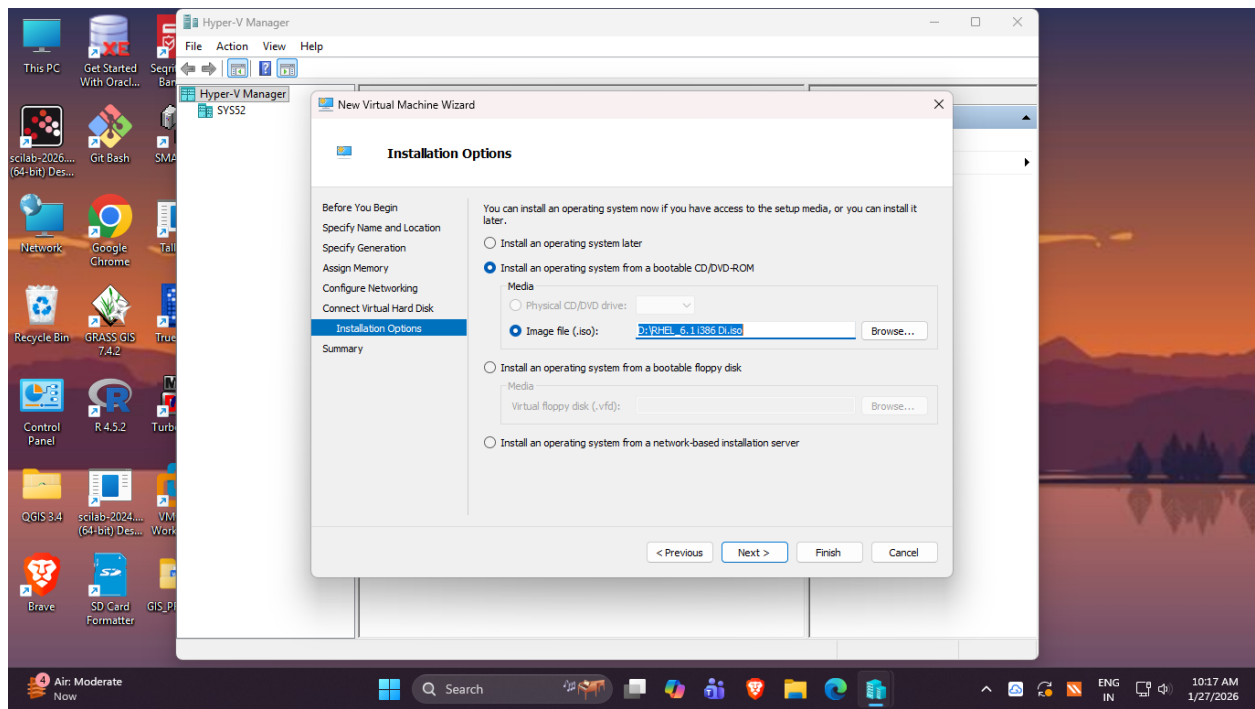
Open "control panel", then "programs", then "programs and features", then "turn windows features on or off" and check on "Hyper v"



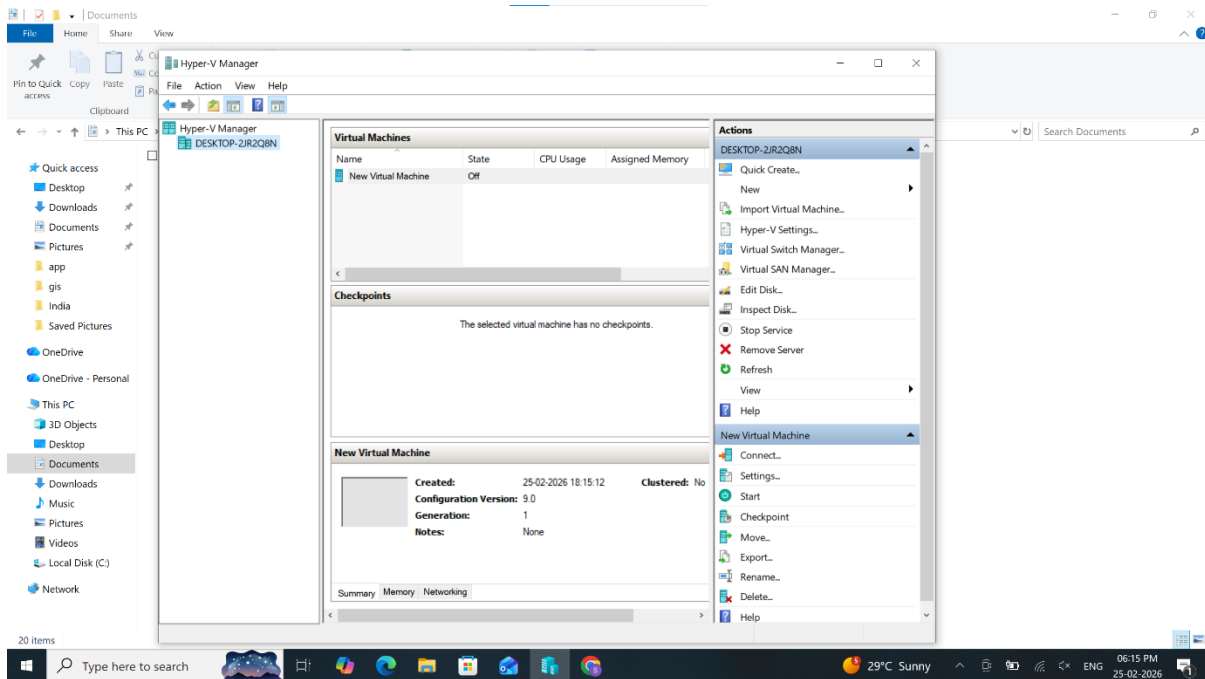
Restart your base machine and open “Hyper V Manager”



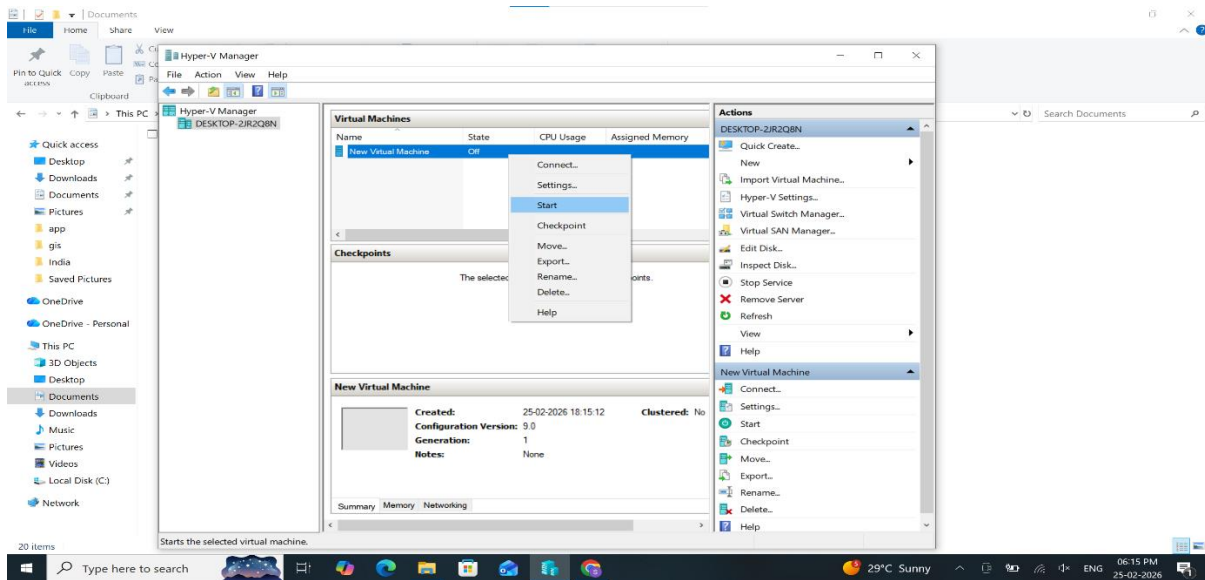




New VM gets created.

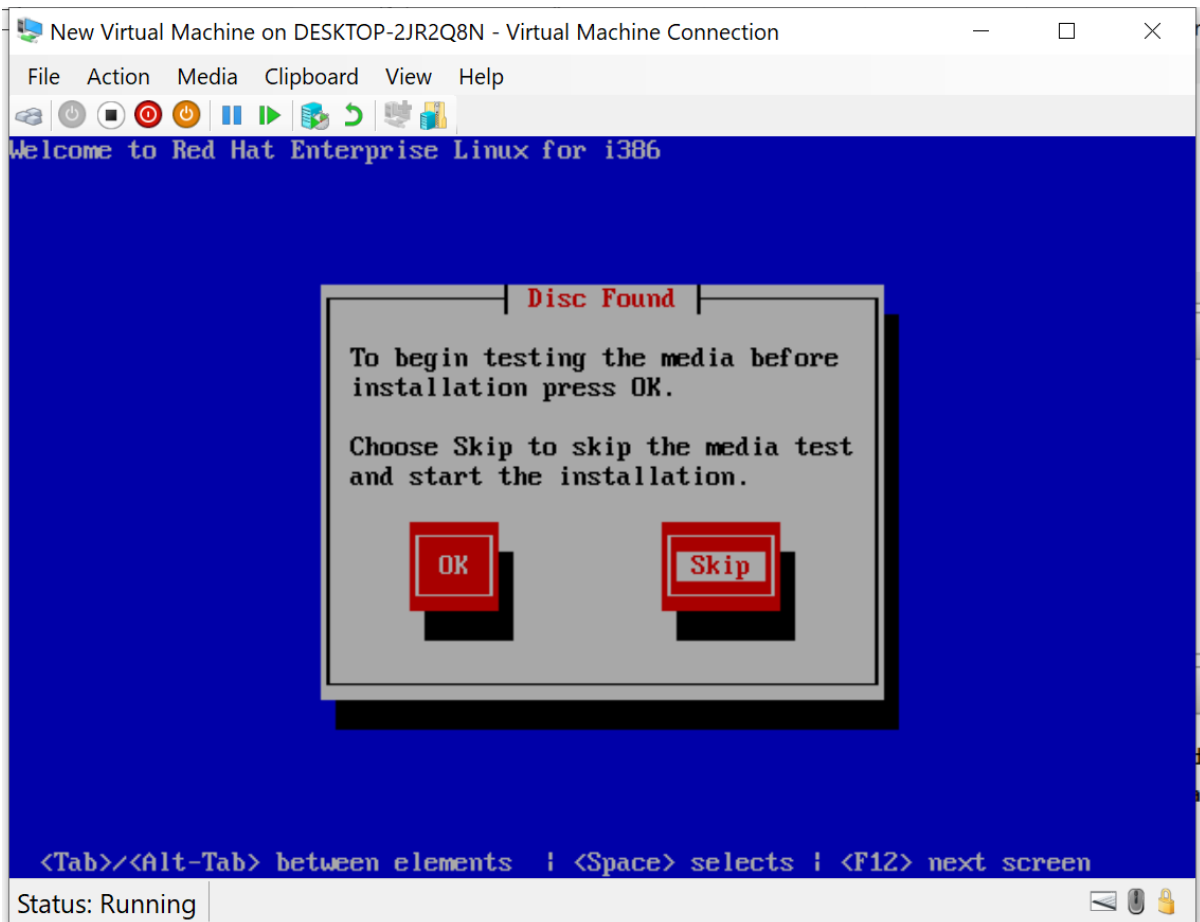


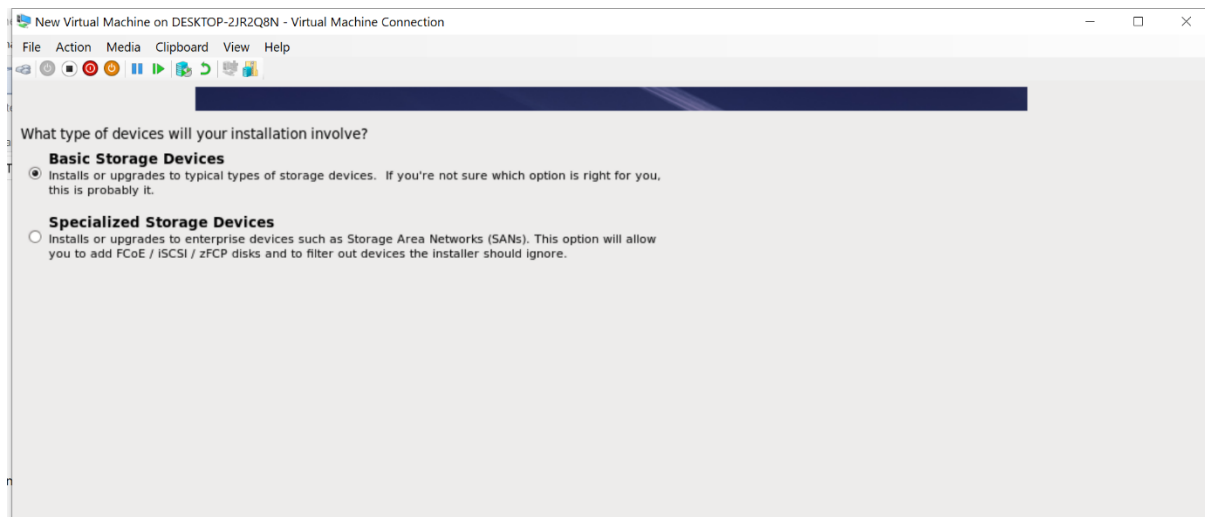
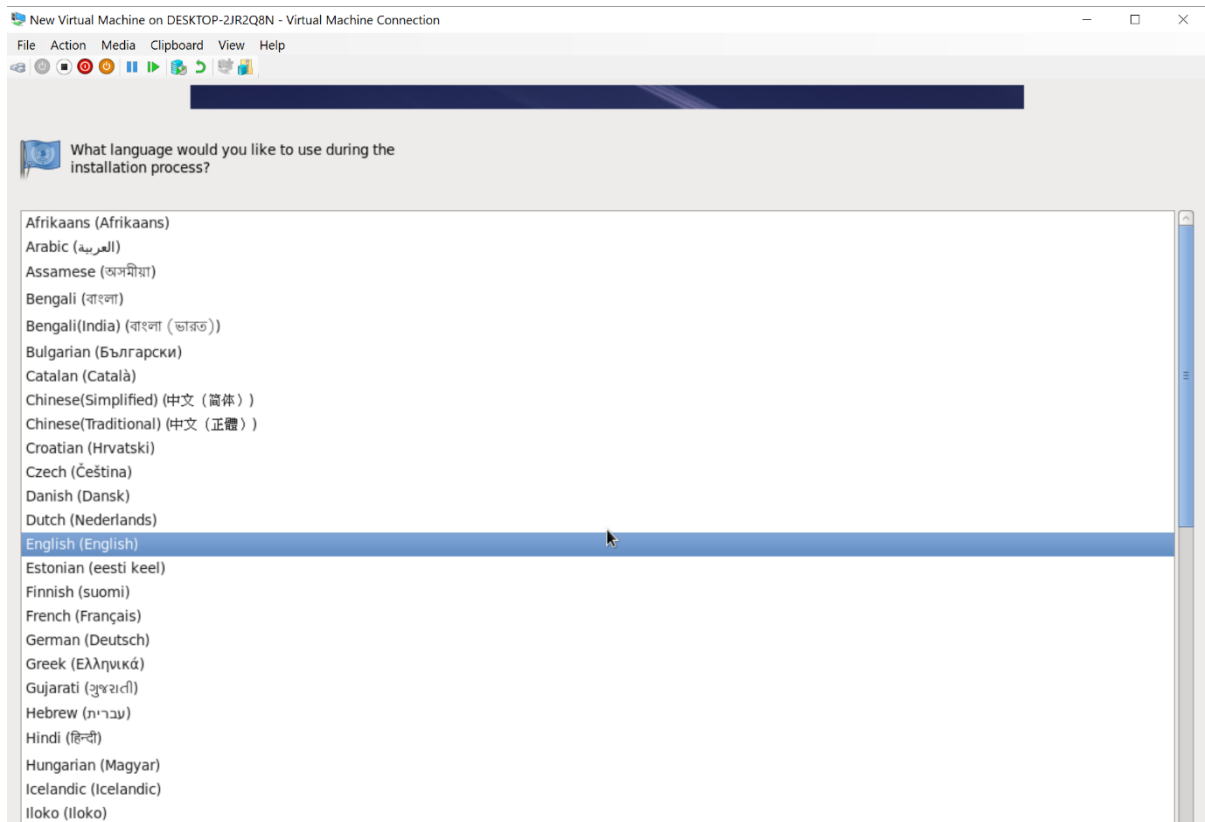
Right click on the VM and then click on “start” option.



New linux machine

New Virtual Machine			
	Created:	25-02-2026 18:15:12	Clustered: No
	Configuration Version:	9.0	Heartbeat: No
	Generation:	1	
	Notes:	None	
SummaryMemoryNetworking			





Storage Device Warning



The storage device below may contain data.



ATA Virtual HD

130048.0 MB

pci-0000:00:07.1-scsi-0:0:0:0

We could not detect partitions or filesystems on this device.

This could be because the device is **blank**, **unpartitioned**, or **virtual**. If not, there may be data on the device that can not be recovered if you use it in this installation. We can remove the device from this installation to protect the data.

Are you sure this device does not contain valuable data?

☒ Apply my choice to all devices with undetected partitions or filesystems

Yes, discard any data

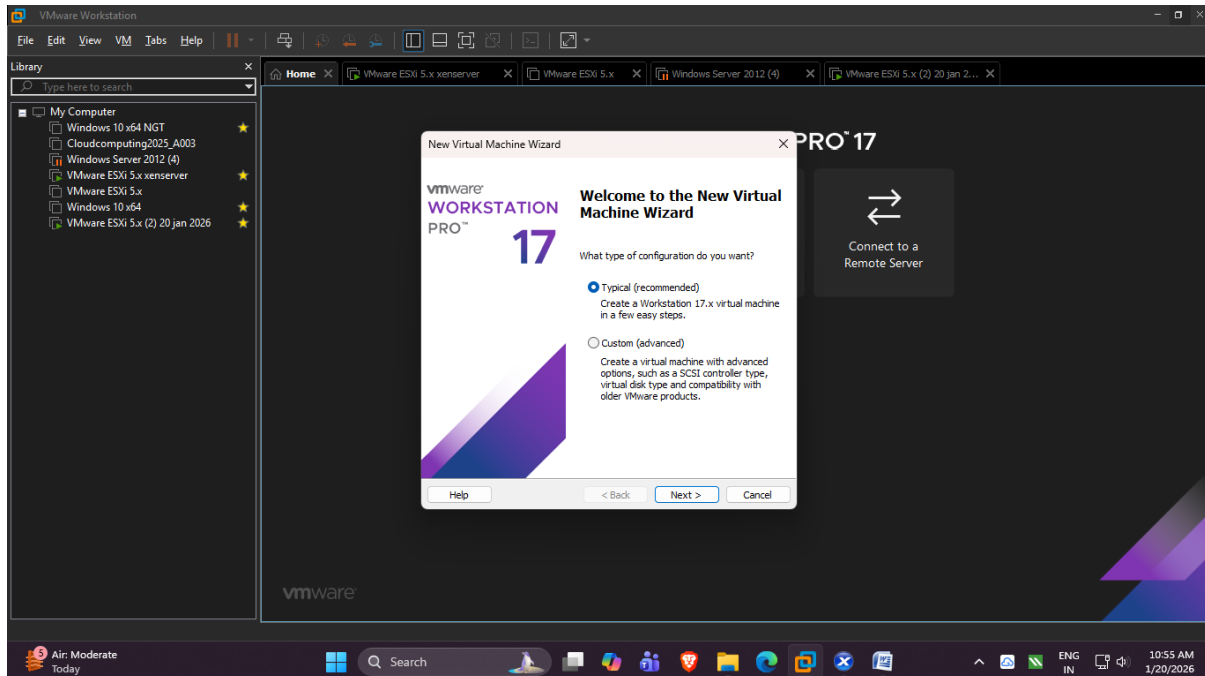
No, keep any data

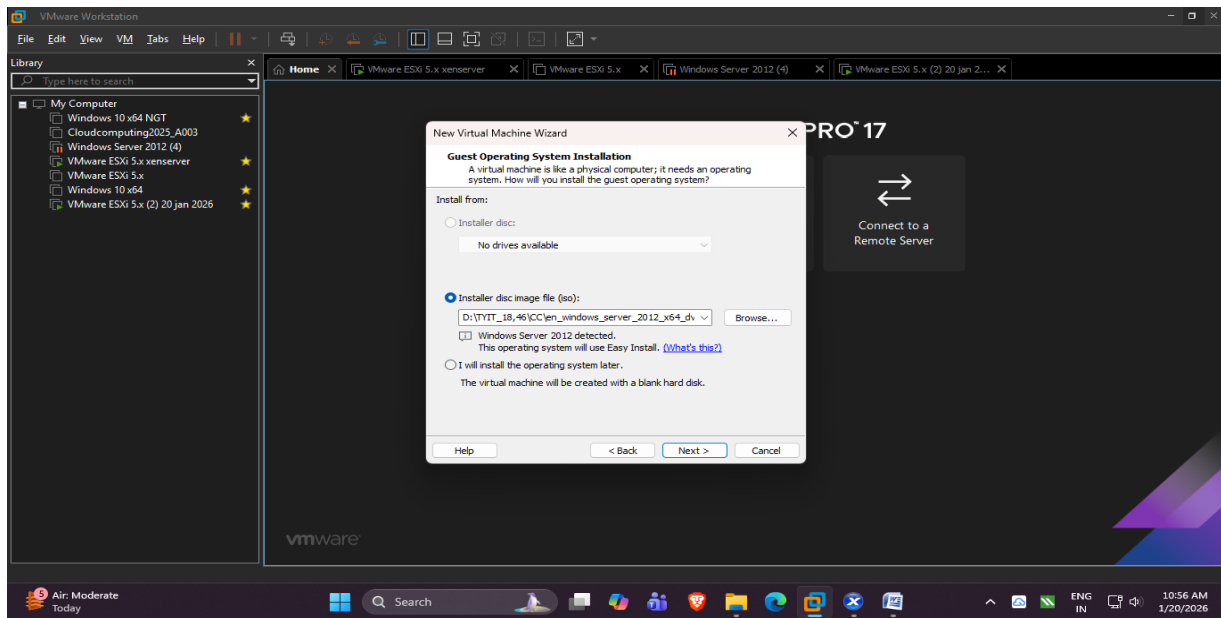
9. Implementing operating system on HyperV.

Implementing an operating system on Hyper-V means creating and running virtual machines on a physical server using Microsoft's built-in Type-1 (bare-metal) hypervisor integrated into Windows Server 2012. Running directly on hardware through the Windows Server kernel, Hyper-V allows a single physical server to host multiple virtual machines, each with its own operating system such as Windows or Linux. Every virtual machine receives dedicated CPU, RAM, storage, and network resources, ensuring isolation and optimal performance while enabling organizations to reduce hardware costs, improve efficiency, and simplify server management.

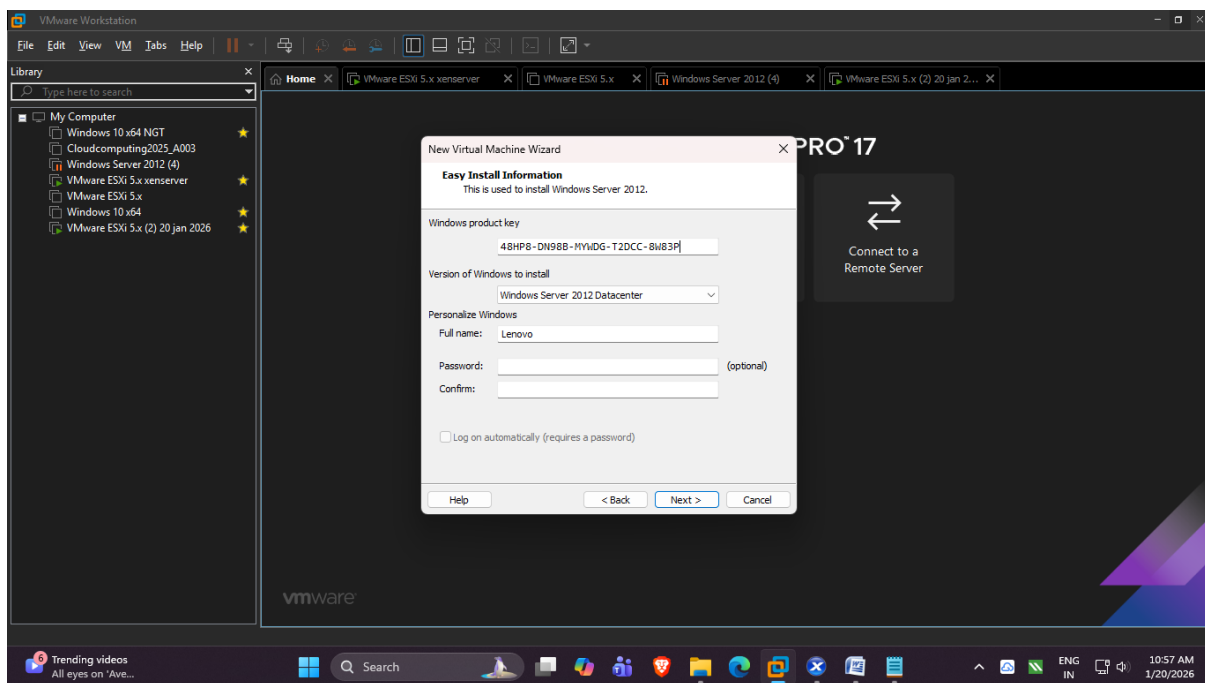
STEPS

Create new virtual machine

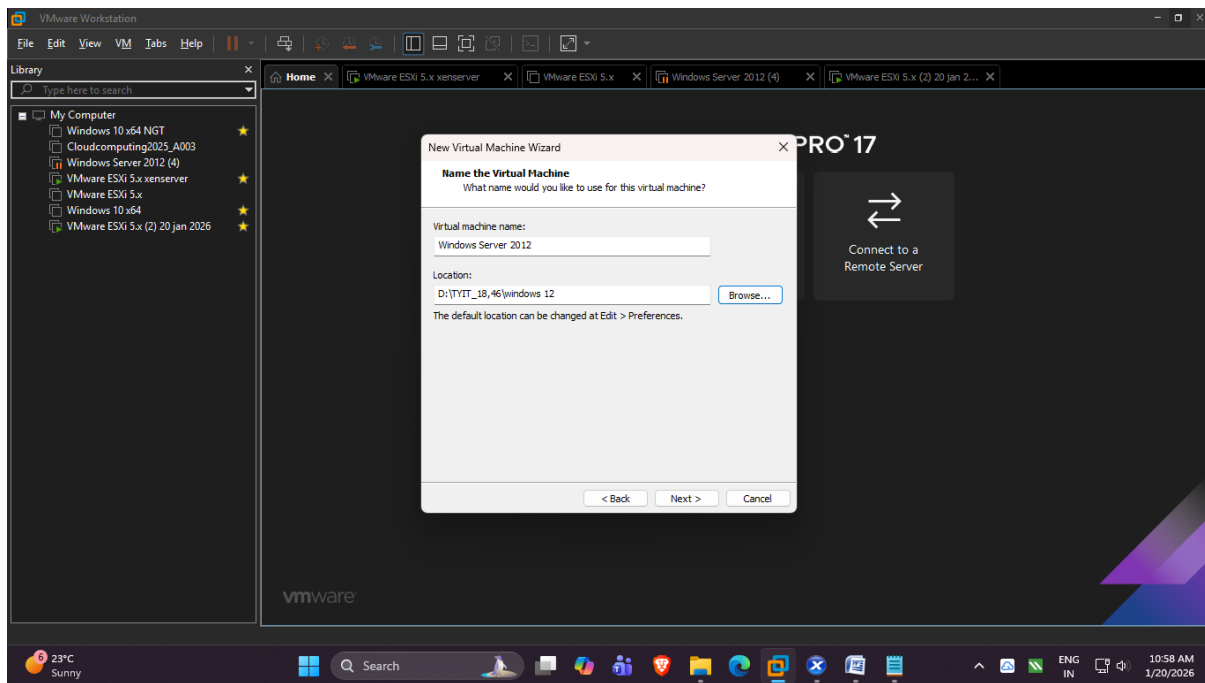




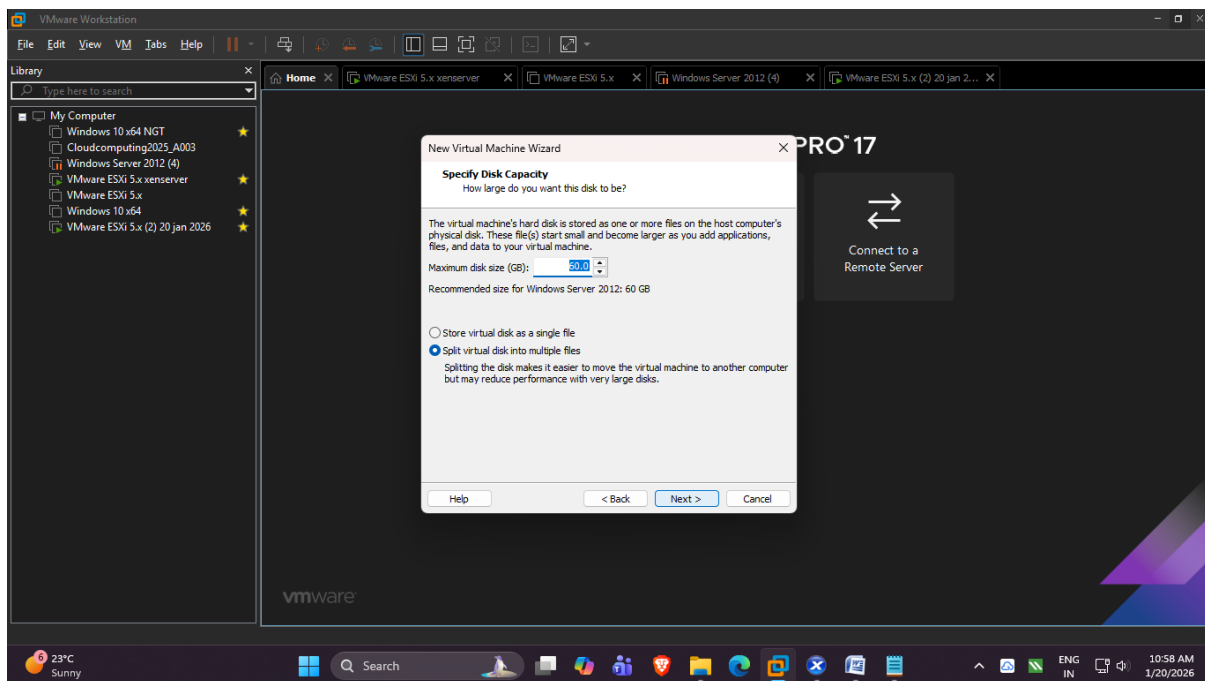
Give windows product key

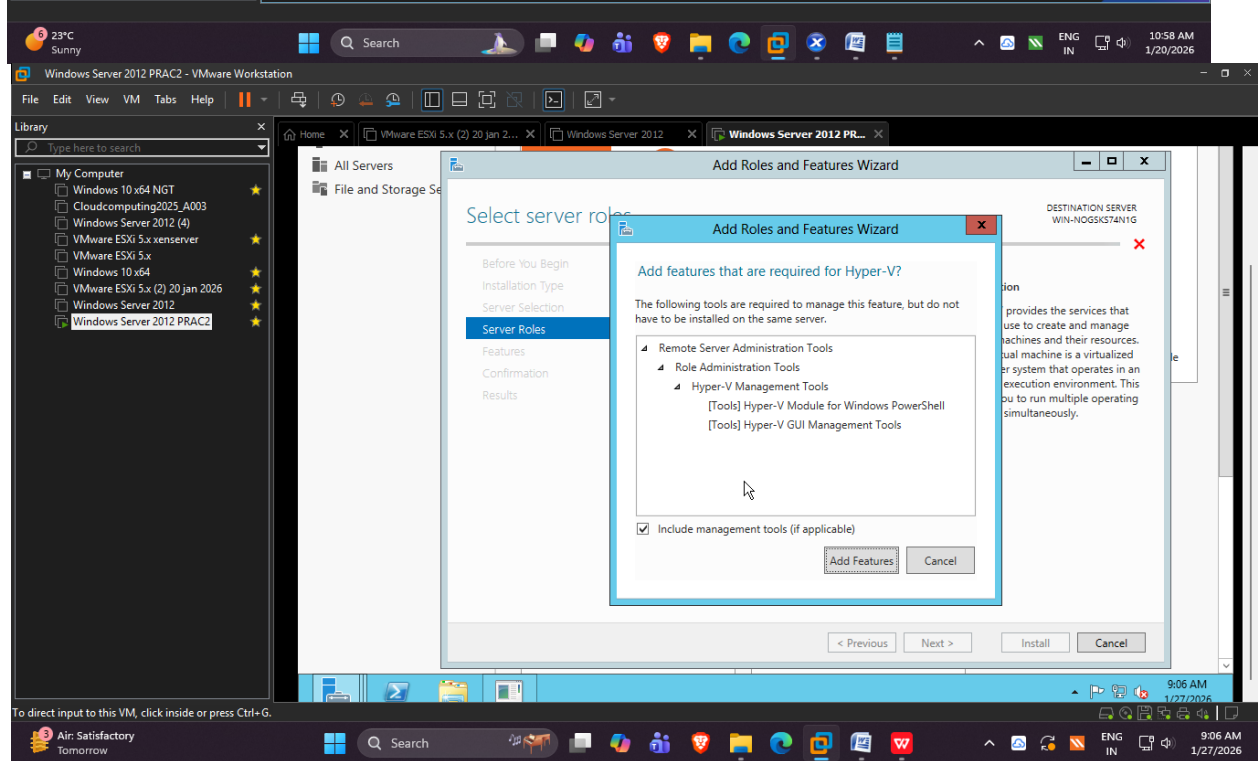
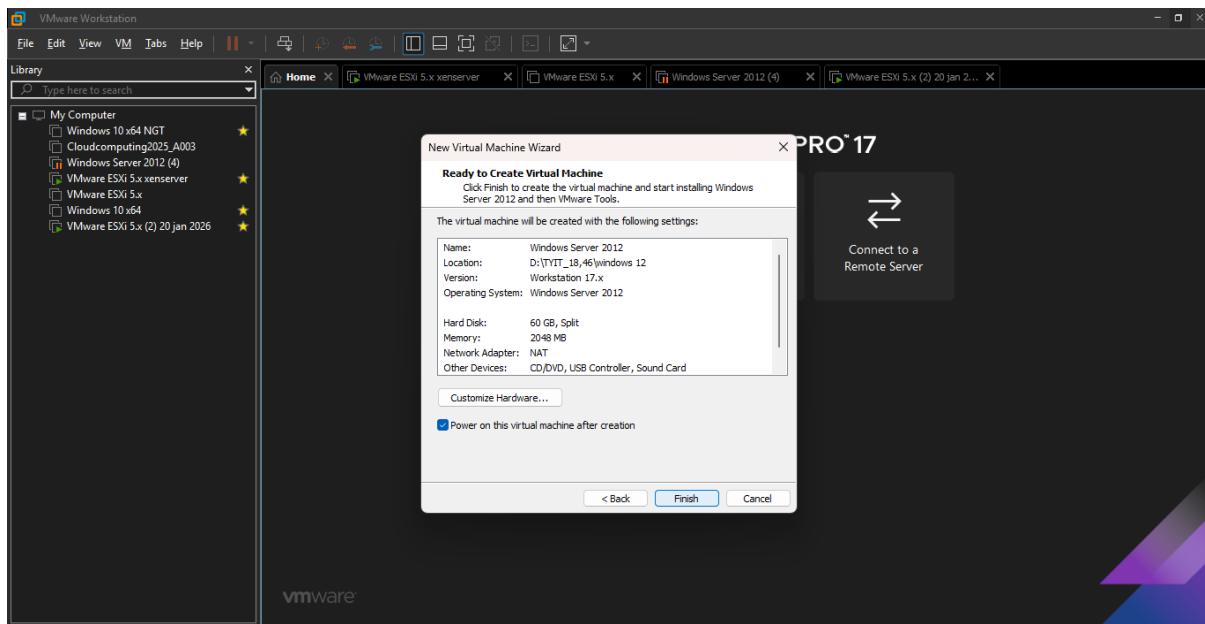


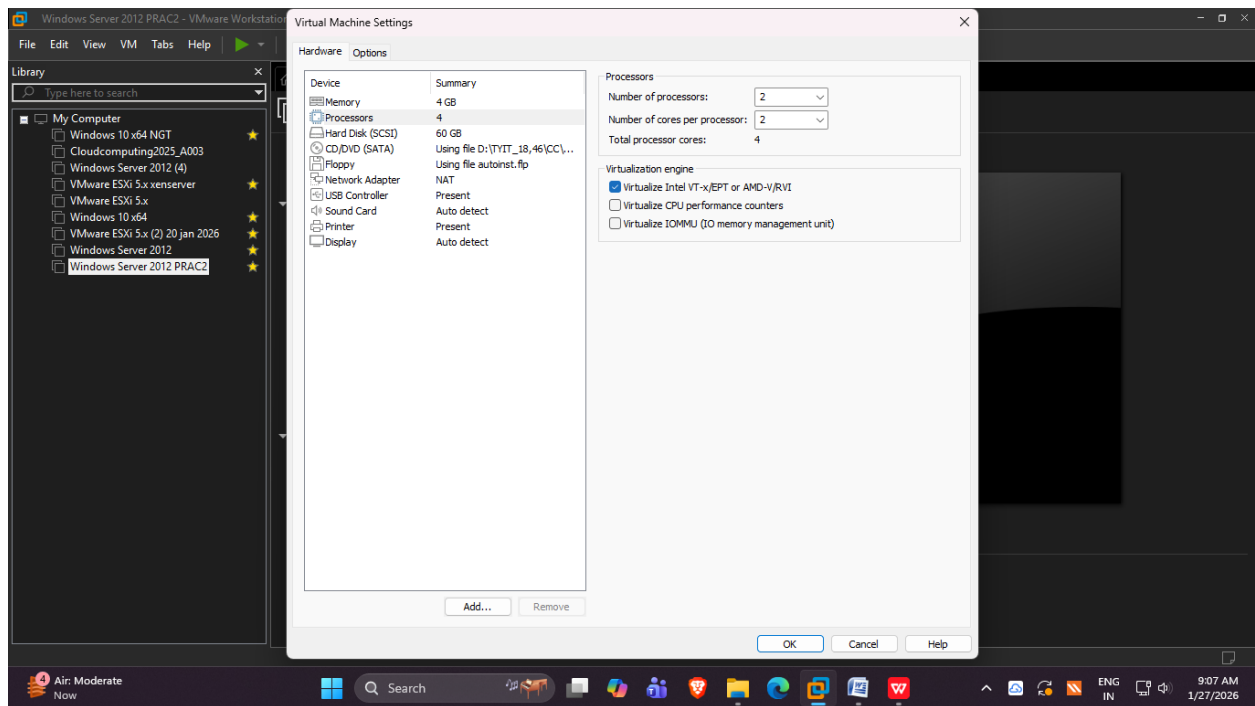
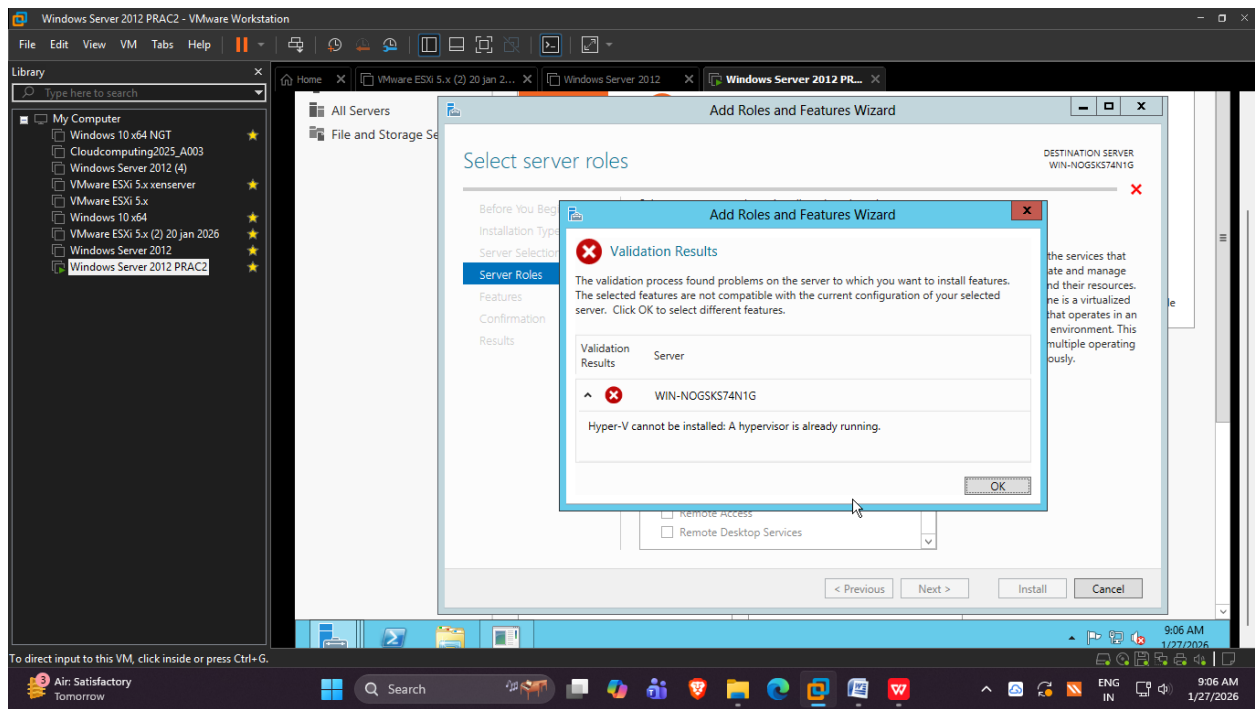
Choose the location to store the file in D drive



Choose the disk size as 60 GB



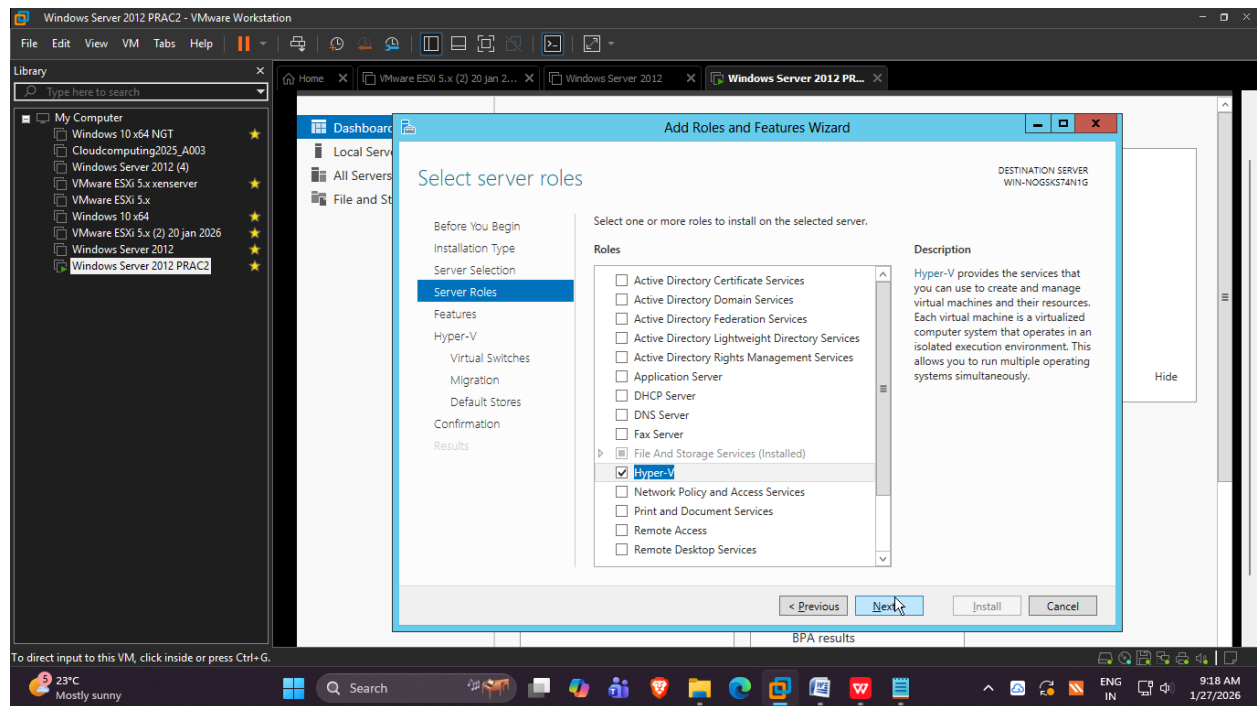




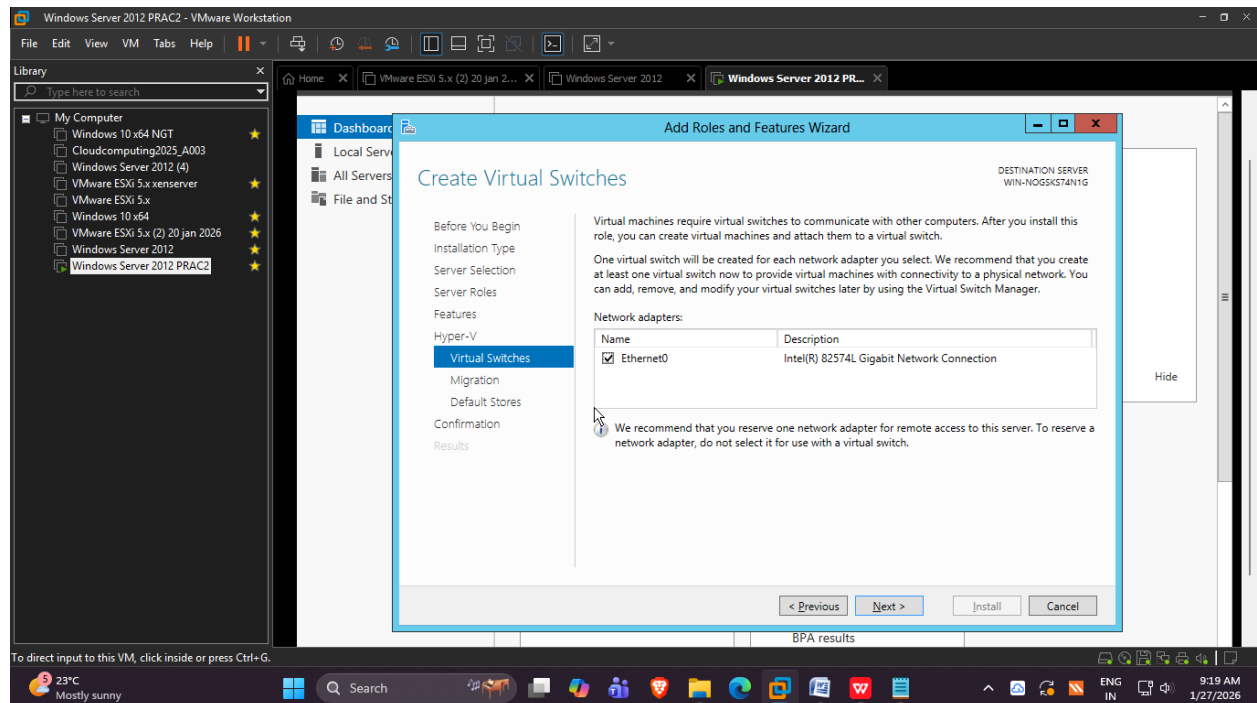
Open “.vmx” file in notepad

Power on machine

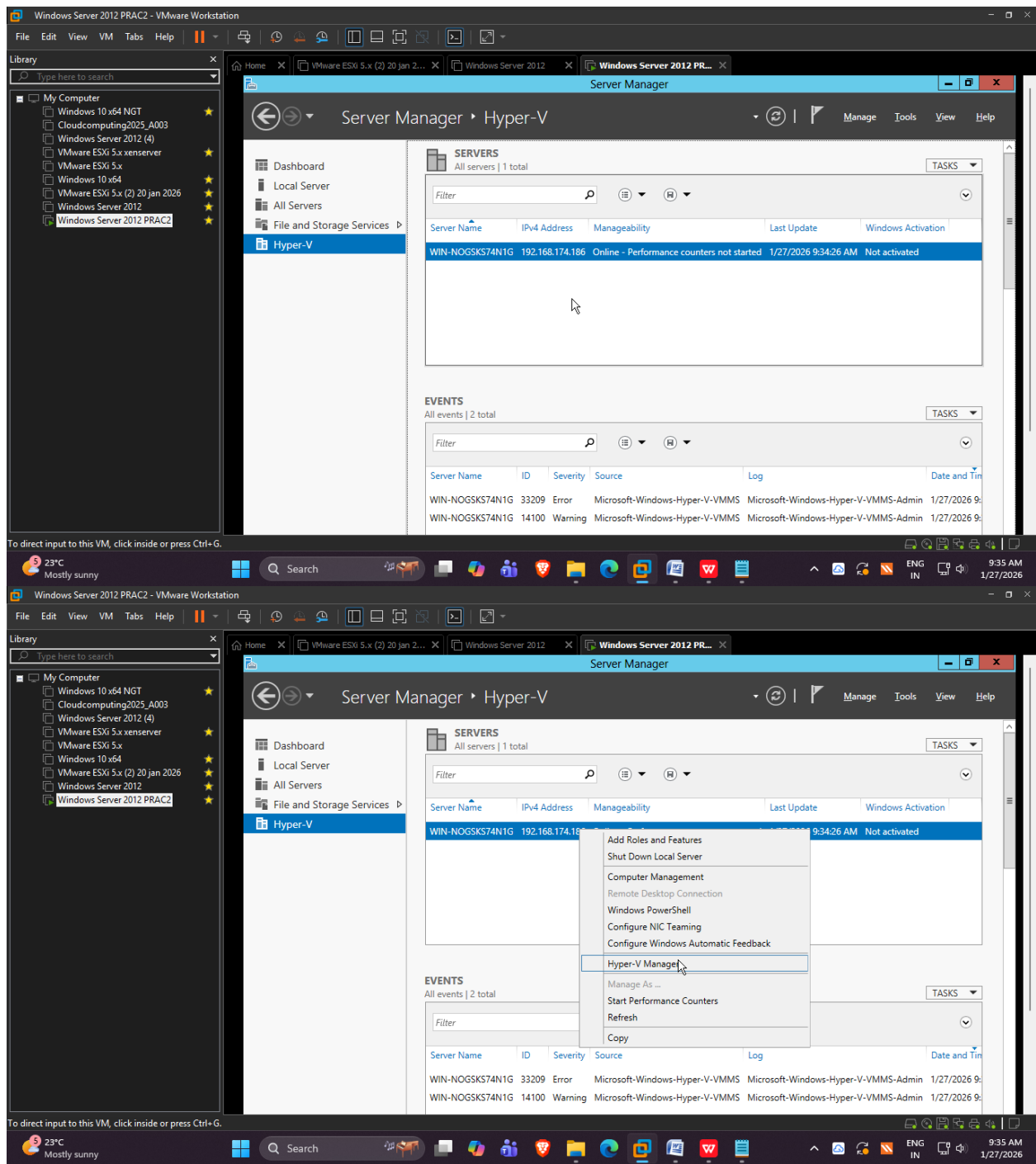
Click on add roles and feature



Select “ethernet0”



Restart guest



The machine is created

